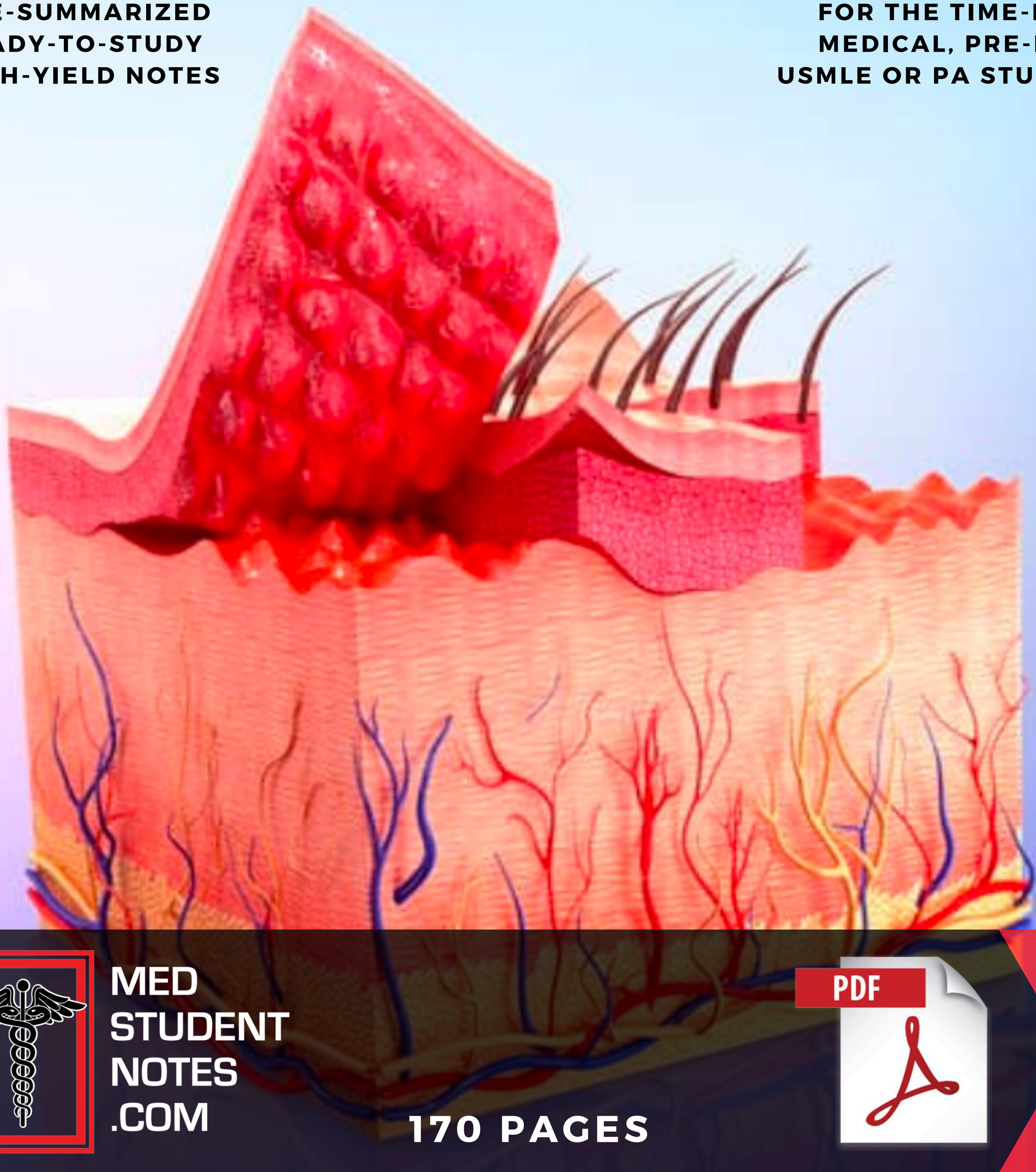
CLINICAL DERMATOLOGY

NOTES

FOURTH EDITION

PRE-SUMMARIZED
READY-TO-STUDY
HIGH-YIELD NOTES

FOR THE TIME-POOR
MEDICAL, PRE-MED,
USMLE OR PA STUDENT



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A Message From Our Team

Studying medicine or any health-related degree can be stressful; believe us, we know from experience! The human body is an incredibly complex organism, and finding a way to streamline your learning is crucial to succeeding in your exams and future profession. Our goal from the outset has been to create the greatest educational resource for the next generation of medical students, and to make them as affordable as possible.

In this fourth edition of our notes we have made a number of text corrections, formatting updates, and figure updates which we feel will enhance your study experience. We have also endeavoured to use only open-source images and/or provide attribution where possible.

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Table Of Contents:

What's included: Ready-to-study anatomy, physiology and pathology notes of the integumentary system presented in succinct, intuitive and richly illustrated downloadable PDF documents. Once downloaded, you may choose to either print and bind them, or make annotations digitally on your iPad or tablet PC.

(BLUE = CLICKABLE HYPERLINK)

OVERVIEW OF SKIN STRUCTURE & FUNCTION

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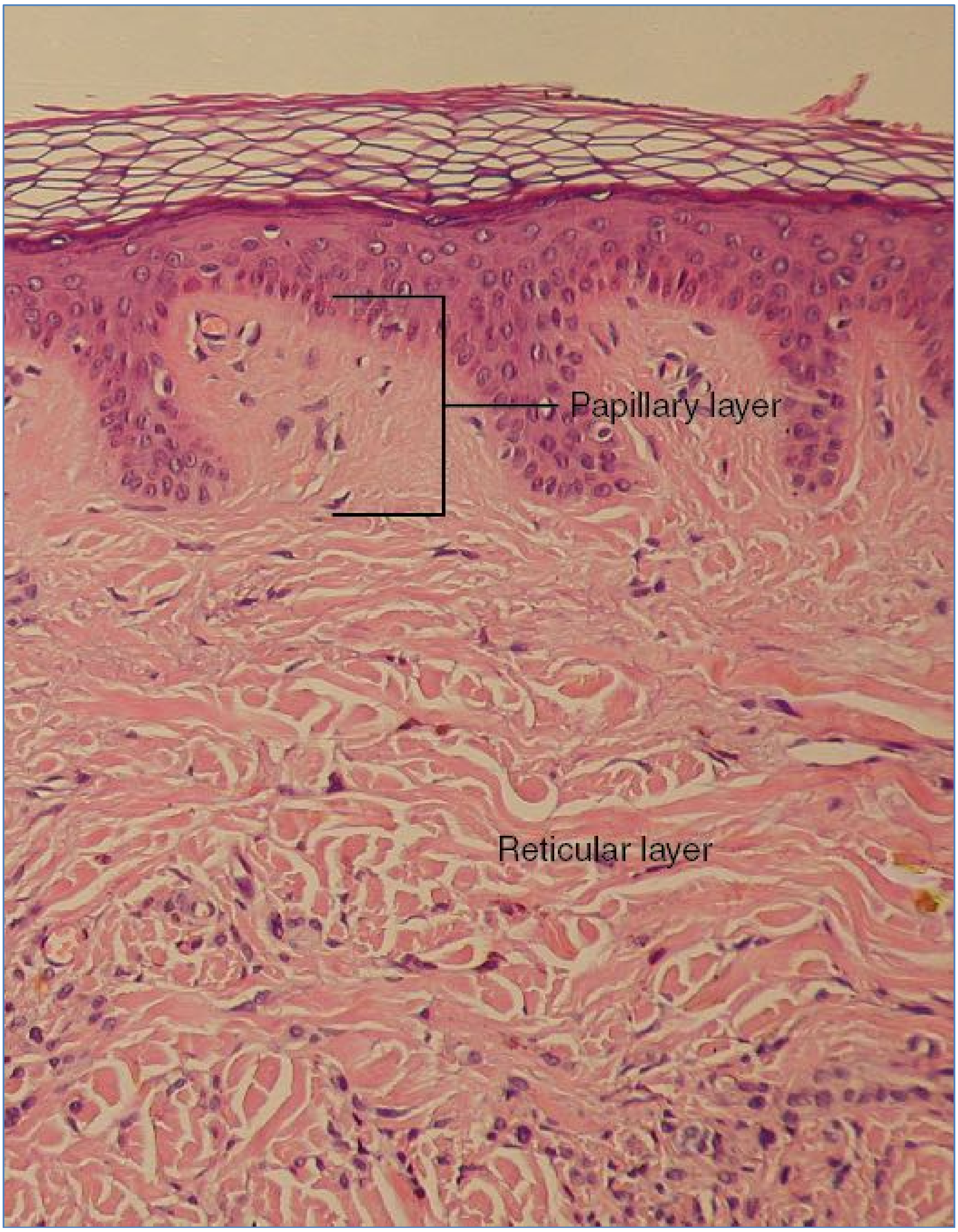
Functions of Normal Skin:

- Mechanical barrier
- Chemical barrier
- Prevent Fluid Loss (Overlapping Cells & Intercellular Lipid → Minimises loss of Water)
- Defence against micro-organisms
- Immunological barrier
- Endocrine organ – (Produces Vitamin-D under UV)
  - Vitamin D → Maintains Calcium Homeostasis (↑Ca^+ Absorption in Gut & ↑Renal Ca^+ Retention)
- Melanin → Protects against UV
- Thermoregulation – (Varying blood flow →Allows heat Conservation or Evaporative Cooling)
- Sensory organ

Basic Structure of Normal Skin:

- **3 Layers:**
  - **Epidermis (Top Cellular Layer):**
    - Keratinocytes
    - Melanocytes
    - **The 5 Layers of the Epidermis:**
      - **1) Stratum Germinativum/Basale:**
        - Single layer of cuboidal/columnar cells
      - **2) Stratum Spinosum (Prickle Cell layer):**
        - Several layers of polyhedral cells
      - **3) Stratum Granulosum:**
        - 3-5 layers flattened cells
      - **4) Stratum Lucidum:**
        - (Present only on very thick layers of epidermis (Eg: Glabrous Skin [palms/feet]))
        - = The Lucid layer of Flattened Cells before Stratum Corneum
      - **5) Stratum Corneum:**
        - 5-50 layers of Flattened, Dead Cells (Squames)
        - Protective Barrier; Holds in Moisture
        - Cytoplasm is filled with **keratin & keratohyalin granules**
  - **Dermis (Middle Fibrous Layer):**
    - Connective Tissue
    - Blood Vessels
    - Nerves (Sensory Receptors & Free Nerve Endings)
    - Hair Follicles + Arrector Pili (the Smooth Muscle)
      - **Note: Skin without hair** – (Eg: Palms & Soles) = ***"Glabrous Skin"***
      - **Note: Skin With Hair** – (Ie: Rest of the body) = ***"Non-Glabrous Skin"***
    - Glands (Sweat, Sebaceous)
    - **Contains Some Cells:**
      - **Fibroblasts**: Synthesis and degradation of connective tissue
      - **Histiocytes/Macrophages**: Phagocytic cells
      - **Mast cells**: Secretory cells →Vasoactive Mediators (histamine) → Skin Allergies
      - **Lymphocytes:** Small number collect around blood vessels in normal skin
    - **2 Layers:**
      - **(R) - Reticular Layer** (thick Collagen, lower layer) - much stronger
      - **(P) - Papillary layer** (fine Collagen, upper layer) – weaker
  - **Hypodermis (Fat Layer):**
    - Adipose Tissue
    - **Functions:**
      - Insulates the body
      - Stores Energy



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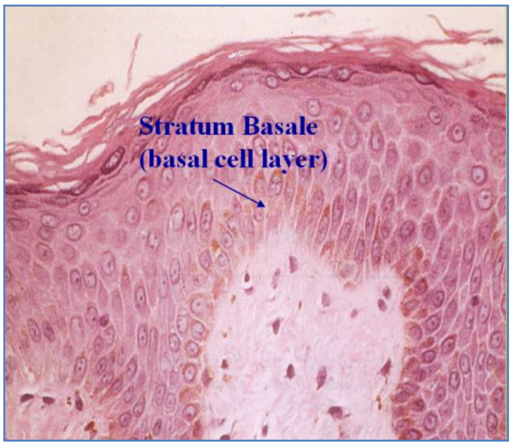
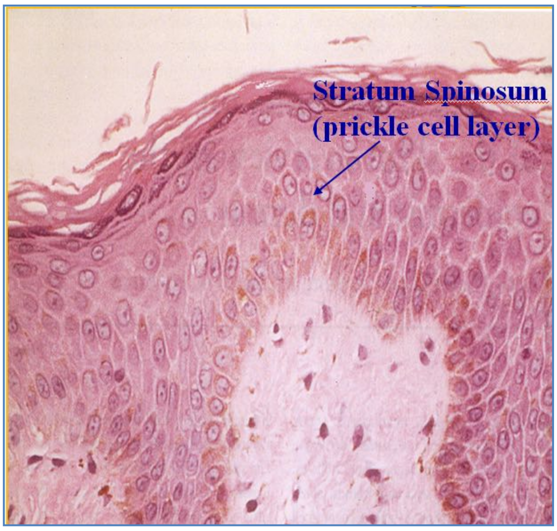
LAYER 1: THE EPIDERMIS:

- **Structure of the Epidermis:**
    - Most Superficial Layer of Stratified Squamous Keratinised Epithelium
    - Avascular
    - Almost Entirely Cellular:
        - 95% of Epidermal Cells are *Keratinocytes* (Produce Keratin)
        - Other Cells include: *Melanocytes, Merkel Cells & Langerhans Cells*
        - Continually regenerating (Rate is just sufficient to replace cells lost from the surface)
    - Protective barrier
    - Approximately 0.1 to 1mm thick

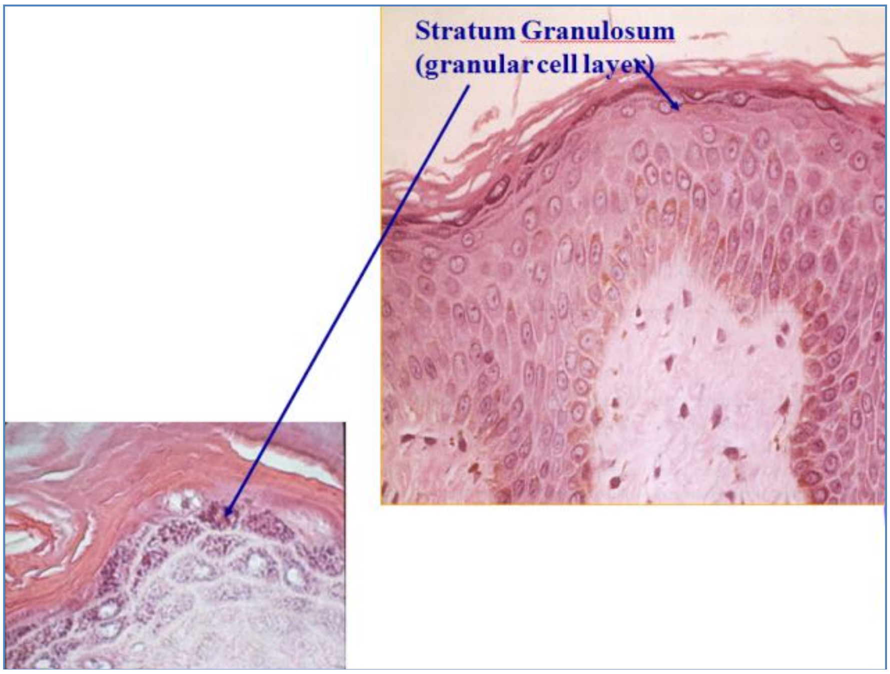
- **The 5 Layers of the Epidermis:**
    - **1) Stratum Germinativum/Basale:**
        - Single layer of cuboidal/columnar cells
        - **Basal Cells** take 14 days to differentiate → Keratinocytes
        - Keratinocytes take 14 days to be shed off
        - Strongly adherent to the Basement Membrane by **Hemidesmosomes**



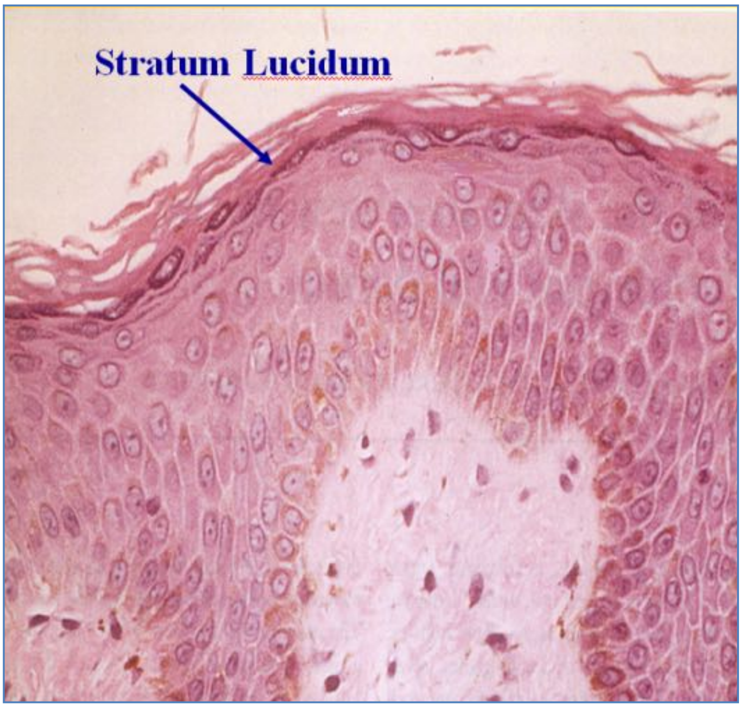
    - **2) Stratum Spinosum (Prickle Cell layer):**
        - Several layers of polyhedral cells
        - Keratin-Tonofibrils Insert into interconnecting desmosomes from adjacent cells
        - **Keratinocytes** Contain **'Lamellar (lipid containing) bodies'**:
            - Are Secreted into the Extracellular spaces (Sticks Cells Together)
                - → Forms a water-barrier
                - →Also acts to cement keratinized squames together in Stratum Corneum
        - **Langerhans Cells:**
            - Like Macrophages
            - Have Antigen Presenting Capacity



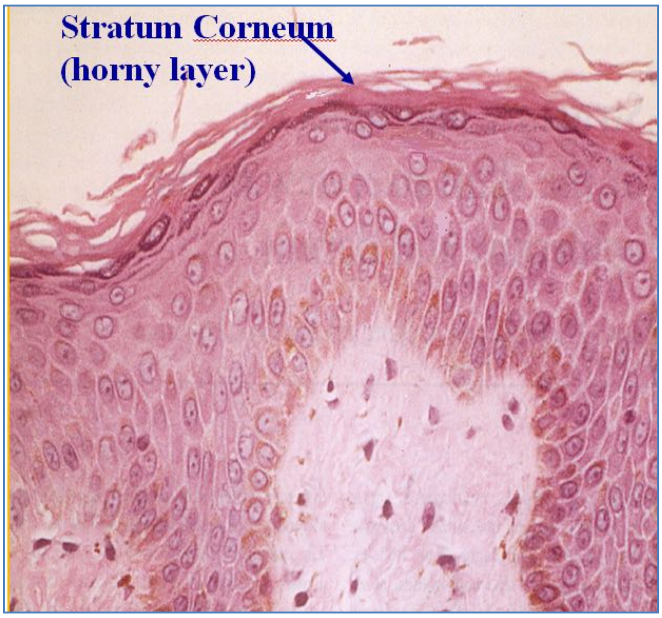
- **3) Stratum Granulosum:**
    - 3-5 layers flattened cells
    - Loss of nuclei
    - **Keratinocytes:**
        - **Keratohyalin granules** (Composed of *Profilaggrin*, *Keratin Filaments*, & *Loricrin*)
            - Keratohyalin = Protein involved in Keratinisation
            - Note: Eczema is probably due to a mutation in Profilaggrin
        - **Lamellar Body secretions:**
            - Lamellar Body Contents Discharged into Intracellular Spaces
                - (Lipids, Enzymes)
            - – Acts like the *"Mortar"* between the cellular *"Bricks"*



- **4) Stratum Lucidum:**
    - (Present only on very thick layers of epidermis (Eg: Glabrous Skin [palms/feet]))
    - = The Lucid layer of Flattened Cells before Stratum Corneum
        - Cells have No nuclei or organelles

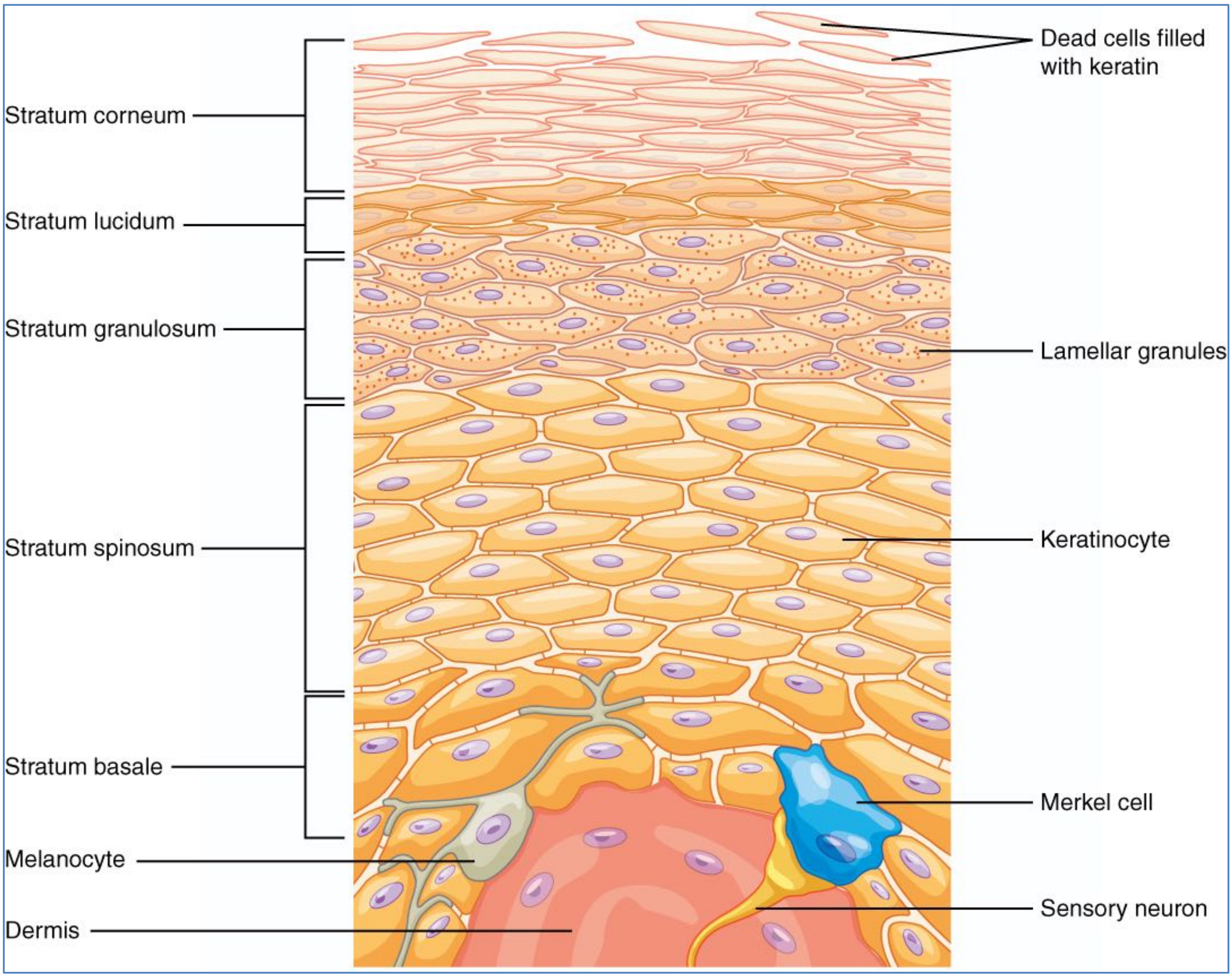


- **5) Stratum Corneum:**
  - 5-50 layers of Flattened, Dead Cells (Squames)
    - Devoid of Nuclei & Organelles
  - Protective Barrier; Holds in Moisture
  - Cytoplasm is filled with **keratin & keratohyalin granules**
  - Cells are stuck together by contents of **Lamellar Bodies**
  - Thicker on Glabrous Skin (Palms and soles)



C = Corneum; L = Lucideum; G = Granulosum; S = Spinosum; B = Basale

- **Cellular Component of the Epidermis:**
  - ***Keratinocytes* – (In all Layers):**
    - Form the *Keratinized* Stratified Squamous Epithelium
    - Contain **Keratin** (Protein) Intermediate Filaments in the Cytoplasm
    - **Keratin:**
      - **Two types of epithelial keratins:**
        - Type I (acidic)
        - Type II (basic)
        - (Note: Each are produced in Equal Amounts)
      - **Keratin Proteins Polymerise as Heterodimers:**
        - Type 1 + type 2 → Polymers
      - **Keratin Polymers aggregate → Keratin Filaments**
      - **Keratin Filaments aggregate → Tonofilaments**
    - **"Keratinisation" – Keratinocytes Differentiate as they Migrate Upwards:**
      - Formation of Keratin proteins in Cytoplasm
      - →Aggregation of the Keratin Filaments into **'Tonofilaments'** (Like a rope)
        - (*Tonofilaments* →Anchor Desmosomes to the Cytoskeleton)
    - **Join to Adjacent Cells via:**
      - **Desmosomes:**
        - Intercellular attachments between keratinocytes
      - **Hemidesmosomes:**
        - Attachment of the Basal Keratinocytes (Epidermis) to the Basement Membrane (Dermis)
        - Very Tight Connection



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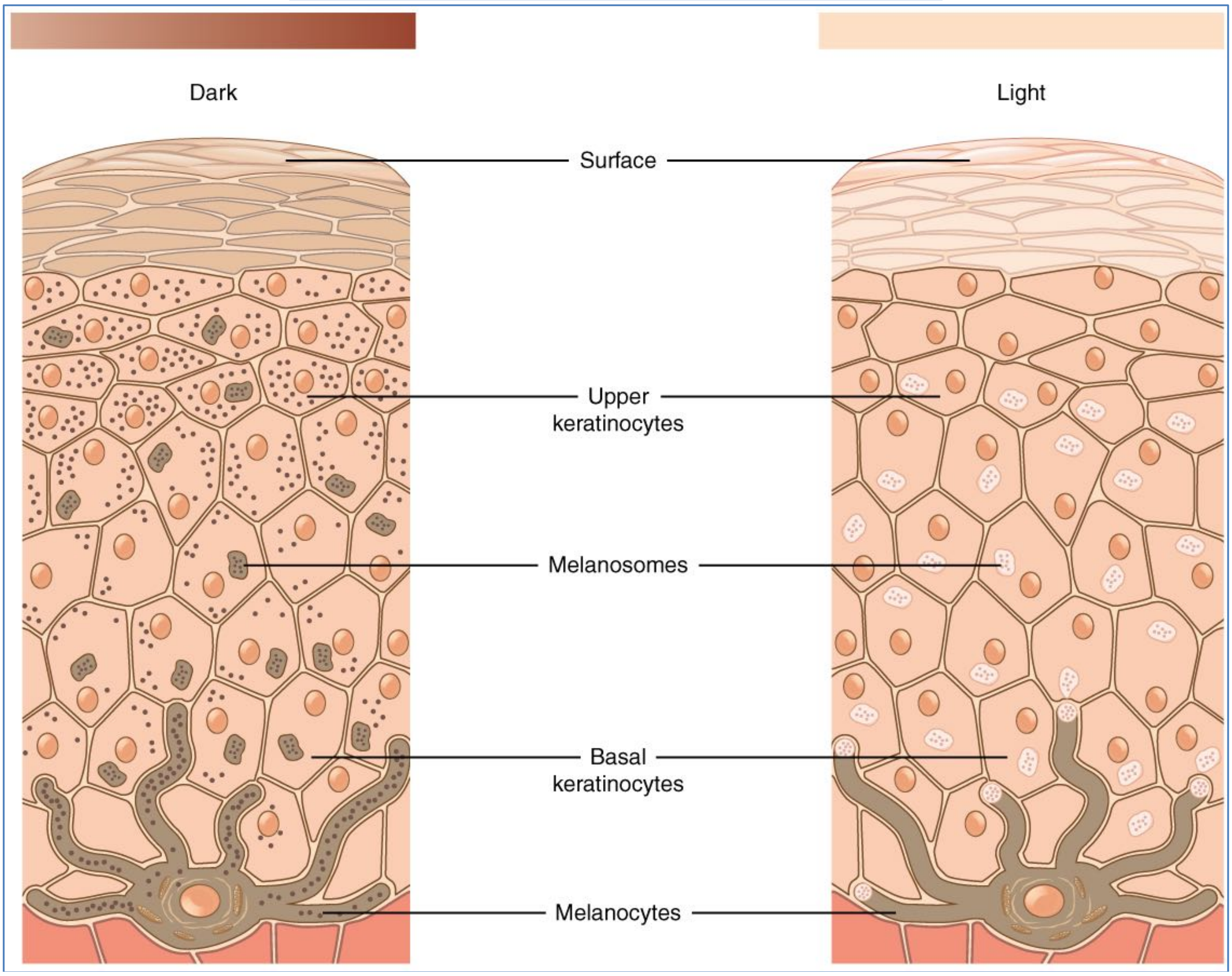
- *__Melanocytes__* **– (In Stratum Basale):**
    - **Pigment Cells – Synthesise Melanin Pigment**
        - Melanin contained in *Melanosomes*
        - Transferred to keratinocytes via Dendritic Processes
        - Melanin Absorbs UV radiation
    - **Located in Stratum Basale, Dermis and Hair Follicles**
    - **Note: Differences in Racial Pigmentation:**
        - Due to ↑Melanocyte *Activity* (NOT Melanocyte *Number*)



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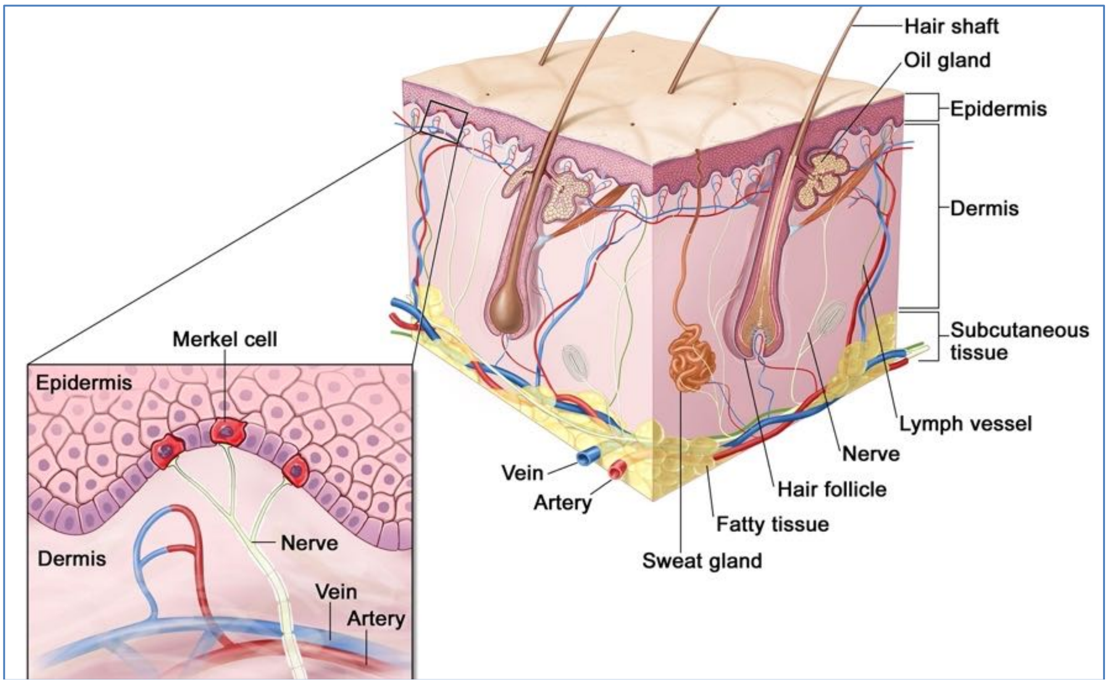
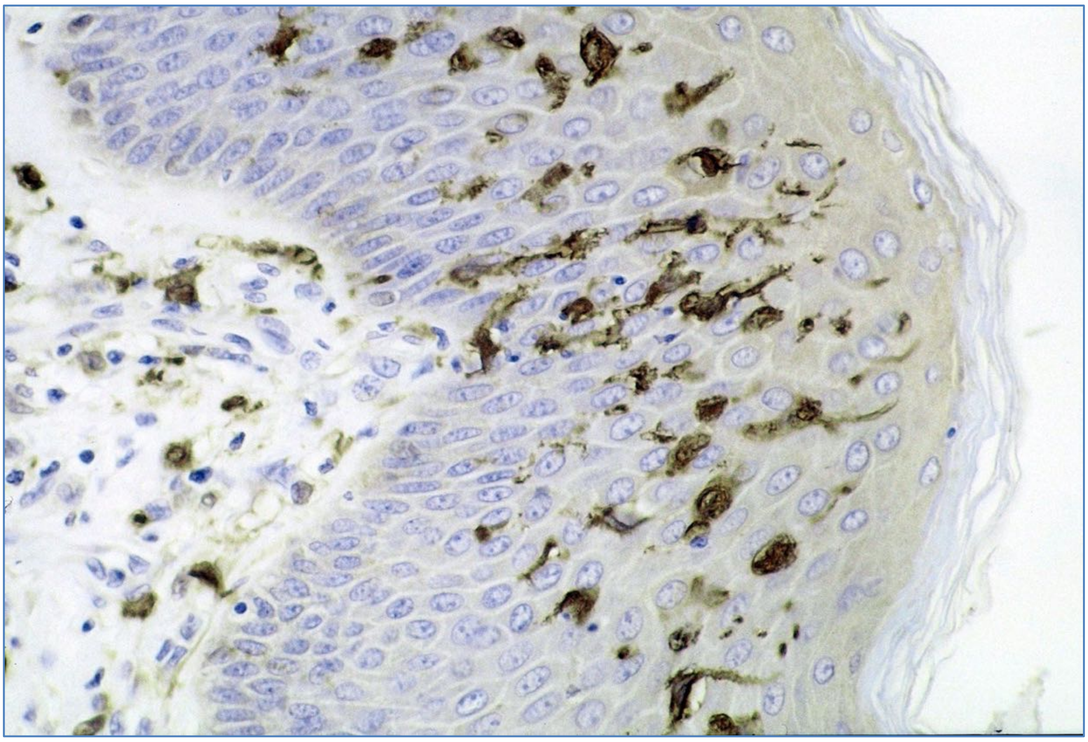
- ***Merkel Cells*** **– (In Stratum Basale):**
    - Nerve Ending Associated
    - Contain small, dense Granules of Catecholamines
    - **Located in Stratum Basale**
    - - Probably Neuro-Endocrine Function
        - Sensory mechanoreceptors
        - Closely associated with Nerve Terminal Filaments (free nerve endings)



Source: https://www.dana-farber.org/merkel-cell-carcinoma/

- ***Langerhans Cells*** **– (In Stratum Spinosum):**
    - Antigen Presenting Cells of the Immune System
    - Have long Dendritic Processes that radiate through the Epidermis
    - **Located in the Stratum Spinosum**
    - Migrate through Epidermis and Dermis to Lymph Nodes
    - - Immune Function



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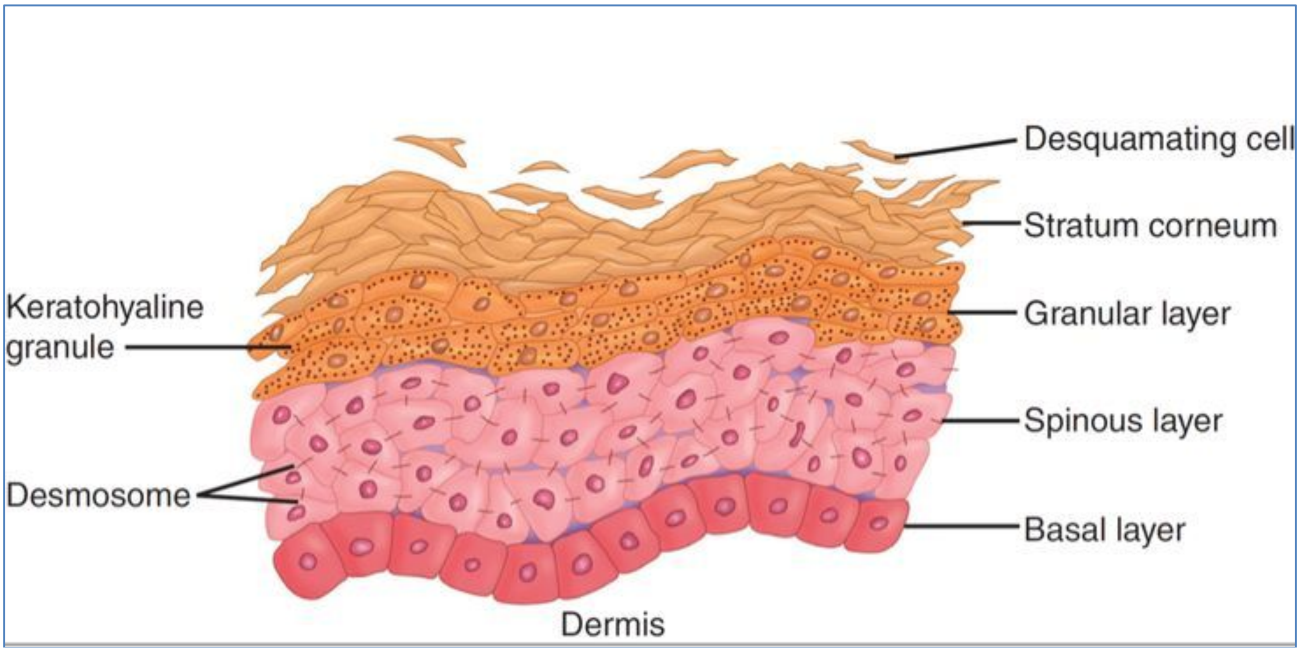
- **<u>Epidermopoiesis (Epidermal Renewal) & Desquamation (Shedding of Skin):</u>**
    - **<u>(Note: Rate of Epidermopoiesis must = Rate of Desquamation)</u>**
        - **Stimulated by Growth Factors:**
            - Epidermal Growth Factor (EGF)
            - Transforming Growth Factor a (TGFa)
            - Fibroblast Growth Factor (FGF)
        - **Inhibited by Cytokines:**
            - Interferon Gamma (IFNy)
            - Tumour Necrosis Factor (TNF)

    - **<u>Epidermopoiesis (Epidermal Renewal):</u>**
        - Epidermis continually renews itself
        - - by Basal (Columnar) Cell Division
        - Turnover rate ≈6 Weeks
        - **Basal Cells Become → Keratinocytes (Epidermal Cells):**
            - Via Keratinization = Synthesis of Keratin Protein
        - Transit time to the stratum corneum is approx 14 days

    - **<u>Desquamation (Shedding of the Skin):</u>**
        - Occurs when desmosomes are degraded (Mediated by Cholesterol Sulphatase)
        - Or by Wear & Tear
        - Desquamation requires another 14 days



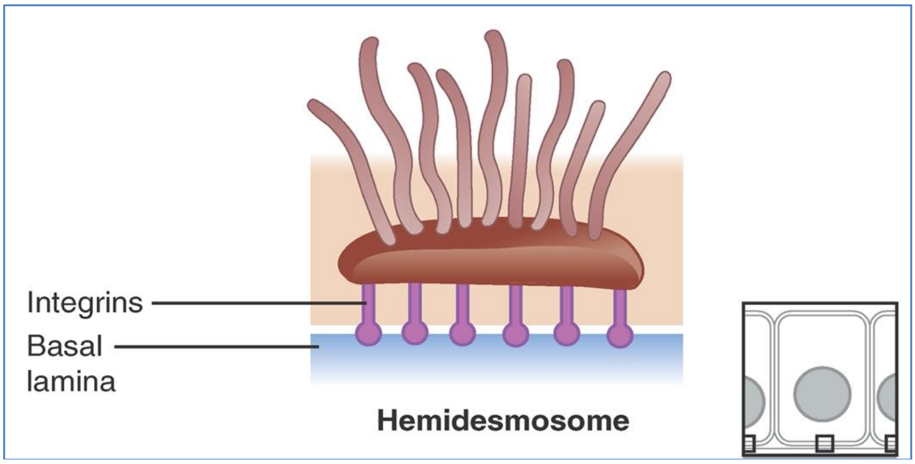
Baumann, L. S., & Baumann, L. (2009). Cosmetic dermatology. McGraw-Hill Professional Publishing.

DERMAL-EPIDERMAL JUNCTION:

- **Dermal-Epidermal Junction (Stratum Basale: Basement Membrane):**
    - Where the Stratum Basale of the Epidermis attaches to the Basement Membrane
    - **Stratum Basale:**
        - **Basal Cells** (Basal Keratinocytes)
            - ***Hemidesmosomes*** on Basal Cells attach the Epidermis to the Basement Membrane
        - **Melanocytes** are Interspersed amongst the Basal Cells
            - Large Dendritic Cells
            - Responsible for Melanin Pigment Production
        - **Merkel Cells:**
            - - Probably Neuro-Endocrine Function
            - Contain small, dense Granules of Catecholamines
    - **Basement Membrane:**
        - **Lamina Lucida** - anchoring transmembrane filaments of the Hemidesmosomes
        - **Lamina Densa** - (Lattice structure of type IV Collagen)

    - **Hemidesmosomes:**
        - Important in maintaining adhesion between Dermis & Epidermis
        - Associated with Keratin Filaments & Keratin Tonofibrils

    - **Epidermis & Dermis join in a ripple-like fashion for extra strength (in addition to desmosomes):**
        - **Rete Ridges** (Epidermis)
        - **Dermal Papilla** (Dermis)



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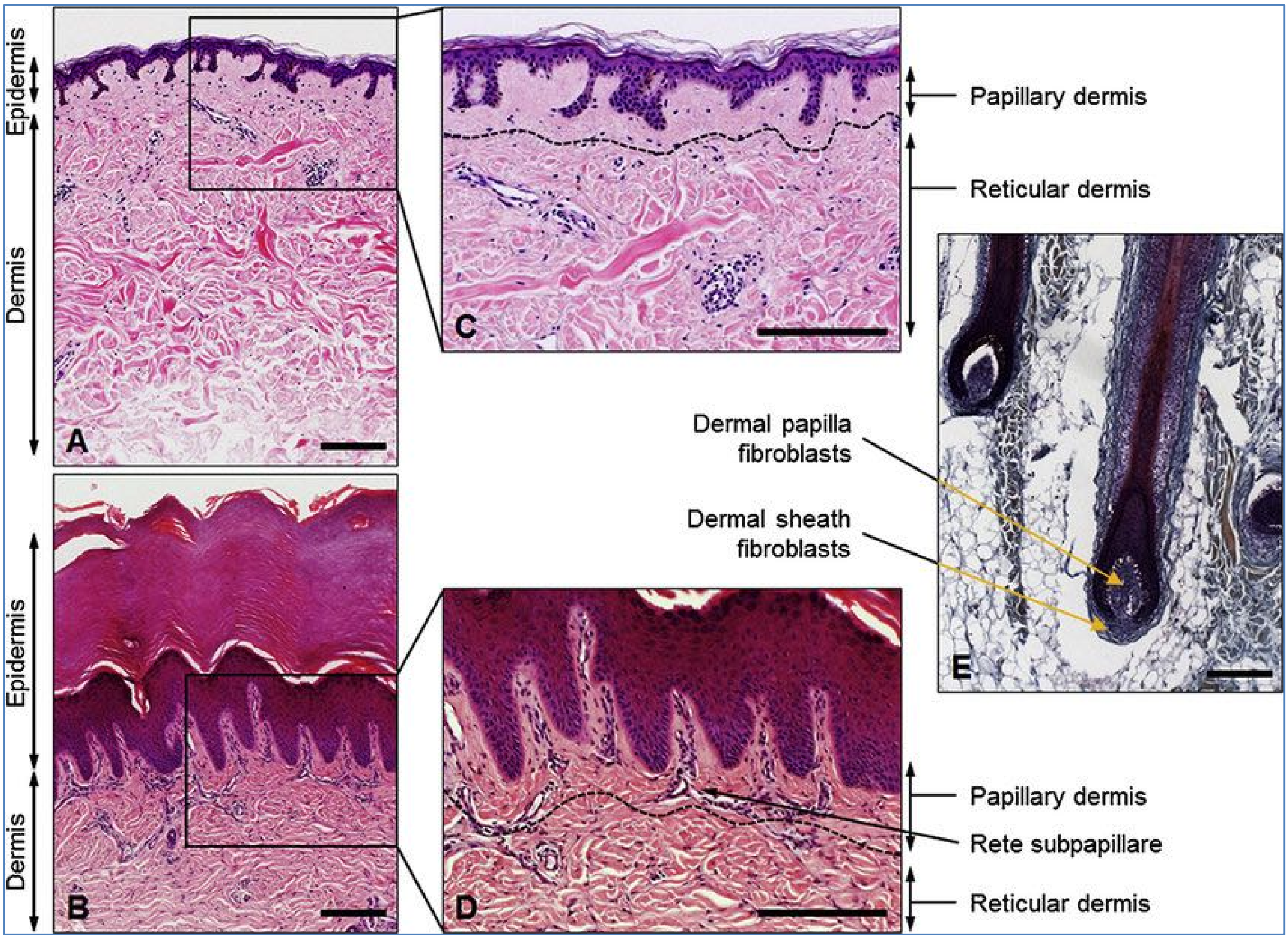
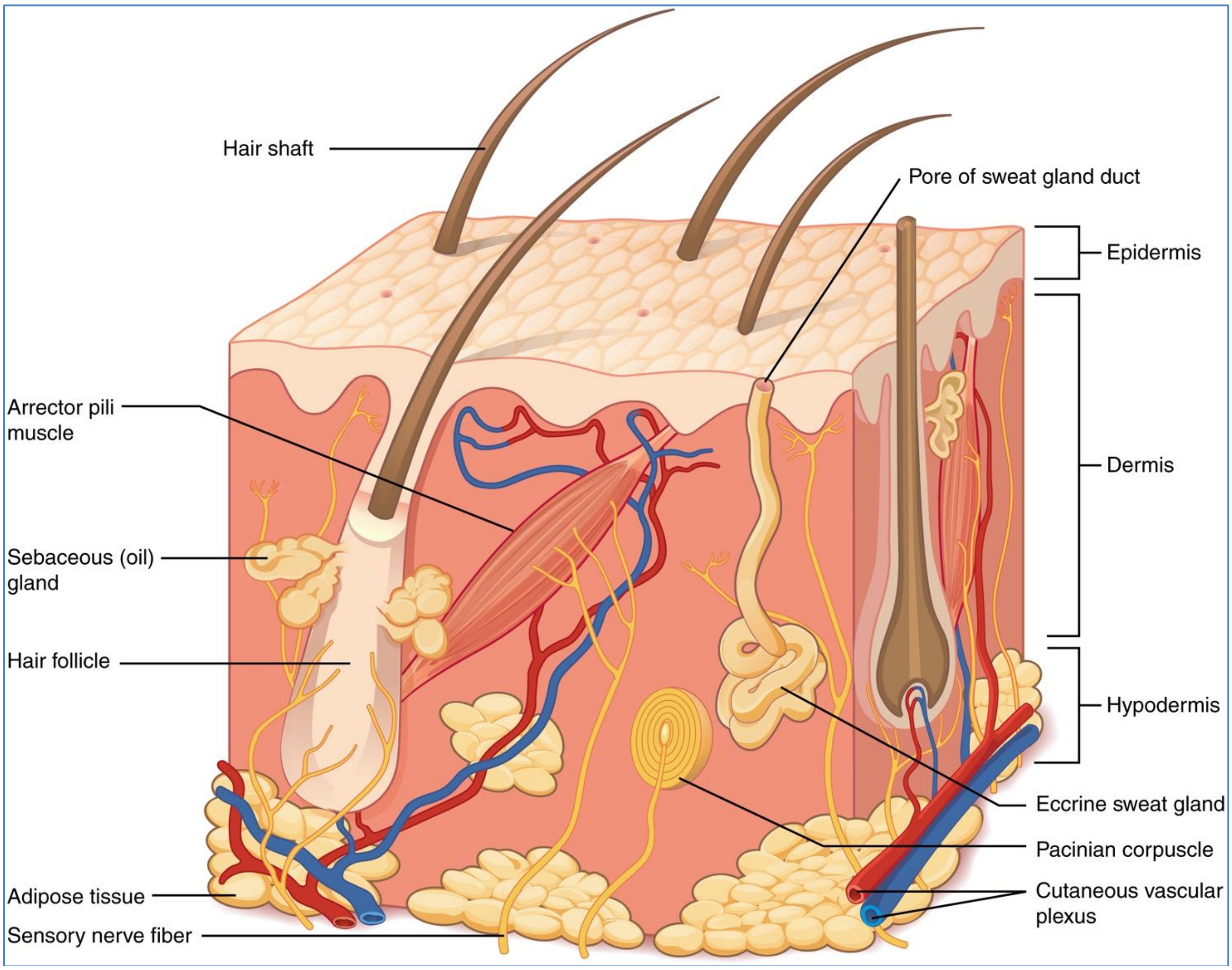
LAYER 2: THE DERMIS:

- **Structure of the Dermis:**
  - **Mostly Extracellular Matrix:**
    - Extracellular Collagen
    - Elastin
  - **Contains Some Cells:**
    - **Fibroblasts**: Synthesis and degradation of connective tissue
    - **Histiocytes/Macrophages**: Phagocytic cells
    - **Mast cells**: Secretory cells →Vasoactive Mediators (histamine) → Skin Allergies
    - **Lymphocytes:** Small number collect around blood vessels in normal skin
  - **Also contains Neurovascular & Other Auxiliary Skin Structures:**
    - Blood Vessels
    - Nerves
    - Sweat Glands
    - Hair Follicles (And Arrector Pili – The Smooth Muscle of the Hair Follicle)
  - **2 Layers:**
    - **(R) - Reticular Layer** (thick Collagen, lower layer) - much stronger
    - **(P) - Papillary layer** (fine Collagen, upper layer) - weaker



Available from: https://www.researchgate.net/figure/Adult-human-skin-consists-of-epidermis-and-dermis-The-epidermis-is-thrown-into-folds_fig2_281588115



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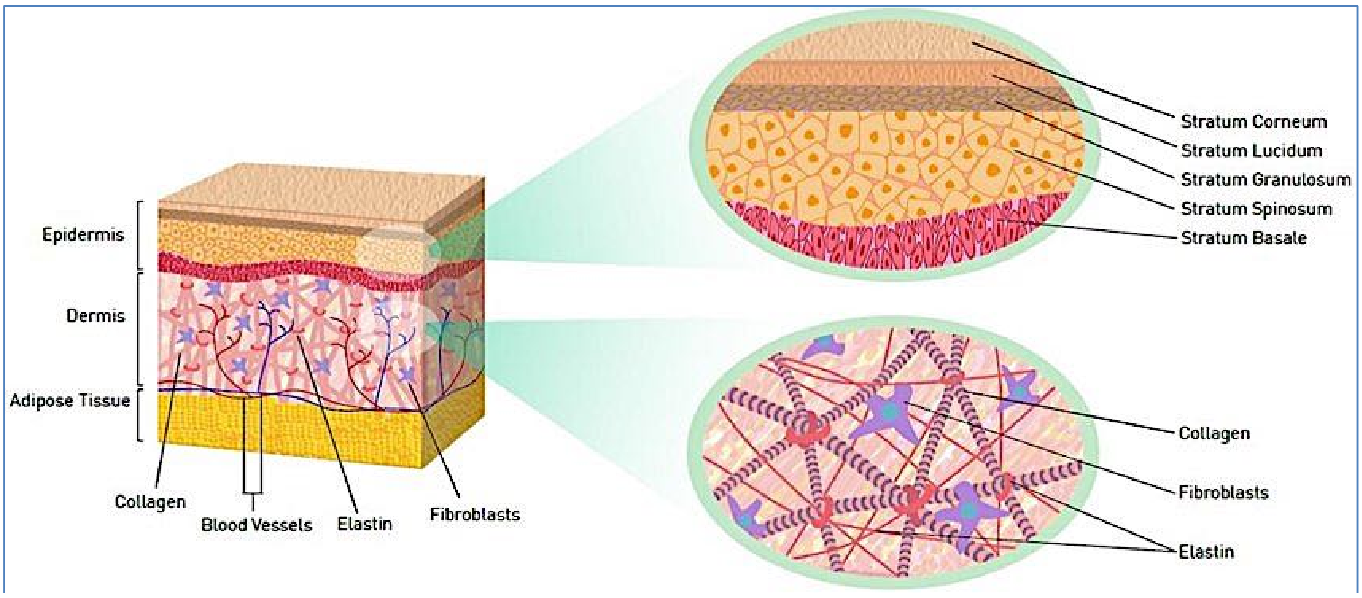
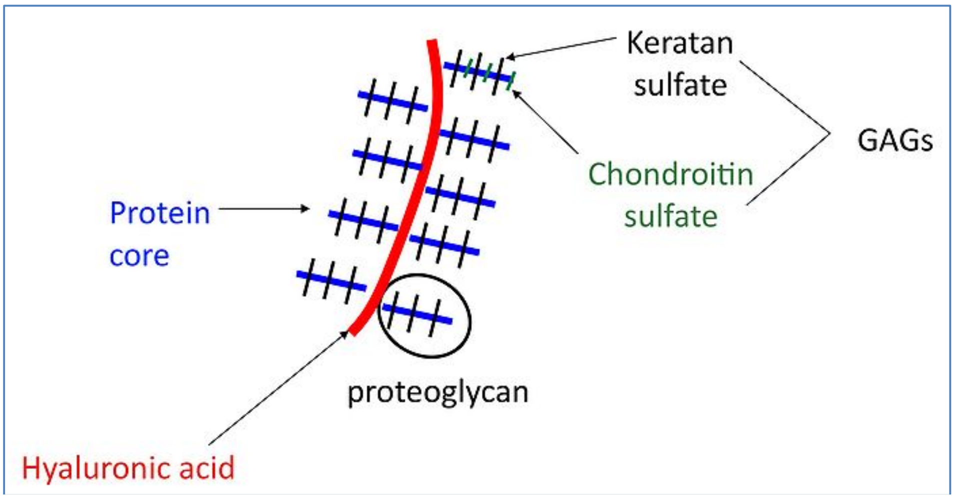
- **Connective Tissue of the Dermis:**
  - **#1 Collagen** = Very tough, fibrous protein – *High Tensile Strength*
    - The Predominant Protein of the Dermis (As opposed to Keratin of the Epidermis)
  - **Elastin** = Provides Elasticity to Skin – *Deformity with Memory*



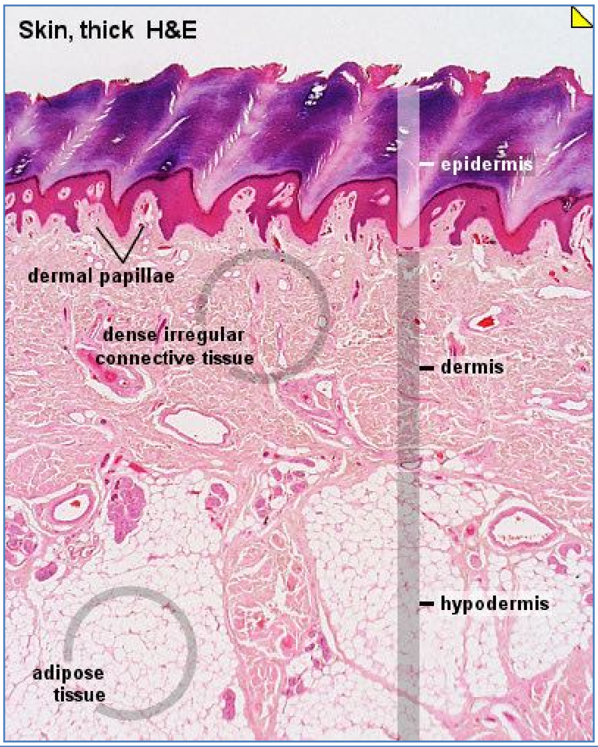
Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Fig-3-Diagram-showing-the-structure-of-healthy-skin-in-which-the-different-layers_fig6_276696375

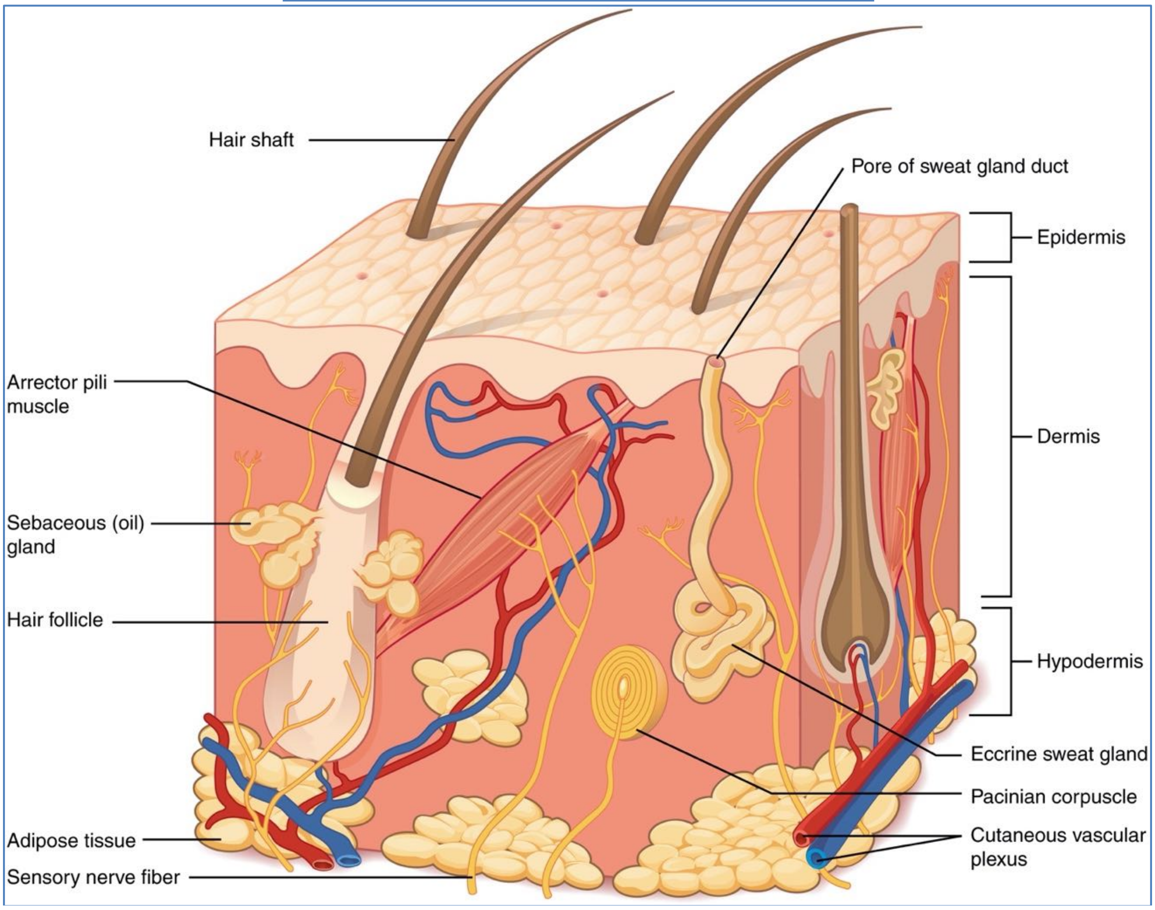
  - **Glycosaminoglycans (GAGs) =** Absorb water – *Provide Viscosity*
    - Hyaluronan backbone with glycoprotein branches



LAYER 3: THE HYPODERMIS:

- **Structure of The Hypodermis:**
  - **Composition:**
    - Mostly Fat (Adipose Cells)-(Thickest in Abdomen)
    - Blood Vessels
    - Nerves
  - **Functions:**
    - Insulates the body
    - Stores Energy





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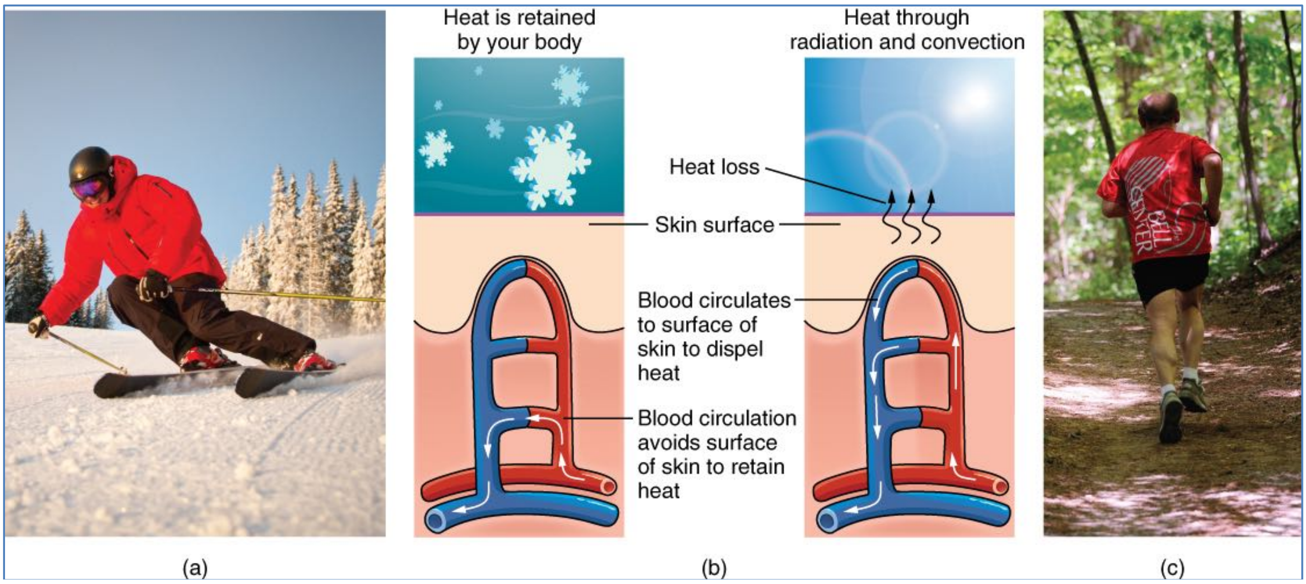
<u>AUXILIARY COMPONENTS OF SKIN:</u>

- **<u>Vasculature:</u>**
  - Abundant network of dermal vessels
  - **<u>Functions:</u>**
    - **Nutrition of Skin Tissue:**
    - **Regulation of Body Temperature:**
      - Conducts heat from Interior to Exterior → Heat loss to Environment
      - Vasodilation/constriction Important in Thermoregulation
    - **Blood Reservoir:**
      - Under conditions of circulatory stress (Eg: Exercise/Haemorrhage/Shock), Sympathetic Stimulation → Vasoconstriction → ↑Circulating Blood Volume

  - **<u>Cutaneous Circulatory Apparatus – 2 Types of Vessels:</u>**
    - **1) Nutritive Vessels:**
      - Arteries
      - Capillaries
      - Veins
      - (Organized into a horizontal ***Subdermal Plexus***):
        - Ascending arterioles extend towards the epidermis
          - ***Subpapillary Plexus*** (In the Papillary Dermis)
            - **Capillaries** loop into the dermal papilla
    - **2) Vascular Structures for Heat Regulation:**
      - **Extensive Subcutaneous Venous Plexi:**
        - Can hold Large quantities of blood
      - **Arteriovenous Anastomoses:**
        - Only present in areas of Maximal Cooling (Hands, Feet, Lips, Nose & Ears)
      - **Note: *BOTH Innervated by Sympathetic Adrenergic Vasoconstrictor Fibers:***
        - **Sympathetic NS** → Vasoconstriction → Minimal Heat Loss
        - (No Stimulation → Vasodilation → Maximal Heat Loss)



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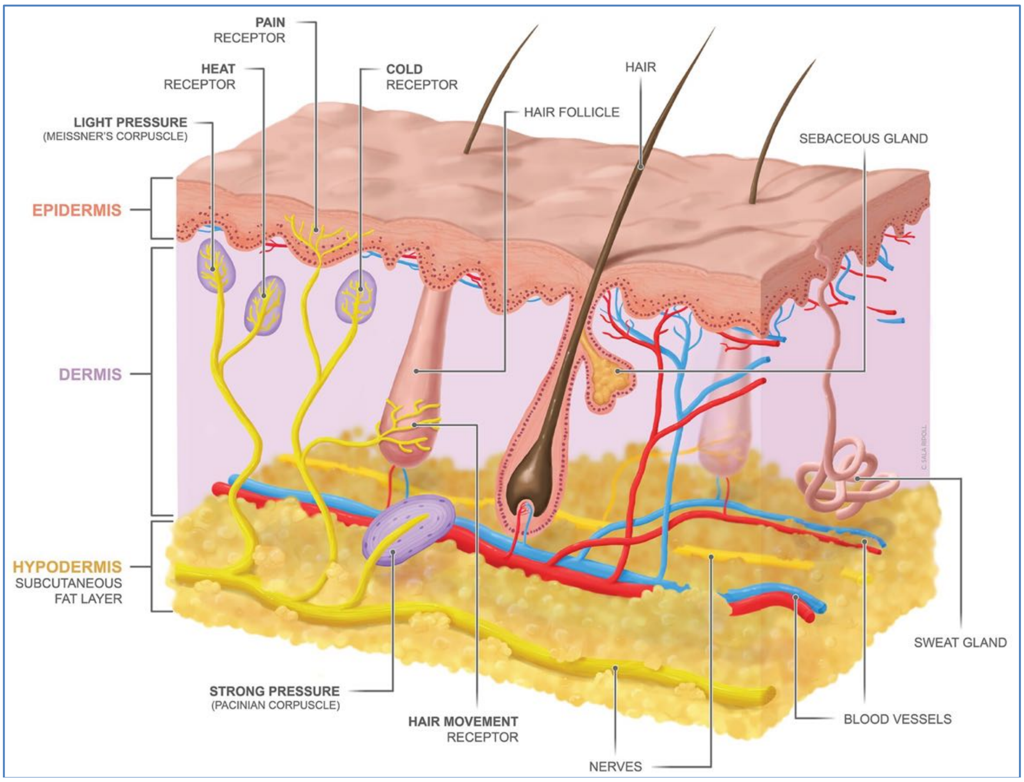
- **Cutaneous Sensory System:**
  - **Sensory Apparatus of the Skin:**
    - Most afferent nerve fibres terminate the Face & Extremities (Few on back)
  - **Sensory receptors:**
    - Receptors of touch, pain, temperature, itch and mechanical stimuli
    - Nerve fibers enter the dermis from the underlying adipose tissue
    - **1) Unencapsulated:**
      - **Free Nerve Endings:**
        - In Superficial Dermis & Epidermis
        - Receptors for Pain, Touch, Pressure, & Temperature
        - **Including Nociceptors:**
          - **A-Delta Fibres:**
            - Fast, Sharp, Pricking Pain
            - Thick, Myelinated → Fast Conduction
          - **C-Fibres:**
            - Slow, Dull, Aching, Burning Pain
            - Thin, Unmyelinated → Slow Conduction
      - **Merkel Cell Discs (Merkel Touch Spots):**
        - Found in the Stratum Basale of the Epidermis
        - Receptors for Light Touch
      - **Hair-Follicle Receptors:**
        - In & Surrounding Hair Follicles
        - Mechanoreceptors
    - **2) Encapsulated (Lamellated Capsule):**
      - **Meissner Corpuscles:**
        - In Dermal Papillae
        - Fine Touch receptors
      - **Pacinian Corpuscle:**
        - In Deep Dermis & Hypodermis
        - Deep Pressure Sensors
      - **Ruffini Endings:**
        - In Deep Dermis & Hypodermis
        - Directly Associated with Collagen Fibrils → Stretch Receptors



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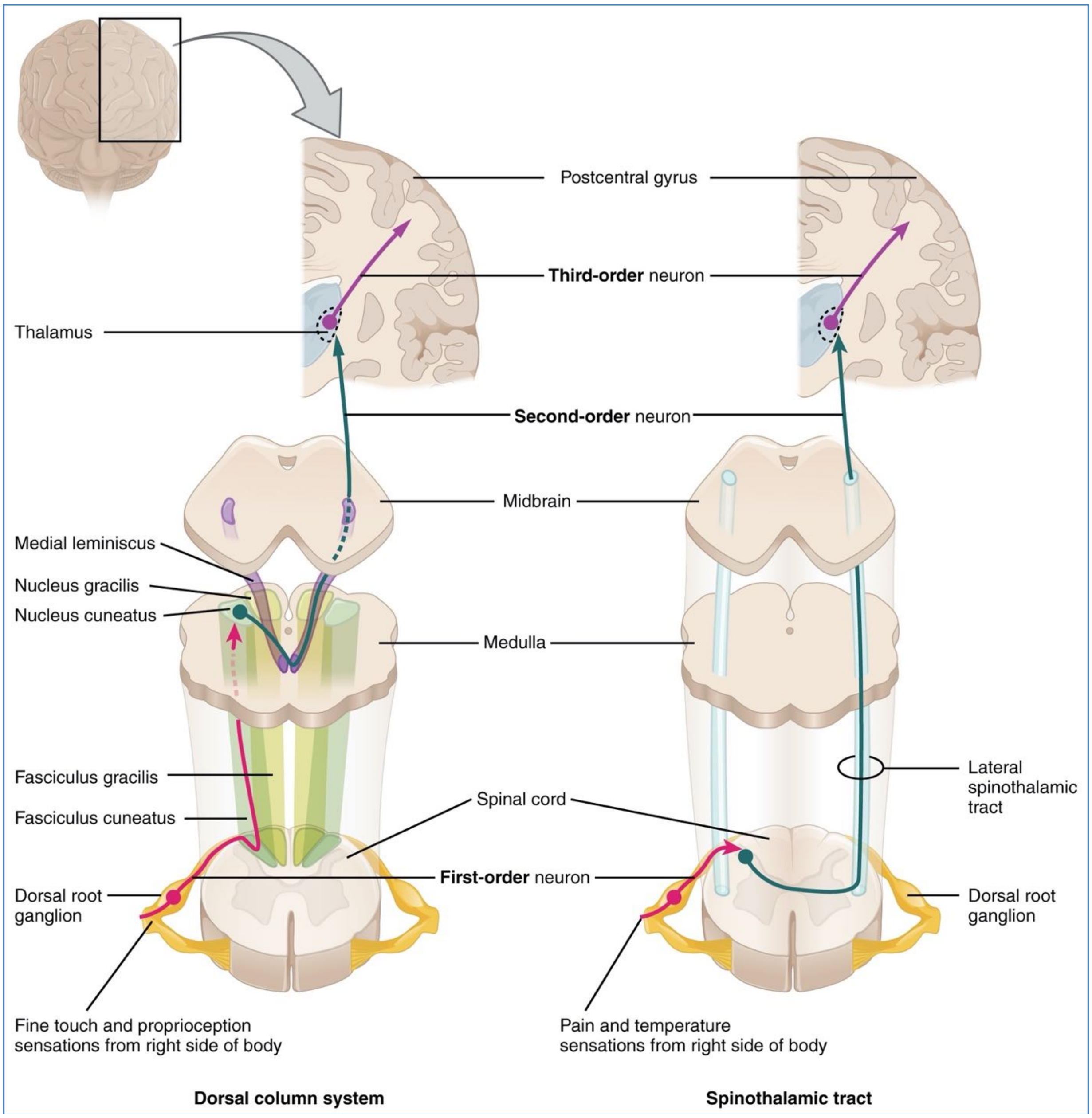
- **Physiology of Sensory Receptors:**
  - **Concept of Adaption:**
    - ***Adaption:*** Under a maintained stimulus of constant strength, the frequency of action potentials declines over time
    - **Slowly-Adapting Receptors (Eg: Nociceptors):**
      - Continue to Transmit Impulses to the brain as long as the Stimulus is Applied
    - **Rapidly-Adapting Receptors (Eg: Pacinian Corpuscles):**
      - Receptors Rapidly Adapt & are stimulated only when the Stimulus Strength has Changed
- **Connection to the CNS:**
  - **First Order Neuron:**
    - Sensory Neuron Nucleus is in the **Dorsal Root Ganglion**
    - Its axon extends from periphery to Dorsal Horn of the Dorsal Root Ganglion
  - **Second Order Neuron:**
    - Neuronal Nucleus is in the Substantia Gelatinosa
    - Its axon Decussates, Then it Ascends in the **Spinothalamic Tracts** → Thalamus
  - **Third Order Neuron (Thalamus):**
    - Neuronal Nucleus is in the Thalamus
    - Its axon passes through the Internal Capsule (behind Pyramidal Fibres) → Sensory Cortex



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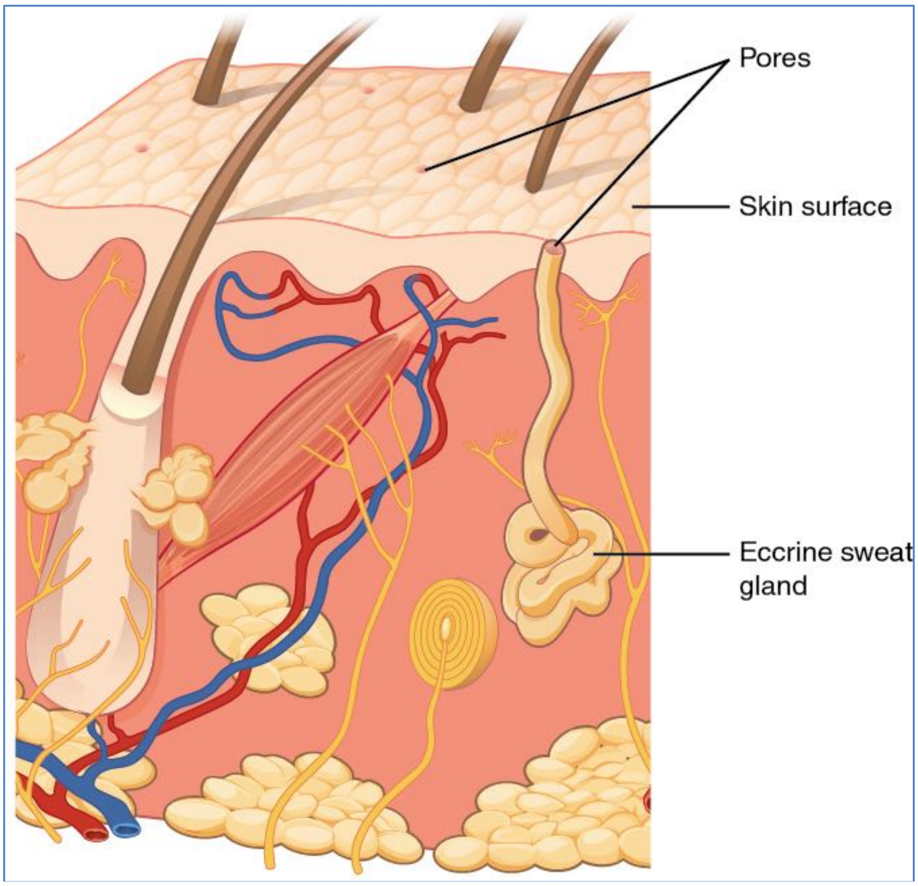
- **<u>Glands:</u>**
  - **<u>Sebaceous (Sebum) Glands:</u>**
    - **Associated with hair follicles (The Pilosebaceous Unit)**
      - (:. Not found on Palms/Soles)(Most found on Face & Scalp)
    - **Holocrine Secretion:**
      - (Ie: Secretion via complete destruction of cells)
      - **Produce oily sebum:**
        - Triglycerides
        - Fatty Acids
        - Wax Esters
        - Cholesterol
        - (Also Antibacterial/Antifungal Action)
    - **Stimulated by Androgens:**
      - Stimulated by Androgens (Inhibited by Estrogens)
      - Very Active at puberty
  - **<u>Eccrine (Sweat) Glands:</u>**
    - On most of the body (Scarce on the back)
    - **Simple, Coiled Tubular Glands:**
      - Secretory Coil (deep in Dermis) – Secrete the Water & Electrolytes
      - Sweat Duct – Reabsorb Na^+ Ions from the sweat
    - **Clear watery secretion**
      - Person can perspire several liters per hour
      - **Process:**
        - 1) Secretion of Electrolyte-Rich Fluid
        - 2) Reabsorption of excess Na^+ by the Duct
    - **Stimulated by High Temperature and Stress**
      - Emotional Sweating doesn't occur during sleep
      - Thermal sweating does occur during sleep
      - Innervated by Sympathetic Nerves



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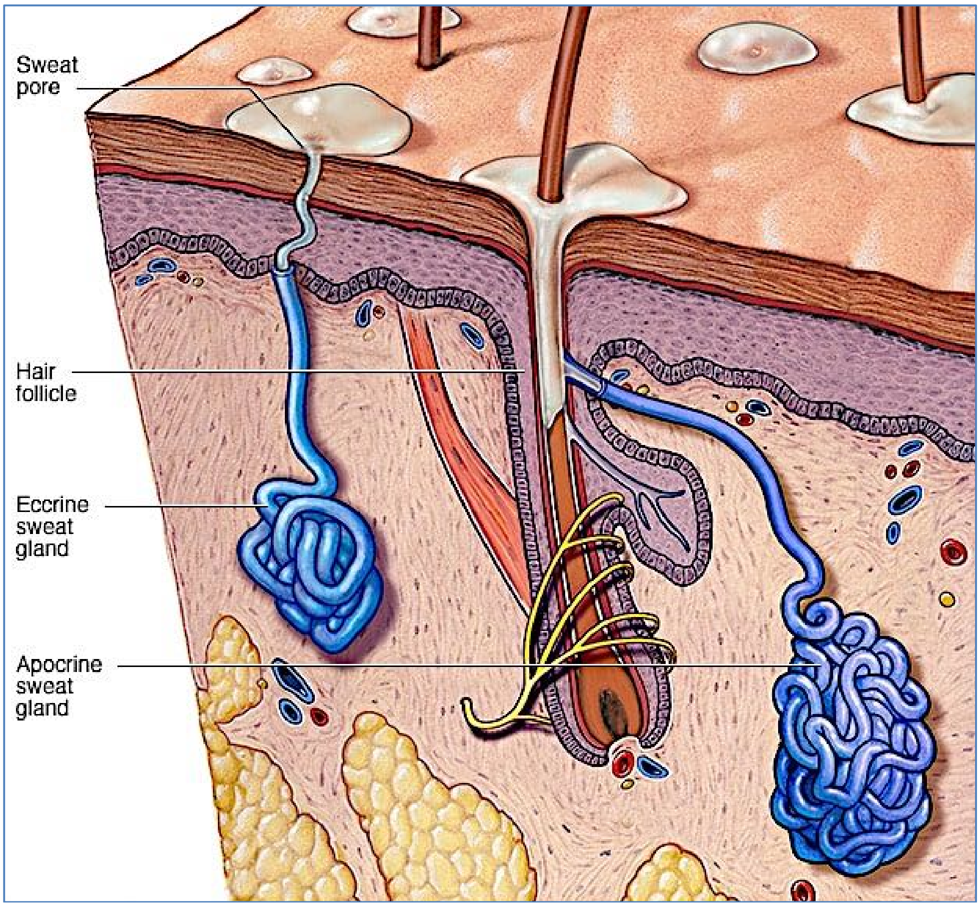
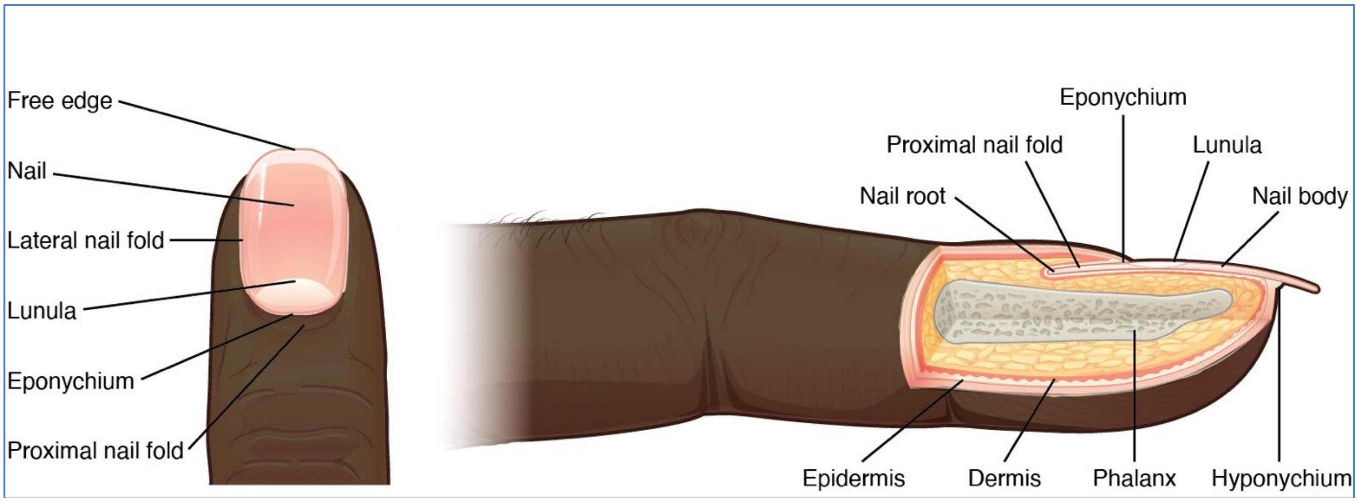
- **Apocrine (Pheromone) Glands:**
    - Associated with hair follicles
    - **Large complex gland:**
        - Located in Dermis
        - Duct Opens into Hair Follicle
    - **Viscous, Milky Secretion – (protein and cellular debris):**
        - **Produces pheromones**
            - Bacterial action is required for odor production
            - Thought to have or had a role as a sexual attractant in humans
            - Respond to emotive stimuli
    - **Stimulated by Androgens:**
        - Most Active at puberty



Source: Mayoclinic.org

- **Nails ("Ungals"):**
  - **Composition:**
    - Plate of hardened and densely packed **keratin (Protein)**
  - **Functions:**
    - Protect distal phalanges of fingers and toes
    - Aid in picking up objects
    - Efficient natural weapon
  - **Structural Landmarks:**
    - **Nail Plate:**
      - Fully keratinized structure
      - Firmly attached to the nail bed
    - **Nail Bed:** Skin underneath the nail
    - **Lunula:** Proximal whitish half-moon-shaped area on Nail Plate
    - **Cuticle:** The dorsal part of the proximal nail fold
    - **Nail Matrix:** Nail growth occurs From Here - by proliferation and differentiation of the nail matrix
    - **Paronychium:** The Skin around the Nail
    - **Hyponychium:** The area where the nail plate detaches from the digit
    - **Proximal Nail Fold:** The fold at the Proximal Edge of the nail (Covers ≈1/4 of the nail)
    - **Lateral Nail Folds:** The folds of skin at the Lateral Edges of the Nail



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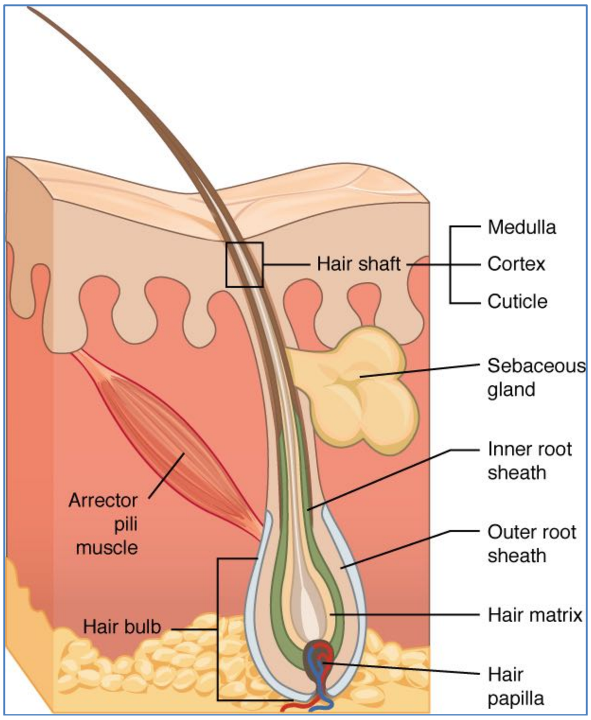
- <u>**Hair/The *'Pilosebaceous Unit'*:**</u>
    - **The Pilosebaceous Unit:**
        - Hair Shaft
        - + Sebaceous/apocrine ducts empty into hair follicles
    - **Arrector Pili Muscle**:
        - Smooth Muscle
        - Supplied by Adrenergic Nerves
        - → Erection of the Hair during Cold/Emotional Stress (Goose-Bumps)
    - **2 Types of Hair:**
        - **Vellus:**
            - Fine, Short & Almost invisible
            - (All over the body)
        - **Terminal:**
            - Thick, coloured & Visible
            - (Scalp, beard, axilla, genital area)
    - **Hair Follicles Respond to Androgen:**
        - Pre-pubertal children don't have *Terminal Hair* in Axilla/Genital Area/Facial Hair
        - During Puberty, hair grows in these areas
        - With age, androgen stimulation decreases → *Terminal Hair* on scalp reverts from *Terminal* to *Vellus* hair (Hair follicles aren't lost, but change to *Vellus* hair)
    - **Hair follicles undergo cycles of Growth, Resting and Shedding:**
        - **Anagen (growing):**
            - Lasts 3 Years
            - Keratinocytes in the follicular bulb proliferate to form the hair shaft
        - **Catagen (Resting):**
            - Lasts 2 Weeks
            - The keratinocytes and melanocytes undergo programmed cell death
        - **Telogen (Shedding):**
            - Last 3 Months
            - Hair but does not grow
            - May Remain Anchored, or Be Shed
            - Following telogen a new growth cycle will begin (Ie: anagen)
    - **Hair Pigmentation:**
        - Colour of hair is determined by the melanocytes – actively pigmented only in Anagen
        - Absence of pigment →white hair
        - Diminished pigment →grey hair



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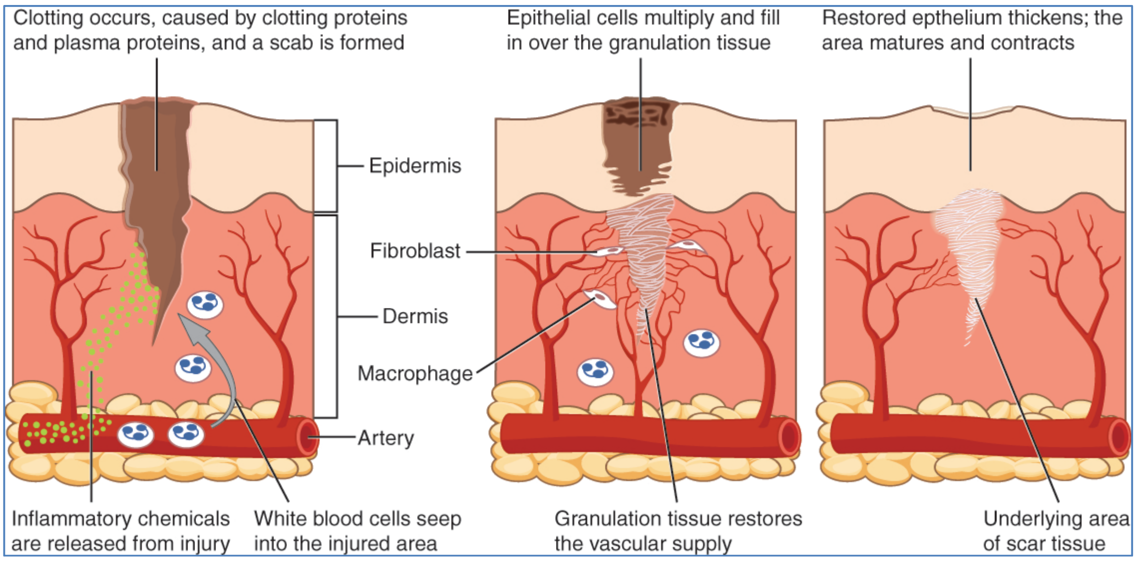
__Causes of Skin Injury:__

- **Physical Agents:**
  - Mechanical trauma
  - Thermal Burns
  - Cold
  - Electrical
  - Radiation (Eg: In Radiotherapy)
- **Hypoxia:**
  - Ischaemia (Eg: In Peripheral Vascular Disease)
- **Chemicals:**
  - Acid/Alkali
  - Phosphorus
- **Infectious:**
  - Bacteria
  - Viruses
- **Autoimmune:**
  - Scleroderma
  - Lupus Erythematosus
- **Genetic:**
  - Histiocytosis X

__Mechanisms of Healing:__

- **Acute Inflammation:**
  - Vasodilation
  - Increased Permeability → Stasis
  - Leukocyte Margination/Migration
  - Phagocytosis of Damaged/Dead Tissue/Organisms + Enzyme Release
- **Granulation Tissue:**
  - **3-5 Days After Injury**
  - **Occurs only in Deeper Cuts/Injuries (Not Superficial Injuries)**
  - **Angiogenesis** – (Migration + Proliferation of Endothelial Cells)
    - Driven by FGF (Fibroblast Growth Factor) + VEGF (Vasculo-Endothelial Growth Factor)
  - **Fibrosis -** Fibroblast Migration + Proliferation
    - Driven by PDGF (Platelet-Derived Growth Factor) + TGFB (Transforming Growth Factor-B)
- **Collagen Synthesis:**
  - A Triple Helix Protein
  - Synthesis is Vitamin C Dependent
  - As time goes by, the Collagen Scar gets Stronger:
    - At 1wk, the scar is weak
    - Strengths peaks @ 3mths
  - Metalloproteinases (collagenases, Gelatinases) → Degrade Collagen



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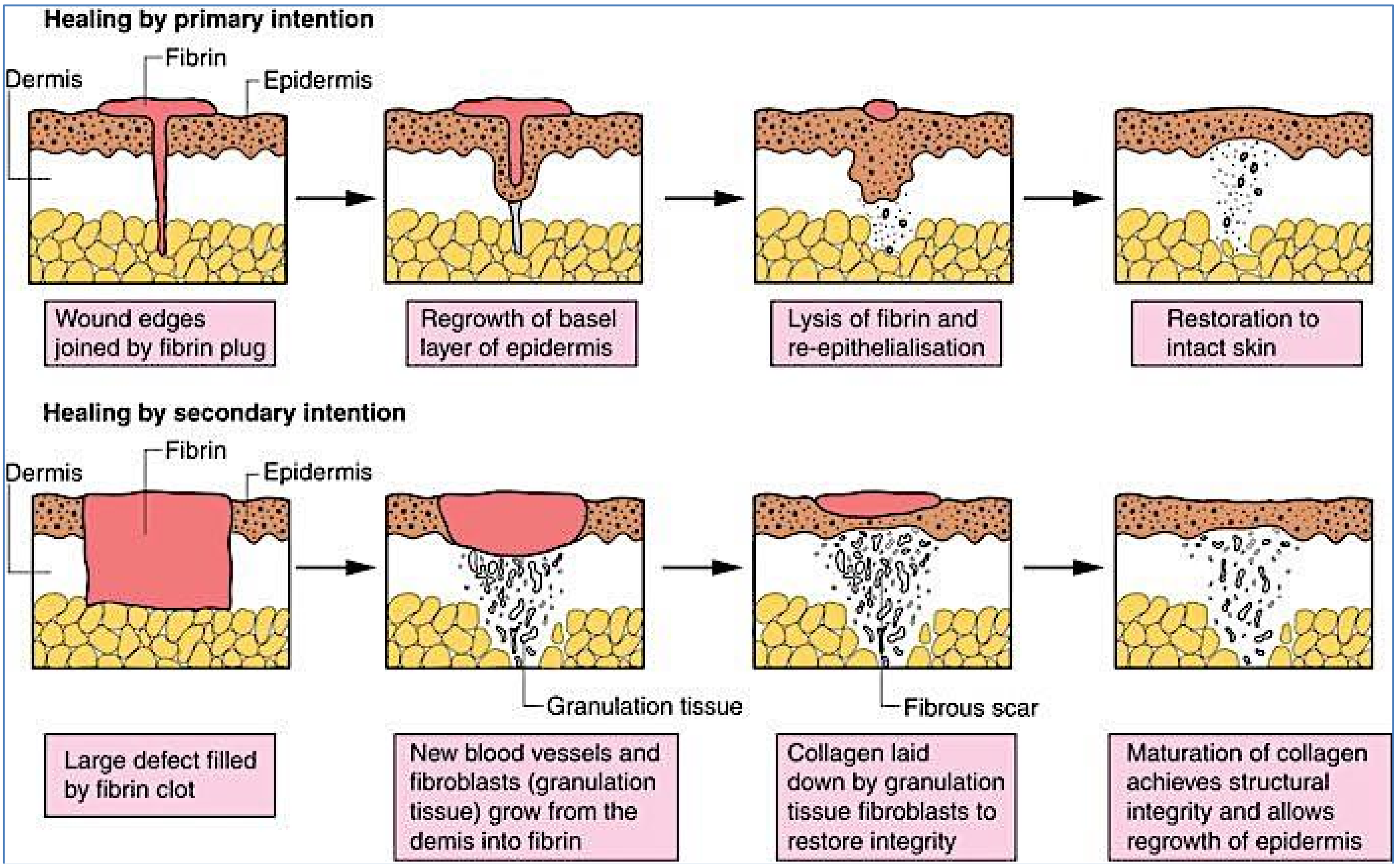
Healing By Primary Intention Vs Secondary Intention:

- **Primary Intention:**
  - Wound edges are CLOSE together
  - Tends to be Quick
  - Cells of the Basal Layer of the Epidermis lines the Surface of the Wound (within 1-2 Days)
- **Secondary Intention:**
  - Wide, open wound (Edges are FAR apart)
  - Takes a lot Longer
  - Granulation Tissue Fills the Wound → Epithelialisation from Wound Margins



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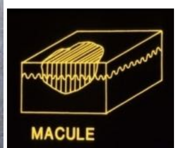
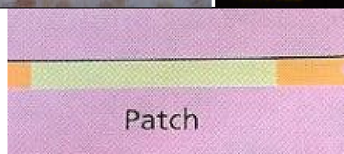
GENERAL PRINCIPALS OF DERMATOLOGY

Key words + Definitions:

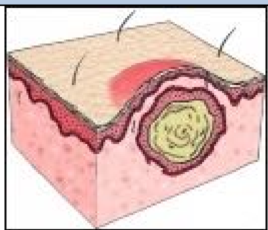
- **Primary Lesion:**
  - The initial lesion that characterizes a condition
- **Secondary Lesion:**
  - Over time, the primary lesion may continue to develop or be modified by regression/trauma, producing a "secondary lesion"

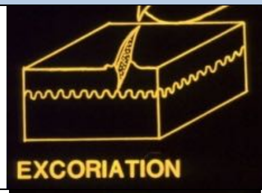
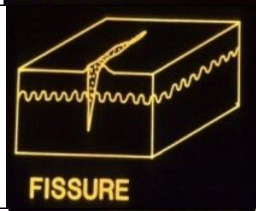
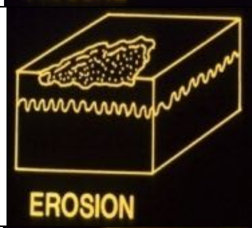
Morphology Terminology – Primary Lesions:

Lesion Name:	Lesion Description:	Picture:
Erythema	Redness due to vascular dilation (Eg: Cellulitis)	
Erythroderma	Red rash covering >90% of the Body	
Telangiectasia	Permanently dilated dermal blood vessels to the point of being visible; Blanches under pressure	
Macule	A macule is a small, flat is of altered colour, without elevation or depression (nonpalpable). (Eg: Lentigo Simplex, Eg: Freckles)	 
Patch	A patch is a large Macule	
Petechiae	Tiny macules of blood in the skin; Small, non-blanching Extravasated RBC's	
Purpura (May Be Palpable)	A Large Macule or Patch of blood in the skin. Doesn't blanch under pressure. (Eg: A Hickey) Larger area of RBC extravasation (The hallmark of vasculitis – severe inflammation of the blood vessels; If it is this bad in the skin, it's probably this bad elsewhere in the body)	

Ecchymosis/ Haematoma	A Larger subcutaneous bleed. The resultant swelling = a 'Haematoma'	

Micropapule	A Very Small raised lesion 1-2mm in diameter.	
Papule	A small raised lesion (less than 0.5cm)	
Plaque	Flat-topped, raised area of the skin. (Usually dermal thickening.) A large Papule more than 2cm in width but NO depth. (Eg: Psoriasis)	
Papilloma	A nipple-like mass projecting from the skin	
Burrow	A linear Papule caused by a burrowing organism. Eg: Scabies	
Nodule	A raised, solid lesion greater than 0.5cm in width AND depth. (Eg: Keloid Scar)	
Tumour	A mass of enlarged tissues of more than 1cm diameter	

Vesicle	A clear, fluid-containing elevation of less than 5mm in diameter. (Eg: Chickenpox) (Typically appear clear, but can fill with pus to become a *Pustule*)	
Bulla	A large Vesicle more than 0.5cm in diameter	
Pustule	A small Pus-filled vesicle of less than 5mm in diameter	
Impetiginized (Impetigo):	Covered in Crust, Pustules, & Often Weeping	
Cyst	A cavity containing liquid, semisolid, or solid material	
Abscess	A Pus-containing cavity of more than 1cm in diameter. (Note: a cyst can become infected and fill with pus, becoming an abscess)	

Excoriation	A Scratch - May be Epidermal or may extend down into the Dermis - Often linear - Often Covered with Crust	
Fissure	A deep crack - extends down to the dermis & blood vessels	
Erosion	A superficial <u>incomplete</u> loss of the epidermis. Eg: Superficial Burn	
Ulcer	An area of <u>complete</u> loss of the epidermis and often portions of the dermis and even subcutaneous fat.	
Wheal	A wheal is an elevated, white, compressible area produced by dermal oedema. It typically disappears within 24 to 48 hours	
Angioedema	Oedema which extends to the subcutaneous tissue	
Comedone (Blackhead)	A plug of keratin or sebum blocking a sebaceous orifice	
Alopecia (Hair Loss)	Can be: - Scarring (Permanent loss of hair follicles) - Or Non-Scarring (follicles are still alive and well)	
Eschar (Necrosis)	Patch of Dead Skin (necrosis) – A Thick Crust over an Ulcer or Erosion: – Typically black; Full-thickness skin loss (Dead skin = something bad – RED FLAG) (Implies something vascular → Ischaemic Necrosis)	

Morphology Terminology – Secondary Lesions:

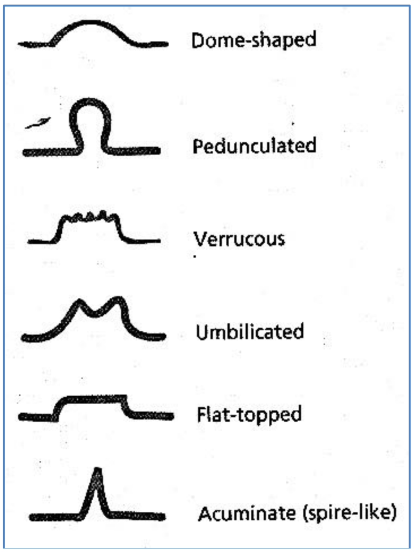
Lesion Name:	Lesion Description:	Picture:
Scale	Dry, laminated masses of keratin that represent thickened stratum corneum. Eg: Psoriasis	
Keratosis	Thickening of the skin. Eg: Solar Keratosis	
Hyperkeratosis	Large are of Thickened skin → Thick Scale.	
Verrucous	Very Hyperkeratotic – Similar to a Wart	
Lichenification	Palpable epidermal thickening of the skin usually due to friction. Eg: Lichen Simplex	
Crust	Dried serum, pus, or blood usually mixed with epithelial and sometimes bacterial debris	
Atrophy	Thinning of the skin (epidermal, dermal, or subcutaneous)	

Erosion	A superficial incomplete loss of the epidermis. Eg: Superficial Burn	
Ulcer	An area of complete loss of the epidermis and often portions of the dermis and even subcutaneous fat. Eg: SCC	
Excoriation	A superficial loss of epidermis from scratching or picking, therefore often linear and often covered by crust	
Scar	A pale, firm raised or depressed lesion resulting from skin injury. Eg: Keloid Scar	
Stria (Stretch Marks)	Streak-like, linear, pink/purple/white marks due to changes in Connective Tissue	
Pigmentation	An Increase or Decrease in skin pigmentation. Eg: Birthmark	

Other Descriptors:

- **Colour:**
  - **Erythema =** Red
  - **Violaceous =** Purple
  - **Slate =** Gray
  - **Hyperpigmented =** Darker than surrounding skin
  - **Hypopigmented =** Lighter than surrounding skin
  - **Bronze =** Red-Brown
  - **Dusky =** Purple-Grey
  - **Variegated =** Many different Colours

- **Relating to Edge of Rash:**
  - **Definition:**
    - Eg: Well Defined
    - Eg: Ill/Poorly Defined
  - **Geometric Shape:**
    - **Nummular** = Round
    - **Annular** = Rink-like
    - **Circinate** = Circular
    - **Arcuate** = Curved
    - **Discoid =** Disc-like
    - **Gyrate/Serpiginous =** Wave-like
    - **Polycyclic =** Edge is like a number of overlapping circles
- **Surface Contour:**



- **Relating to Symptoms:**
  - **Weeping =** Oozing clear fluid from the skin surface
  - **Crusted =** Covered in scabs
  - **Pruritis =** Itchy
  - **Dysaesthesia =** Tingling, Burning, Numbness

- **Symmetry:**
  - Eg: Dermatitis is typically symmetrical
  - Eg: An infective lesion (Eg: Abscess) is likely to be Unilateral

- **Is it in Typical Distribution:**
  - Eg: Seborrheic Dermatitis (Dandruff) typically occurs on the scalp, forehead, eyebrows & chest
  - Eg: Atopic Dermatitis typically in the cubital & popliteal fossae

- **Localised or Universal:**
  - **Universal** – Eg: Chicken Pox
  - **Localised** – Eg: Herpes Zoster (Shingles) localised to a dermatome

BENIGN CYSTIC SKIN LESIONS

BENIGN CYSTIC SKIN LESIONS

EPIDERMAL CYST

- **Clinical Presentation:**
  - Round, Firm, Mobile
  - Slow-growing
  - Range from 1cm-several cm across
  - May have an enlarged pore in the centre
- **Pathophysiology:**
  - Cystic enclosure of epithelium within the dermis
  - → Becomes filled with keratin and lipid-rich debris (Cream coloured/odorous)
  - Cyst wall is thin → can rupture → inflammation & pain
- **Epidemiology:**
  - Young-middle age
  - Most common type of cyst
- **Clinical Course:**
  - May rupture → inflammation
  - May increase in size/number over time
- **Management:**
  - Usually self-limiting
  - Can require excision

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PILAR CYST (TRICHILEMMAL CYST):

- **Clinical Presentation:**
    - Smooth, firm, dome-shaped
    - 0.5-5cm nodules
    - No central punctum
    - >90% occur on scalp
- **Pathophysiology:**
    - Cyst of keratin → very dense
    - → becomes calcified
    - → pressure by cyst causes atrophy of overlying hair follicles
- **Epidemiology:**
    - 2^{nd} most common cyst
    - typically middle-age
    - mostly females
- **Clinical Course:**
    - Rupture can cause pain & inflammation
- **Management:**
    - Cyst can be removed intact by excision

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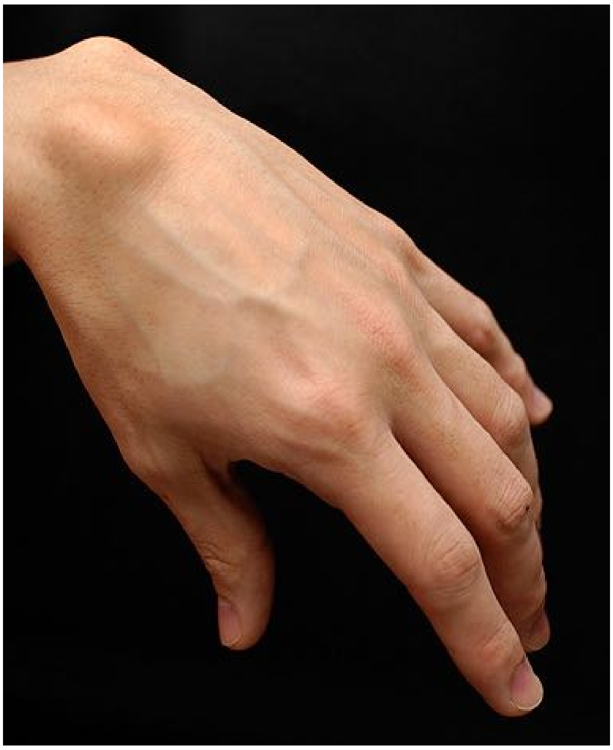
GANGLION CYST

- **Clinical Presentation:**
    - Solitary & Rubbery
    - Originates from a joint/tendon sheath
- **Pathophysiology:**
    - Associated with arthritis
- **Epidemiology:**
    - Occurs in older age
- **Clinical Course:**
    - Stable
    - May impede joint range/mobility
- **Management:**
    - Nil required
    - May be incised to relieve contents



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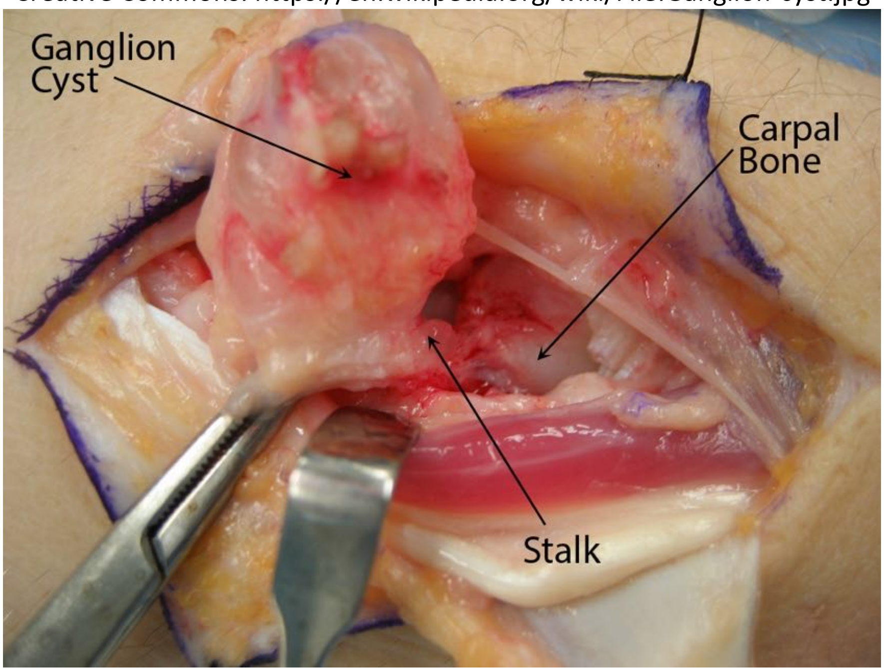

Credit: https://orthoinfo.aaos.org/

MILIUM

- **Clinical Presentation:**
  - 1-2mm superficial papules
  - white/yellow
  - typically on eyelids/cheeks/forehead
- **Pathophysiology:**
  - Epidermoid Keratin accumulation
  - Can be associated with other dermatoses:
    - Eg: Pemphigoid, porphyria, cutanea tarda, lichen planus, epidermolysis bullosa)
  - Can be associated with skin trauma:
    - Eg: Abrasion, burns, dermabrasion, radiation therapy
- **Epidemiology:**
  - Very common in infants
  - Can occur at any age
- **Clinical Course:**
  - Self-resolving in ~4wks in newborns
- **Management:**
  - No treatment required
  - Contents may be expressed

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BENIGN FIBROUS SKIN LESIONS

BENIGN FIBROUS SKIN LESIONS

DERMATOFIBROMA

- **Aetiology:**
  - Thought to originate from minor trauma (Eg: Shaving/insect bites)
  - Some association with SLE
- **Epidemiology:**
  - Most common in adults
  - Most common in females
- **Pathophysiology:**
  - Benign tumour of fibroblast proliferation in Dermis
- **Clinical Features:**
  - Firm Dermal Papule/Nodule
  - Skin coloured/darkened
  - May be itchy/tender
  - Fitzpatrick's Dimple Sign – lateral compression causes dimpling of lesion
- **Management:**
  - Excision if desired

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SKIN TAGS

- **Aetiology:**
  - Familial/older age
  - Most common in females
- **Pathophysiology:**
  - Benign outgrowth of the skin
- **Clinical Features:**
  - Harmless
  - Small 1-10mm skin coloured papule
  - Soft
  - Often multiple clustered around eyelids, neck, axillae, groin
- **Management:**
  - Excision/cryotherapy
  - Banding/tying

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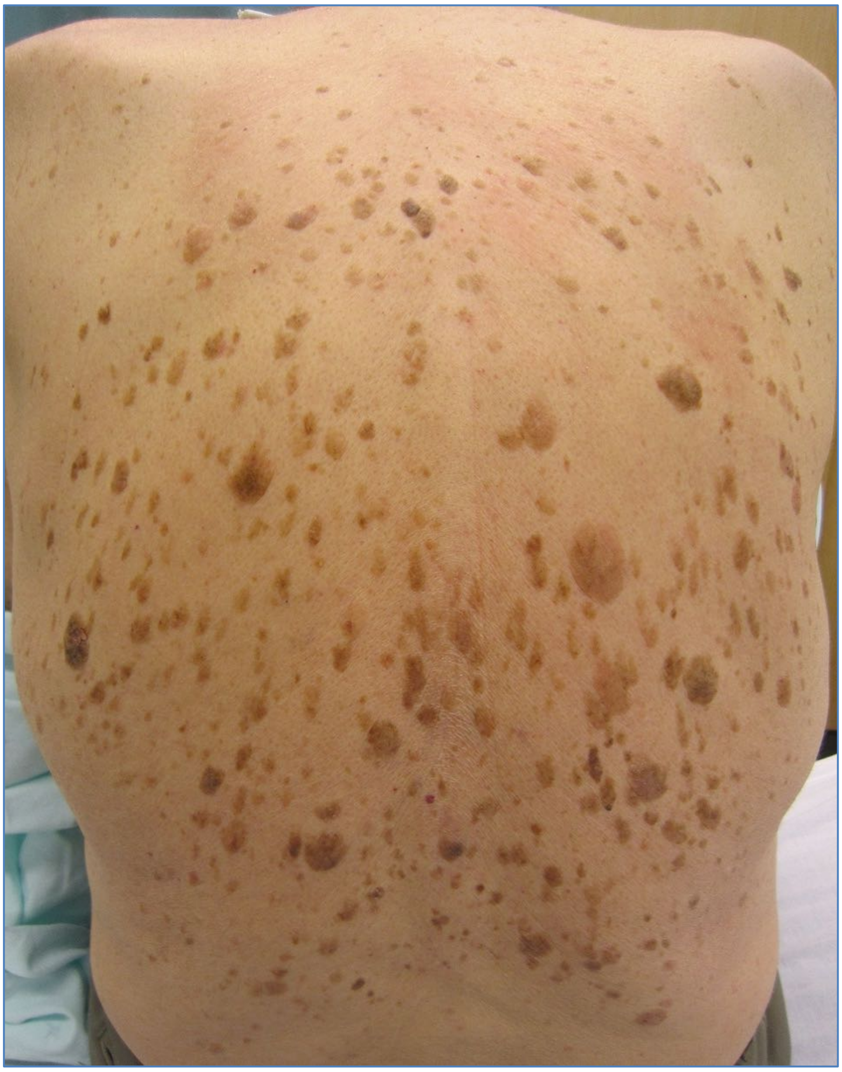
BENIGN HYPERKERATOTIC LESIONS

BENIGN HYPERKERATOTIC LESIONS

SEBORRHEIC KERATOSIS

- **Aetiology:**
    - Totally Benign Tumour in Old Age (Common in >40yrs)
- **Pathogenesis:**
    - Proliferation of keratinocytes & melanocytes → Benign Epithelial tumour
- **Morphology:**
    - **Gross:**
        - Sticky, Oily Plaques
        - Round, Flat
        - Sharply Defined Borders
        - May be Pigmented
        - rough, warty surface
        - Raised, 'Stuck-on' Appearance
    - **Microscopic:**
        - Thick Hyperplastic Epidermis
        - Keratin Cysts
- **Clinical Features:**
    - Often referred to as "old-age spots"
    - Rarely pruritic
    - Treated only if inflamed
    - No Malignant Potential
- **Treatment:**
    - None required; cryotherapy/excision if desired



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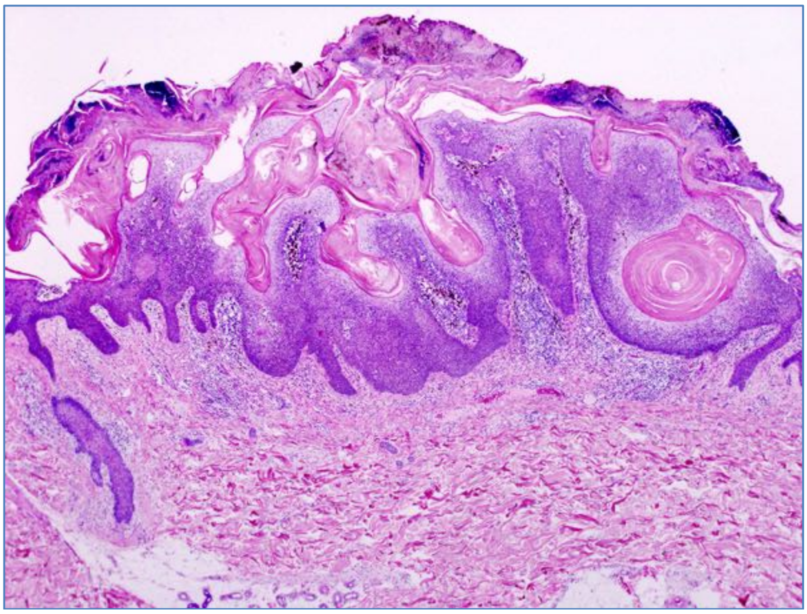

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CORNS / 'HELOMATA'

- **Aetiology:**
  - Pressure
- **Epidemiology:**
  - More common in females
- **Pathophysiology:**
  - Localised hyperkeratosis induced by pressure
- **Clinical Features:**
  - Firm papule with hard keratin core
  - Non-innervated but painful with direct pressure
  - Common on toes and hands
- **Management:**
  - Relieve pressure (padding/orthotics)
  - Paring

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BENIGN PIGMENTED LESIONS

BENIGN PIGMENTED LESIONS

CONGENITAL NEVOMELANOCYTIC NEVI (‘BIRTH MARK’)

- **Aetiology:**
    - Congenital
- **Pathophysiology:**
    - Nevomelanocytes in epidermis
- **Clinical Features:**
    - Present at birth
    - Rare malignant transformation
- **Morphology:**
    - Well demarcated pigmented papule/plaque
    - +/- overlying hairs
    - May have surrounding satellite nevi
- **Management:**
    - Take baseline photo and observe for change in shape/colour/size
    - Excision if suspicious

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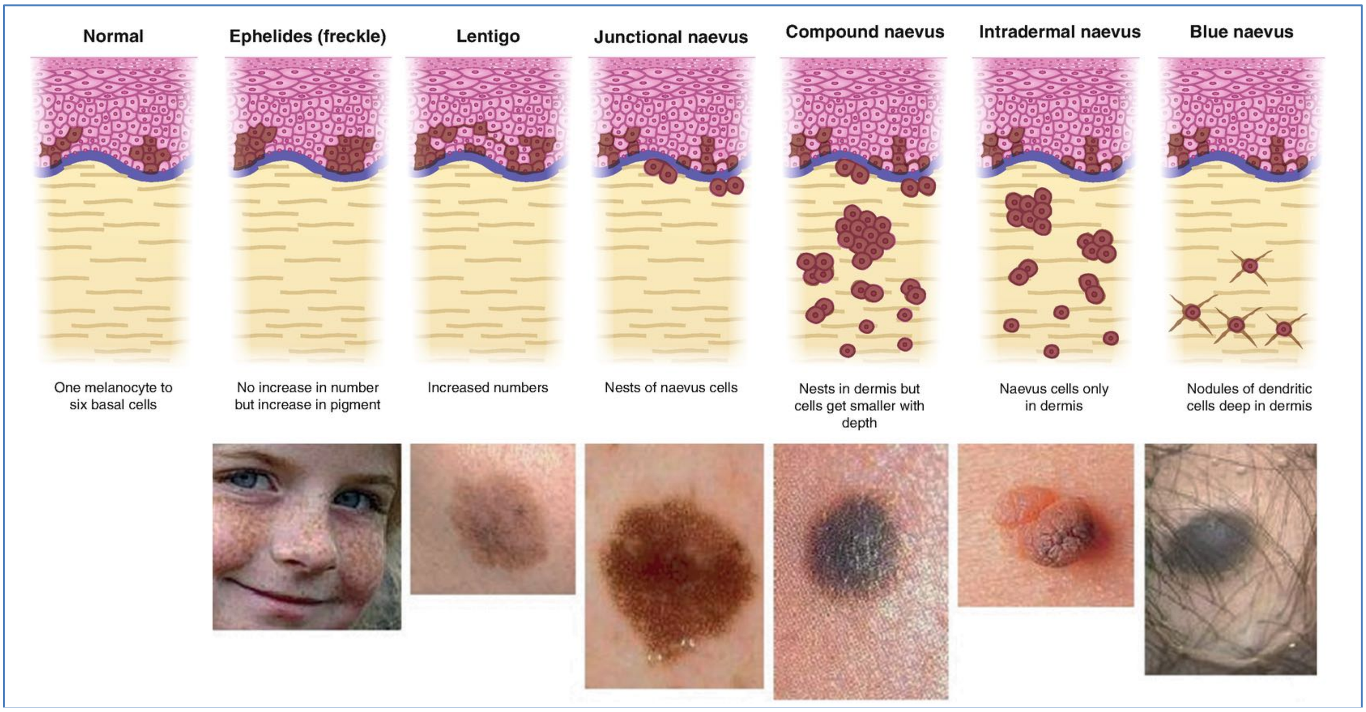
ACQUIRED NEVOMELANOCYTIC NEVI (COMMON MOLE)

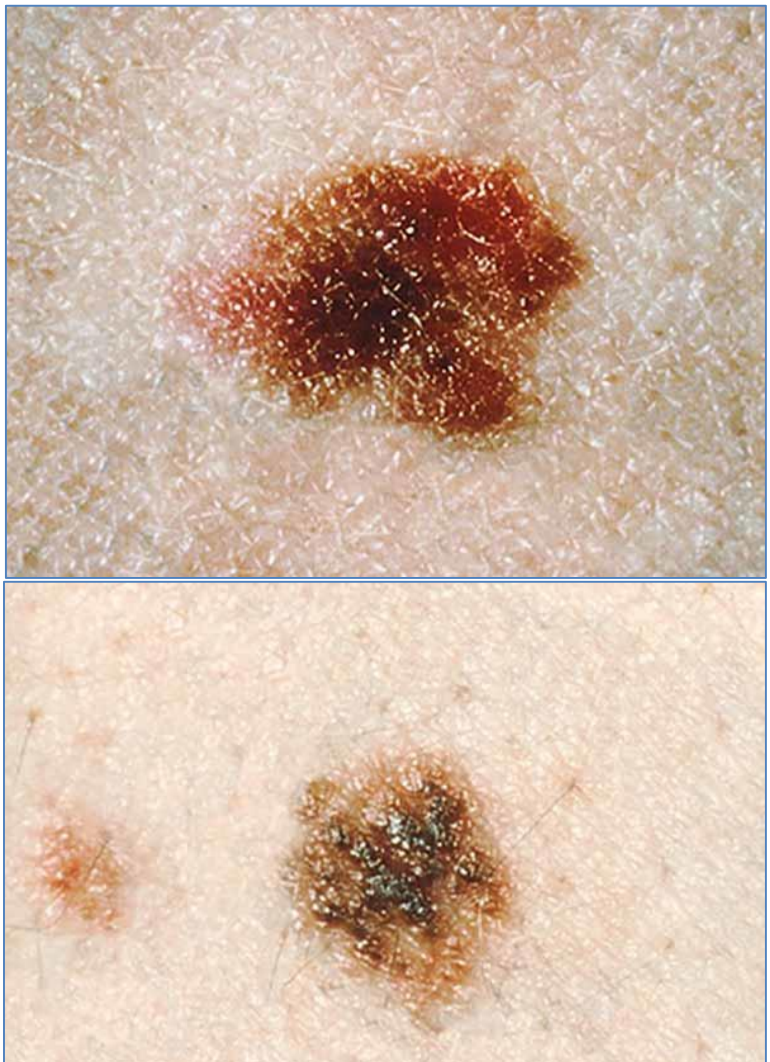
- **Aetiology:**
  - Thought to be initiated by UV Exposure
  - Familial association
- **Epidemiology:**
  - Are normal
- **Pathophysiology:**
  - benign tumours of melanocytes
- **Clinical Features:**
  - Well demarcated, round, pigmented macule
  - Uniform pigmentation
  - <1)5cm diameter
  - Usually Benign (But can become malignant)
- **3 Stages of Evolution:**
  - **Junctional Naevus (Typical Birthmark) =** Excess Melanocytes *AND* Excess Pigment
    - Small, Flat, Symmetric, Uniform Lesions
    - Cluster of Clear Cells (Melanocytes) @ the Dermo-Epidermal Junction
  - **Compound Naevus:**
    - Small, Raised, Dome-Shaped, Symmetric, Uniform Lesions
    - Big Clusters of Melanocytes in Dermis & Dermo-Epidermal Junction
  - **Dermal:**
    - Dome-shaped papule/nodule
    - Melanocytes are exclusively in the Dermis
- **Management:**
  - New/evolving lesions should be evaluated for melanoma
  - Consider Excisional biopsy if Asymmetric/non-uniform colours/irregular borders/itch/bleeding



Buckley D. (2021) Congenital Nevi, Melanocytic Naevi (Moles) and Vascular Tumours in Newborns and Children. In: Buckley D., Pasquali P. (eds) Textbook of Primary Care Dermatology. Springer, Cham. https://doi.org/10.1007/978-3-030-29101-3_29

ATYPICAL NEVUS (DYSPLASTIC NEVUS)

- **Aetiology:**
  - UV exposure
  - Progression of a pre-existing common mole
- **Epidemiology:**
  - Family history
  - High number of common moles
- **Pathophysiology:**
  - DermoEpidermal-Junctional Cluster of Dysplastic (Larger/Irregular/Darker) Melanocytes
  - Clinically and histologically as intermediates between normal naevi and melanoma
  - **Small risk of transformation to melanoma (about 1 in 1000)**
- **Clinical Features:**
  - Pigmented, Raised Lesion, with Central Darker Shade
  - Macular & Papular
  - Uneven pigmentation
  - Fairly regular borders
  - Fair symmetry
- **Management:**
  - Baseline photography/monitoring
  - Excisional biopsy



Source: https://www.cancer.gov/types/skin/moles-fact-sheet

EPHELIDES (FRECKLES)

- **Aetiology:**
  - Sun exposure
- **Epidemiology:**
  - Most common in fair-skinned people (Fitzpatrick 1 & 2 skin types)
  - Appear in childhood
- **Pathophysiology:**
  - UV Exposure → Increased melanin within basal layer keratinocytes
- **Clinical Features:**
  - Small <5mm well-defined light brown macule
  - May Regress if Exposure is Avoided
- **Management:**
  - Sun-safe practices (cover skin/sunscreen)

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SOLAR LENTIGO

- **Aetiology:**
  - UV-Exposure
- **Epidemiology:**
  - common in middle aged to elderly fair skinned people
- **Pathophysiology:**
  - Benign melanocytic proliferation in dermal-epidermal junction due to chronic sun exposure
- **Clinical Features:**
  - brown, well-defined macules on sun exposed skin
  - 1-3cm diameter
- **Management:**
  - Laser therapy
  - Shave excision
  - Cryotherapy

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VASCULAR LESIONS

VASCULAR LESIONS

HAEMANGIOMAS

- **Aetiology:**
  - congenital
- **Epidemiology:**
  - Appears shortly after birth
- **Pathophysiology:**
  - Benign vascular tumour
- **Clinical Features:**
  - Approx half resolve spontaneously by around 5yrs
  - Red/blue subcutaneous mass
  - Soft/compressible
  - Blanches with pressure
- **Management:**
  - Consider treatment if not gone by school age
    - Topical Timolol
    - Propanolol
    - Corticosteroids
    - Laser treatment
    - surgery

Source: https://www.dermcoll.edu.au/atoz/infantile-haemangiomas/

PORT-WINE STAINS

- **Aetiology:**
    - Congenital
- **Epidemiology:**
    - Present at birth
    - Rarely associated with Sturge-Weber syndrome
- **Pathophysiology:**
    - Vascular malformation of dermal capillaries
- **Clinical Features:**
    - Red-blue macule
    - Follows a dermatomal distribution
    - Rarely crosses midline
    - Commonly on nape of neck
    - Doesn't spontaneously regress
- **Management:**
    - Laser therapy

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ACNEIFORM ERUPTIONS:

ACNEIFORM ERUPTIONS:

ACNE VULGARIS (COMMON ACNE)

- **Aetiology & Pathogenesis:**
    - **Genetic Basis (Familial)**
        - Chronic inflammation of the sebaceous glands
    - **Abnormalities of the *Pilosebaceous Unit*:**
        - Blockage of the Sebaceous Duct (By ↑Keratin @ Duct Opening)
        - Increased Sebum Production (Often under influence of ↑Androgen)
        - → Bacterial Infection of the Pilosebaceous Unit (*Propionibacterium Acnes*)
        - → Rupture of the Sebaceous Duct
        - → Inflammatory Effects
    - **Note: Combined Oral Contraceptives** (Ie: With Progesterone) can contribute to Acne
    - (IF Severe, Damage to Dermis Occurs → Scarring. Note: Scarring is a FAILURE of the Doctor to Adequately treat Acne)
- **Epidemiology:**
    - Common Skin Condition in Teenagers (Although Can occur at any age)
    - Most Severe in Males
- **Presentation:**
    - Wide Variation in Severity
    - Most often on Face, Upper Chest & Back
    - ****Presence of Comedones** – Both Open (Blackheads) & Closed (Whiteheads)
        - (Note: If no comedones are present, consider Differential Diagnoses)
    - **Inflammatory Papules & Pustules**
    - Severe Cases →
        - **Cysts**
        - Scarring (Hypertrophic or Atrophic)
- **Diagnosis:**
    - **The 'Acne Triad':**
        - Papules + Pustules + ***Comedones*** = ACNE
    - **Possible Differential Diagnoses:**
        - ***Rosacea:**
- **Treatment:**
    - Topical Anti-inflammatories & Antiseptics
    - Antiseptics (Eg: Benzoyl Peroxide)
    - Antibiotics (If Severe)(Note: Tetracycline also has an Anti-Inflammatory Effect – Good)
    - Keratolytics (To ↓the overproduction of keratin – Eg: Salicylic acid)
    - Retinoids (Derived from Vitamin A) (Isotretinoin/Roaccutane) (In Severe Nodulo-Cystic Acne)
    - ≈1yr after symptoms disappear
- **Prognosis:**
    - Usually Good (Most cases are Self-Limiting)
    - Treatments are safe & well-tolerated
    - NO Patient should have to Suffer!

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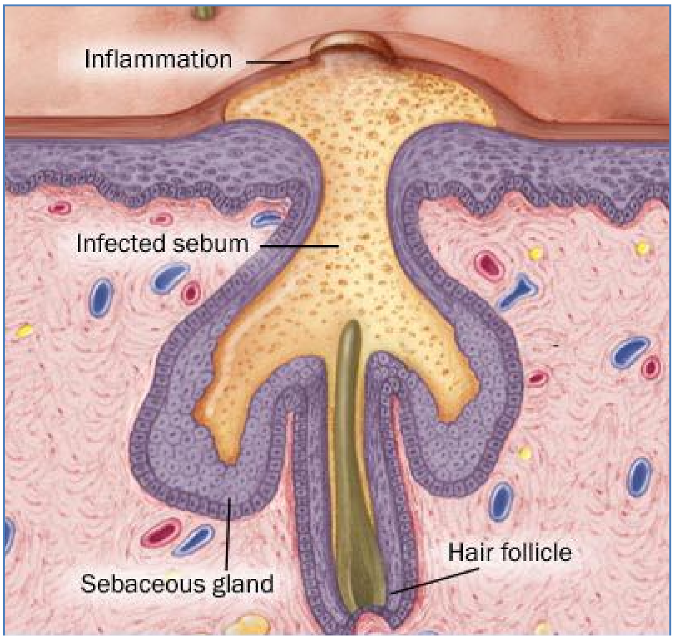


Source: Unattributable

ROSACEA

- **What is it?:**
  - Predominantly a Facial Rash easily confused with Acne
  - = Pustules and Papular Rash on Face
  - *Typically a Disease of Middle Age (30-40yrs)*
- **Aetiology:**
  - **Unknown**
  - (But Familial Association)
  - **Aggregating Factors:**
    - Heat and steam
    - Hot, spicy food
    - Alcohol consumption
    - Emotional stress
    - Sun exposure
- **Presentation:**
  - Wide Variation in Severity
  - **Initial Signs:**
    - Tendency to Flush easily + Burning/Stinging/Itching
  - **Distinguishing Features:**
    - Facial Flushing (Erythema)
    - Dilated, visible Capillaries (Telangiectasia)
    - Papules
    - Pustules
    - But NOT Comedones
  - **If Severe:**
    - Disfiguring Facial Rash
    - + Bulbous Enlargement of the Nose
    - Possible Facial Oedema
  - ***NO Comedones* (:. NOT Acne)**
  - Does not cause Scarring
  - Very Chronic (Not Self Limiting – May last for many years)
  - (Note: Often significant Psychosocial Impact – Eg: Depression)
- **Diagnosis:**
  - **Differential Diagnoses:**
    - Acne (has all features + Comedones)
    - Sun Damage (has Telangiectasia, but No other Features)
    - Lupus Erythematosus (Have Telangiectasia & Erythema, But NO Pustules or Papules)
    - Menopause (Flushing)
- **Treatment:**
  - Avoidance of Aggravating Factors
  - Similar Treatment to Acne
    - Antibiotics (Some have Anti-Inflammatory Effects)
    - Retinoids
  - Laser surgery (for Telangiectasia & Erythema)
  - **Topical corticosteroids make rosacea worse and should never be used to treat it!**
    - → Cause Perioral Dermatitis (Should NEVER be used in Rosacea)

FACIAL REDNESS
Flushing and persistent redness. Visible blood vessels may also applear.

BUMPS AND PIMPLES
Persistent facial redness with papules and pustules (bumps and pimples.)

SKIN THICKENING
Skin thickening and enlargement (phymatous changes), usually around the nose.

EYE IRRITATION
Watery or bloodshot appearance, irritation, burning or stinging. Also known as ocular rosacea.

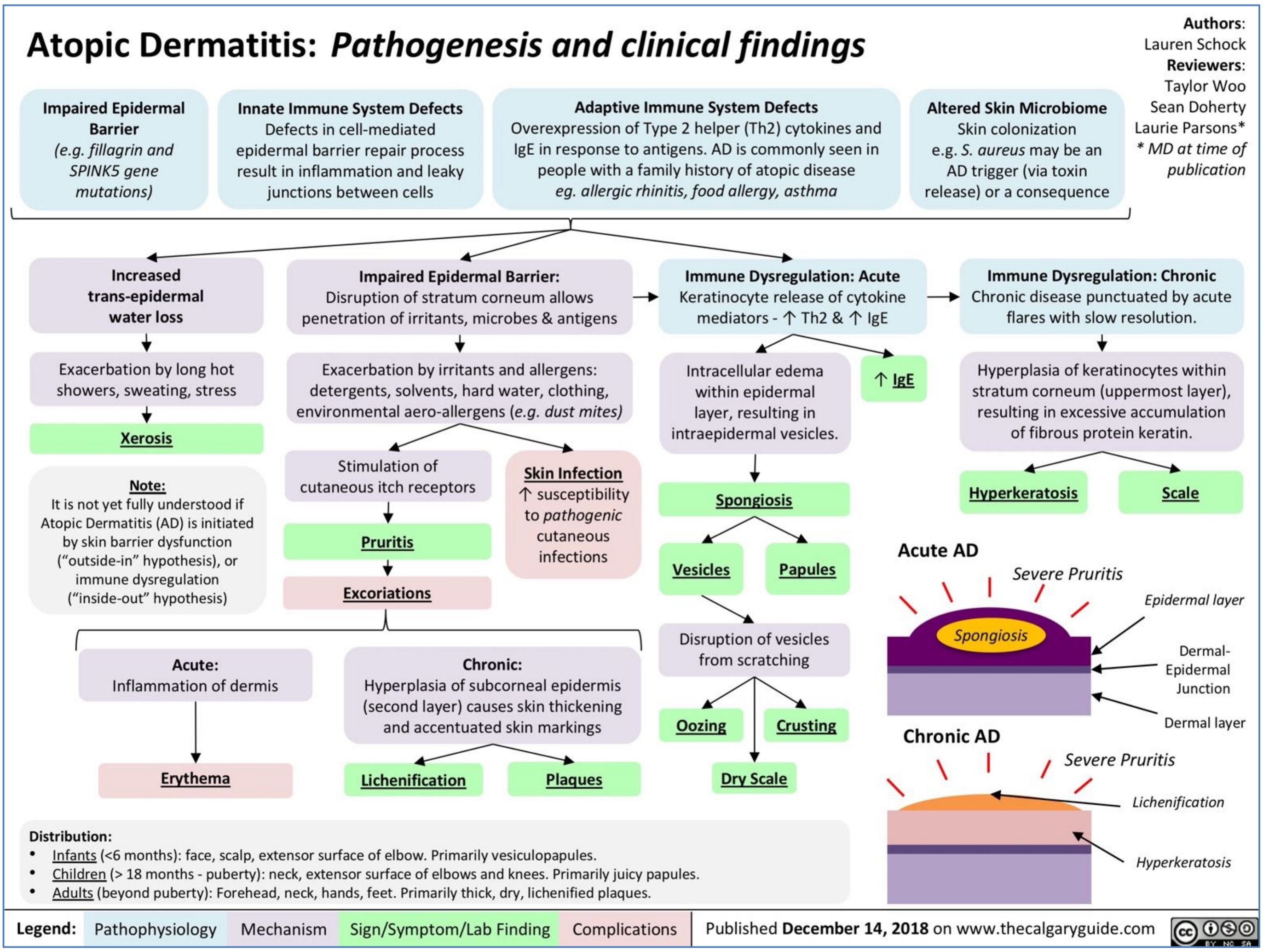
Source: https://www.rosacea.org/patients/all-about-rosacea

DERMATITIS (ECZEMA)

DERMATITIS (ECZEMA)

ACUTE DERMATITIS/ECZEMA:

- **Types of Acute Eczema:**
  - CONTACT DERMATITIS (Due to prolonged exposure to allergen)
  - ATOPIC DERMATITIS
  - (Drug eczema)
  - (Photoeczema)
  - (Primary irritant dermatitis)
- **Aetiology:**
  - Type IV Hypersensitivity (T-Cell Mediated) to Allergen
  - Prolonged Contact with Allergen - Urine, Soaps, Antiseptics, Deodorants, Creams, Foreign Body, etc
- **Epidemiology:**
  - Family history – (in 70% of cases)
  - Most Common in Children
  - Genetics – (50% of pts have a deficiency of the Epidermal Protein "Filaggrin")
  - Hypersensitivity – (Associated with other Atopies (Hay Fever [Allergic Rhinitis], and Asthma))
- **Pathogenesis:**
  - **Initial Exposure:**
    - Ag Processed by Langerhans Cells →Presented to T-Cells in LN → T-Cell Activation
  - **Re-Exposure:**
    - Type IV Hypersensitivity (T-Cell Mediated) reaction to Allergen
    - → Epidermal Oedema + Small Blisters → **"Wet Eczema"**
  - **Chronic Exposure:**
    - →Hyperplasia, hyperkeratosis (lichenification) – **"Dry Eczema"**
- **Morphology:**
  - **Gross:**
    - Erythema
    - Small Blisters – (Weeping & Crusting Blisters, Papules and Plaques)
    - Hyperkeratosis (If Chronic)
  - **Microscopic:**
    - Intra-*Epidermal* Oedema (*Spongiosis*)
    - Epidermal Blistering
    - Perivascular Inflammatory Infiltrate
- **Clinical Features:**
  - Severe Itching
  - Patchy, Erythematous, Poorly Defined Rash
    - Usually in the Popliteal/Cubital Fossae & Face
    - Can be Generalised
  - Dry Skin
  - Excoriation (loss of the surface of the skin from scratching) due to itching and scratching
  - Lichenification (thickening of the skin with accentuated skin lines)
  - Crusting (scabbing) and weeping (loss of fluid through the surface of the skin) due to bacterial infection
  - (Contact dermatitis often has a regular shape (Eg: Square from Band-Aid))
- **Treatment:**
  - Modification of lifestyle to avoid exacerbating factors
  - Remove Allergen
  - Avoid Soaps
  - Use of moisturisers and bath additives
  - Treated with topical Corticosteroids (Symptomatic)
  - Antihistamines (For Itch)



Source: thecalgaryguide.com

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SEBORRHEIC DERMATITIS

- **Aetiology:**
  - Some heredity
  - Some association with commensal Malassezia yeast (fatty acid metabolites may cause inflammatory reaction)
- **Risk Factors:**
  - Oily skin
  - Familial history of dermatitis/psoriasis
  - Immunosuppression
  - Parkinson's disease
  - Sleep deprivation/stress
- **Epidemiology:**
  - Bimodal onset; Infancy (within the first months), or between 20-50yrs old
  - Most common in males
- **Pathophysiology:**
  - Not completely known
- **Clinical Features:**
  - Chronic/relapsing eczema (improves in summer; flares in winter)
  - Primarily affects sebaceous-rich regions (Scalp/face/trunk)
  - Ill-defined, localised scaly patches/scale
  - Blepharitis (scaly red eyelid margins)
  - Salmon-pink thin scaly and ill-defined plaques in skin folds
  - **Infantile Seborrheic Dermatitis:**
    - Numerous Dermatoses in the 1^{st} 3 Months of Life
    - Erythematous but *Non-Itchy* Rash involving the face, scalp, neck, axillae and nappy area. The lesions are well defined and covered in greasy scale
  - **Adult Seborrheic Dermatitis:**
    - Erythema and fine, greasy scale on the cheeks, nose and nasolabial folds
    - Scale and itching of the scalp and eyebrows
    - Well defined but non-scaly erythema of the axillae, groin, scrotum and perianal skin
- **Management:**
  - **Keratolytics** (Eg: Salicylic Acid)
  - **Topical antifungals** (Eg: Ketoconazole)
  - **Mild topical steroids**
  - (Isotretinoin for moderate/severe seborrheic dermatitis)

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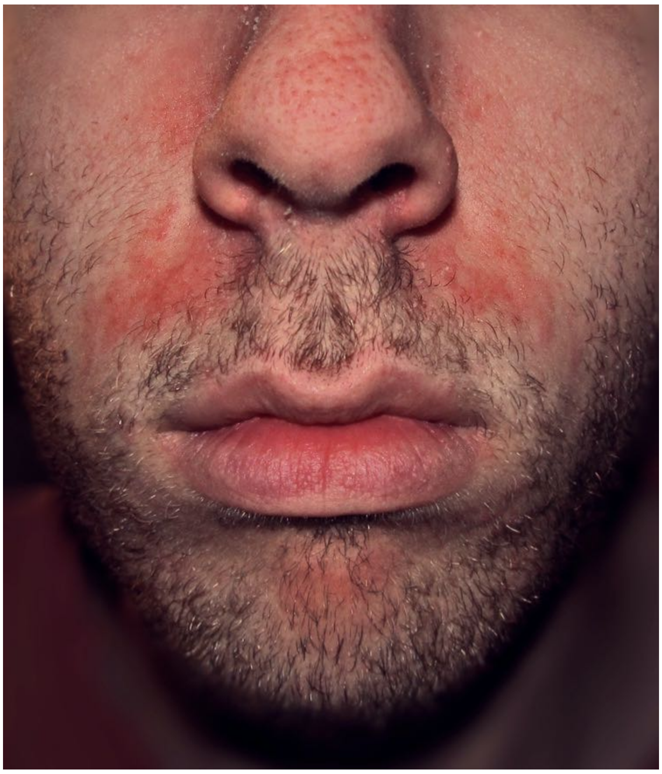

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STASIS DERMATITIS (VENOUS ECZEMA)

- **Aetiology:**
  - Venous insufficiency + Gravity
- **Epidemiology:**
  - Middle-older-aged patients
  - 20% of over 70yr olds
  - Risk factors:
    - History of DVT in affected limb
    - Varicose veins
    - Venous leg ulcers
- **Pathophysiology:**
  - Venous insufficiency → Failure of calf pump mechanism
  - → blood pools in leg veins → increased venous hydrostatic pressure → oedema
  - Fluid collecting in tissues → Activation of innate immune response
- **Clinical Features:**
  - May form discrete patches or become confluent/circumferential
  - Itchy red, blistered & crusted plaques
  - On one or both lower legs
  - Orange-brown macular pigmentation due to Haemosiderin deposition
  - Champagne bottle shape of lower leg
- **Management:**
  - **Reduce Dependent Oedema:**
    - Elevate feet when sitting
    - Walk regularly
    - Elevate feet above bed level overnight
    - Graduated compressing stockings long-term
  - **Treat Eczema:**
    - Dry up oozing patches with potassium permanganate or dilute vinegar on gauze compress
    - Oral antibiotics to avoid secondary infection
    - Topical steroid
    - Moisturising cream
    - Avoid injury to skin
  - **Treat Varicose Veins:**
    - Consult vascular surgeon
    - Sclerotherapy/laser

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LICHEN SIMPLEX CHRONICUS

- **Aetiology:**
    - Repetitive scratching and rubbing, which arises because of chronic localised itch. The primary itch can be due to:
        - Atopic Eczema
        - Contact eczema
        - Venous eczema
        - Psoriasis
        - Lichen planus
        - Fungal infection
- **Epidemiology:**
    - Males and females
    - Unusual in children
    - Association with anxiety/OCD
- **Pathophysiology:**
    - Repetitive scratching/rubbing → hyperkeratosis & inflammation
- **Clinical Features:**
    - Chronic, lichenified eczema/dermatitis
    - May be single or multiple plaques
    - Dry/scaly surface
    - Leathery induration
    - Scratch marks
- **Management:**
    - Potent topical steroids 4-6wks until plaque is resolved
    - Coal tar preparations
    - Moisturisers
    - Antihistamines for itch
    - Antidepressant for psychosomatic aetiologies

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ERYTHEMA MULTIFORME:

- **Aetiology:**
  - Hypersensitivity reaction
  - Some genetic tissue-type associations (HLA-B15, -B35, -A33, -DR53, -DQB1*0301)
- **Pathogenesis:**
  - Hypersensitivity Response to:
    - Infections (Herpes simplex, *Mycoplasma*) *Most common
    - Drugs (Sulphonamides, penicillin, barbiturates)
    - Malignancy (Carcinoma, lymphoma)
    - Auto-Immune (Eg: Lupus, SS, Dermatomyositis)
- **Morphology:**
  - **Gross:**
    - Variable Lesions: Papules/Plaques/Nodules/Blisters/Ulcers
    - Characteristic "*Targetoid*" Lesions
      - Central Grey Necrosis
      - Erythematous raised border
  - **Microscopic:**
    - Dyskeratosis
    - Necrotic Keratinocytes
    - Spongiosis (Epidermal Oedema)
    - Epidermal Lymphocytes
    - Destruction of Basal Epidermal Layer
- **Clinical Features:**
  - Acute & Self-limiting
  - Commonly affects young adults (20-40yrs)

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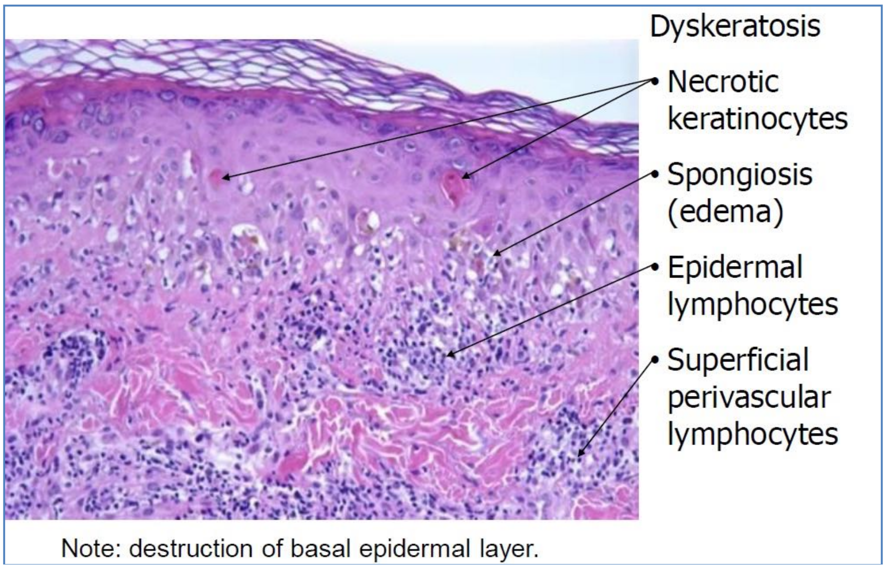


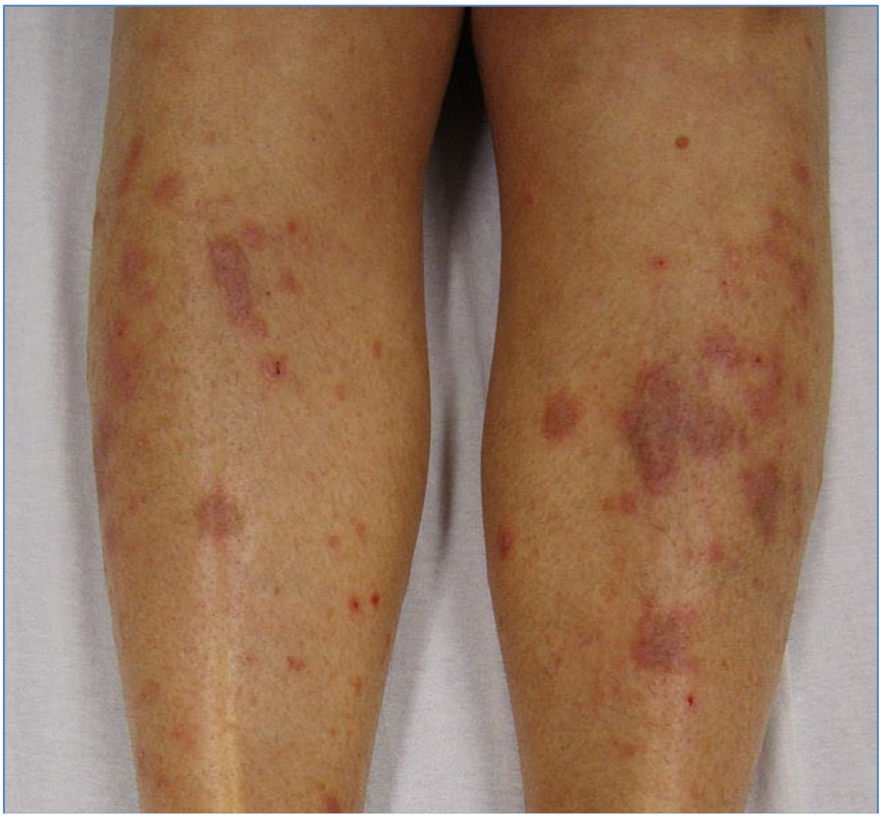
Source: Unattributable

PAPULOSQUAMOUS DISEASES

PAPULOSQUAMOUS DISEASES

LICHEN PLANUS:

- **Aetiology:**
    - Genetic predisposition
    - Physical/emotional stress
    - Localised skin disease (Eg: Herpes zoster)
    - Systemic viral infection (Eg: Hep C)
    - Contact allergy
    -
- **Pathogenesis:**
    - Chronic inflammatory skin condition
    - Similar to Erythema Multiforme (Hypersensitivity), but more chronic
- **Morphology:**
    - **Gross:**
        - Purple, polygonal, planar, Papules & Plaques
    - **Microscopic:**
        - Anucleate Dead Cells in Basal Layer ("Civatte Bodies")
        - Hyperplasia
        - Hyperkeratosis (Scaling)
- **Clinical Features:**
    - Itchy, Purple Polygonal Papules/Plaques
    - Shiny, flat-topped and firm on palpation
    - on Skin (commonly wrists & ankles), Mucosa, Genitals, Oral
    - Self-limiting (1-2 yrs)
- **May affect skin and mucosal surfaces; Several clinical types:**
    - Cutaneous lichen planus
    - Mucosal lichen planus
    - Lichen planopilaris
    - Lichen planus of the nails
    - Lichen planus pigmentosus
    - Lichenoid drug eruption



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PITYRIASIS ROSEA

- **Aetiology:**
  - Viral infection
- **Epidemiology:**
  - Mostly teenagers & young adults
- **Pathophysiology:**
  - Associated with reactivation of HHV6/7 (which cause roseola in infants)
    - → Influenza viruses & vaccines can also trigger pityriasis rosea
- **Clinical Features:**
  - Post-viral rash that occurs several days after an URTI (cough/cold/sore throat etc)
  - lasts 6-12 weeks
  - Characterised by a 'herald patch' followed by similar smaller oval red patches on chest and back
  - **Herald patch:**
    - Appears 1-20 days before generalised rash
    - Oval pink/red plaque
    - 2-5cm diameter
    - scale
  - **Secondary Rash:**
    - Appears a few days after the herald patch
    - More scaly patches/plaques
    - Mostly on chest and back
    - Some on thighs/upper arms/neck
- **Management:**
  - Bathe/shower with plain water & bath oil/aqueous cream/other soap substitute
  - Apply moisturisers
  - Some sun exposure
  - May treat with acyclovir
  - Topical steroid cream may reduce itch
  - Phototherapy

https://www.healthdirect.gov.au/pityriasis-rosea

PSORIASIS

- **Aetiology:**
  - Multifactorial (Genetic & Immune-mediated inflammatory disorder)
  - Emotional stress and physical trauma both exacerbate and precipitate psoriasis
- **Pathogenesis:**
  - Important role of ***T-cell immunity*** is recognised
  - Sensitized T cells infiltrate the skin and secrete cytokines and growth factors
    - →Continuous stimulation of basal cells → Increased cell turnover
    - + →Inflammation, Vascular Proliferation Angiogenesis
- **Morphology:**
  - **Gross:**
    - Plaque covered with Silvery Scales (Due to *Hyperkeratosis* & *Parakeratosis*)
      - Bilateral
      - Well-Demarcated plaques (Differentiates it from Dermatitis)
      - Erythematous Based
    - Abnormalities of the Nails (Dystrophy/Pitting/Onycholysis)
    - ***Auspitz Sign*** - Microbleeding when crusts are removed
  - **Microscopic:**
    - Acanthosis (Deepening of Rete Ridges & Thickening of the Epidermis)
    - Parakeratosis (Stratum Corneum still has Nuclei)
    - Neutrophilic Microabscesses in the Epidermis
    - Tortuous Papillary Dermal Vessels

- **Clinical Features:**
  - Psoriasis may present at any age from infancy onwards
  - Generally persists lifelong; fluctuating in extent & severity
  - NOT itchy
  - Symmetrical distribution
  - Can cause Multi-System Disorder:
    - Psoriatic Arthritis
    - Myopathy
    - Enteropathy
    - Immunodeficiency
- **Diagnosis:**
  - Clinical diagnosis made on the basis of morphology
  - Punch biopsy can be definitive
- **Treatment:**
  - Emollients
  - Coal tar preparations
  - Dithranol
  - Salicylic acid
  - Vitamin D Analogue (Calcipotriol)
  - Topical CorticoSteroids
  - Sunshine/Phototherapy
  - (Systemic if severe: Eg: Methotrexate/Ciclosporin/Acitretin)
- **Prognosis:**
  - Chronic
  - NOT Curable
  - Can be exacerbated by Stress

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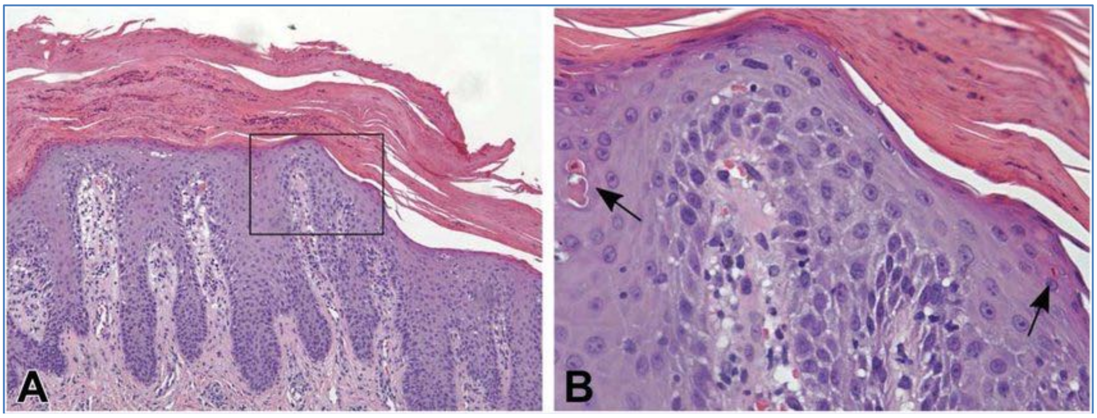

Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Psoriasis-A-Histologic-section-100-demonstrating-classic-findings-of-psoriasis_fig1_333137128

VESICULOBULLOUS DISEASES

VESICULOBULLOUS DISEASES

	PEMPHIGUS	PEMPHIGOID	Dermatitis Herpetiformis
age	mid - older	elderly	30-40
antibody	IgG	IgG	IgA
location	Suprabasilar	Subepidermal	subepidermal
inflammation	mixed	eosinophils	neutrophils
Site of dysfunction	Desmosomes	Basement membrane and hemidesmosomes	Anchoring fibrils
Antibody against:	Desmoglein	Bullous pemphigoid antigen	reticulin
Immuno	"fish net"	Linear basement membrane	Dermal tip

BULLOUS PEMPHIGOID

- **Aetiology:**
    - Autoimmune subepidermal blistering disease
    - Some association with psoriasis, malignancy, HLA (Human leukocyte antigen)
    - Can be associated with PD1-Inhibitor-Immunotherapies (Eg: Pembrolizumab, nivolumab) used to treat metastatic melanoma
- **Epidemiology:**
    - Old Age Patients (Usually >80yrs)
    - M=F
- **Pathogenesis:**
    - Antibody against Basal Layer → Inflammation → Destroys Sub-Epidermal Anchoring Proteins
    - Dissociation of the dermal-epidermal layer → Blisters
- **Morphology:**
    - **Gross:**
        - Separation of the whole epidermis from the dermis
        - Large, Tense, ***Subepidermal*** Bullae
        - Sometimes Haemorrhagic Blisters
    - **Microscopic:**
        - NO Acantholysis
- **Clinical Features:**
    - Non-specific rash for several weeks before blisters appear
    - Eczematous areas
    - Urticaria-like red skin
    - Appearance of fluid filled blisters (clear or cloudy, yellowish/bloodstained fluid)
- **Complications:**
    - Staphylococcal/streptococcal skin infection → Sepsis
- **Treatment:**
    - Ultra-potent topical steroids for localised disease
    - Systemic steroids (Eg: Prednisone) for generalised disease
    - Tetracycline antibiotics (Eg: Doxycycline) may be effective as monotherapy for very mild disease
    - Antibiotics for secondary bacterial infection

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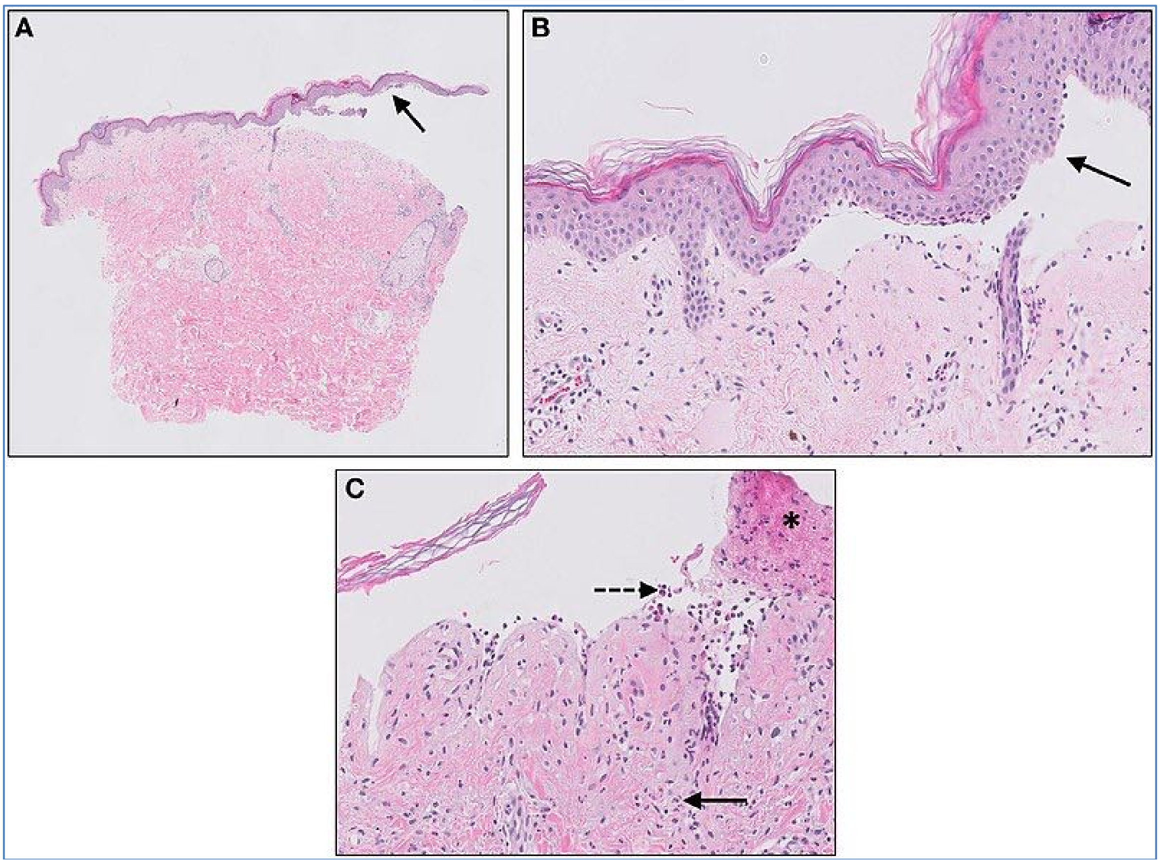
Subepidermal blistering [solid arrows in (A,B)] and influx of inflammatory cells including eosinophils and neutrophils in the dermis [solid arrow (C)] and blister cavity [dashed arrows (C)]. In (C) also deposition of fibrin is noted (asterisks).



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PEMPHIGUS VULGARIS

- **Aetiology:**
  - Autoimmune
  - May also be drug-induced (Eg: Penicillamine, ACE-inhibitors, ARB's, & Cephalosporins)
- **Epidemiology:**
  - Onset 30-60yrs
  - More common in Jewish and Indian descent (genetic association)
- **Pathogenesis:**
  - Antibodies against Desmosomes (that hold keratinocytes together)→ Destroys Desmosomes
  - Keratinocytes separate from each other → Acantholysis (Intraepidermal Blisters)
- **Morphology:**
  - **Gross:**
    - Thin, Flaccid Blisters (Easily Popped)
    - Look similar to burns
    - Many have popped → Crusting
  - **Microscopic:**
    - Acantholysis (***Intraepidermal*** Bullae)
    - Loose cells inside the Bullae
- **Clinical Features:**
  - Common on mucous membranes (Eg: Mouth/genitals)
  - Blisters typically develop after a few weeks/months
  - Popped blisters lead to itchy, painful erosions
- **Treatment:**
  - Systemic corticosteroids (moderate to high dose oral prednisone/prednisolone)
    - Or pulsed IV methylprednisolone
  - Other immunosuppressants (Eg: Azathioprine/Cyclophosphamide/Rituximab)
- **Complications:**
  - Secondary bacterial infection
  - Secondary fungal infection (Eg: Candida)
  - Viral infections (Eg: Herpes simplex)
  - Complications of long term corticosteroid use

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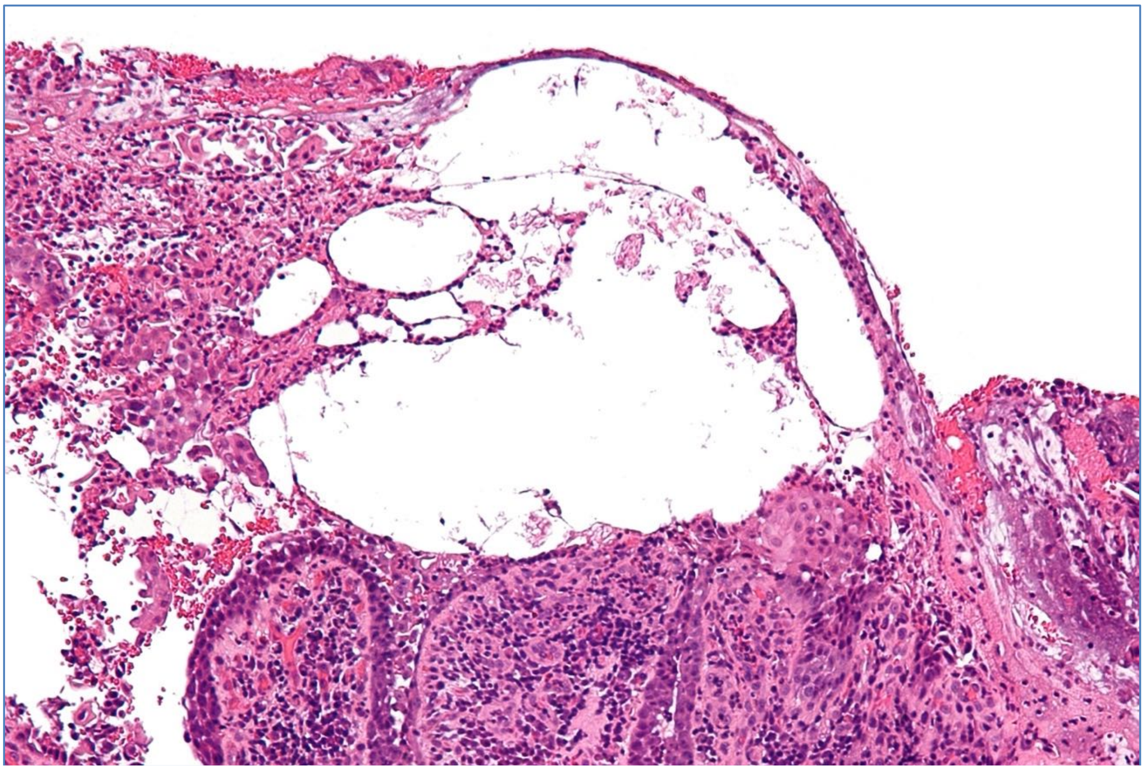

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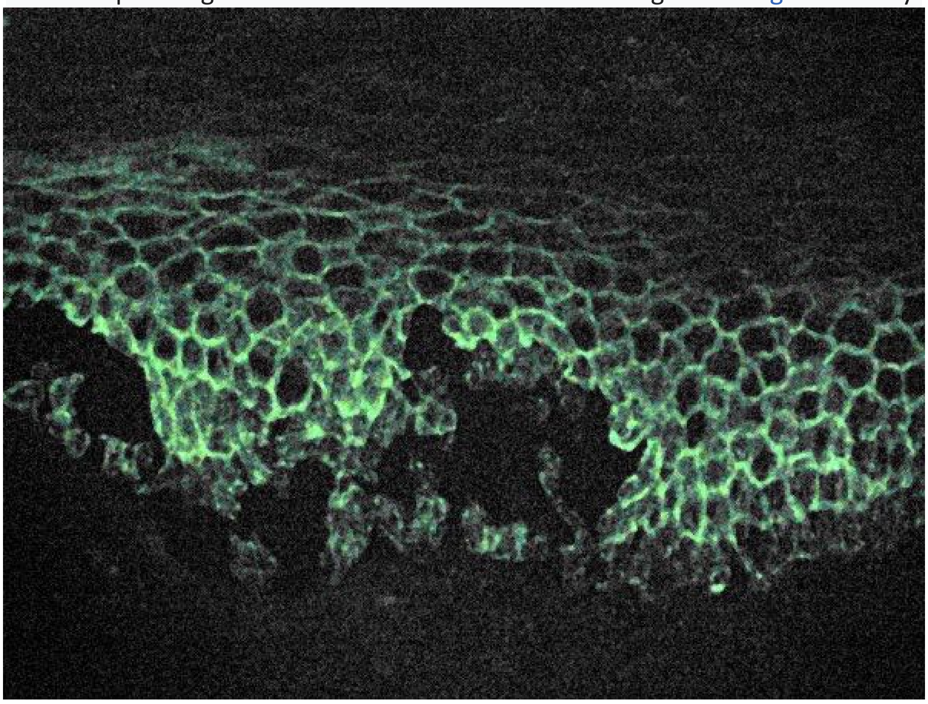
Microscopic image of direct immunofluorescence using an anti-IgG antibody.



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DERMATITIS HERPETIFORMIS

- **Aetiology:**
    - Associated with Coeliac Disease (Gluten Sensitivity)
    - (Note: 'Herpetiformis' is a misnomer derived from the tendency for blisters to appear in clusters, similar to herpes simplex)
- **Epidemiology:**
    - Caucasians age 15-40yrs
    - M:F = 2:1
    - Genetic association with HLA (human leukocyte antigens)
    - Occurs in 15-25% of coeliac patients
- **Pathogenesis:**
    - Gluten triggers IgA antibodies in an autoimmune process that targets the skin and gut → Inflammation
- **Morphology:**
    - **Gross:**
        - VERY Itchy, Small, Erythematous Papules
        - Vesicles
        - Occasional Bullae
    - **Microscopic:**
        - Subepithelial, Neutrophilic Microabscesses in Dermal Papilla
- **Clinical Features:**
    - Associated with Coeliac Disease
    - Symmetrical distribution
    - Commonly on scalp/shoulders/buttocks/knees
    - Lesions often resolve to leave hypopigmentation or hyperpigmentation
- **Treatment:**
    - Gluten free diet
    - Medication:
        - Dapsone (Gold standard)
        - Otherwise topical steroids/systemic steroids/sulfapyridine/Rituximab if intolerant to Dapsone

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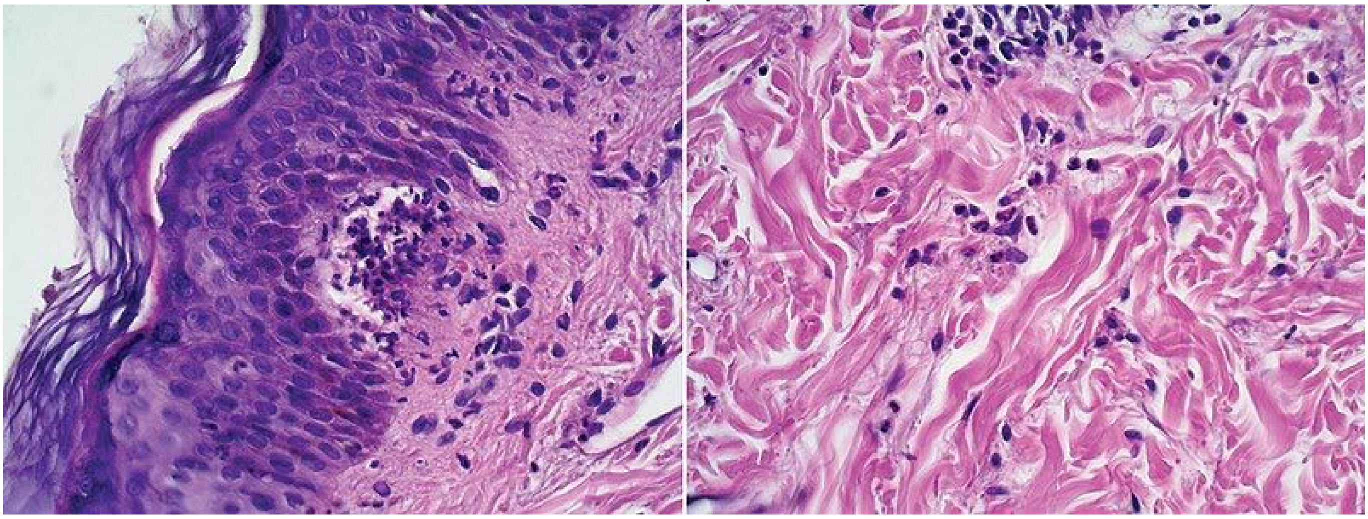
Subepidermal vesicles, with papillary neutrophil microabcesses, with neutrophil, eosinophil and lymphocytes infiltrates in the superficial dermis



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PORPHYRIA CUTANEA TARDA

- **Aetiology:**
  - Genetic or acquired defect in Liver enzyme '**uroporphyrinogen decarboxylase' (UROD)**
    - 30% of patients are hereditary
    - Other patients, cause is underlying liver disease (Eg: Alcoholism, hep B/C, hemochromatosis)
    - Dialysis patients also at risk as cannot excrete porphyrins
- **Epidemiology:**
  - The most common type of porphyria
- **Pathophysiology:**
  - A deficiency or block of one of the enzymes in the porphyrin pathway of haem synthesis → results in a build-up of precursor protein or intermediate molecule
- **Clinical Features:**
  - Photosensitivity
  - Increasingly fragile skin on back of hands/forearms
  - Other sun-exposed areas (Eg: Face/scalp/neck) may also be affected
  - Slow-to-heal crusted erosions following minor injuries
  - Fluid/blood-filled blisters
  - Post inflammatory pigmentations & milia
  - Characteristically dark reddish/tea-coloured urine
- **Management:**
  - Reduce alcohol consumption
  - Cease oestrogen/HRT
  - Avoid excessive iron intake
  - Hep C treatment (If relevant)
  - Sun-protective clothing
  - Venesection (approx 500mL of blood every 2-4 weeks) to reduce iron stores to normal
  - Hydroxychloroquine can be useful in elderly/anaemic patients; aids porphyrin excretion

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Change in urine colour before and after sun exposure Left figure is urine of the first day. Right figure is urine after sun exposure for 3 days. Urine colour changed to “port wine” colour after sun exposure. This colour change is due to increased concentrations of porphyrin intermediates in the urine, indicating an abnormality in production and a partial block within the enzymatic porphyrin chain with metabolite formation. The urine colour usually becomes darker with acute illness, even dark reddish or brown after sun exposure.

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DRUG ERUPTIONS

DRUG ERUPTIONS

MORBILLIFORM DRUG REACTIONS

- **Aetiology:**
  - Type IV Hypersensitivity
  - (Typically caused by antibiotics)
- **Epidemiology:**
  - Most common type of drug reaction
- **Pathophysiology:**
  - Allergic reaction mediated by Cytotoxic T-Cells → Cytokines → Inflammation
- **Clinical Features:**
  - Maculopapular exanthema
    - Typically on trunk → limbs & neck
    - Bilateral & symmetrical
    - Annular, targetoid, urticarial-like or polymorphous lesions
  - Typically 1-2 weeks after starting a drug & up to 1wk after stopping it
  - More rapid drug eruption to subsequent exposure
- **Management:**
  - **Identify causative drug and stop it!**
  - Monitor for complications
  - Emollients
  - Potent topical steroid creams
  - Antihistamines have some limited benefit

Credit: https://dermnetnz.org/topics/morbilliform-drug-reaction

URTICARIAL DRUG REACTIONS

- **Aetiology:**
  - Type I hypersensitivity – Allergy (Food/drug/plant/etc)
- **Pathogenesis:**
  - Antigen is Re-Exposed to a sensitized Mast-Cell/Basophil → IgE-Bound Mast Cell *Degranulates*:
    - → Releasing Inflammatory Mediators (Histamine) of Type-1-Hypersensitivity Reactions
    - →Perivascular inflammatory infiltrate: lymphocytes, neutrophils or eosinophils
- **Morphology:**
  - **Gross:**
    - Erythematous Papules & Plaques (Reddish, raised areas)
  - **Microscopic:**
    - 1) Dermal Oedema (Not Spongiosis – which is *Epidermal* Oedema)
    - 2) Dilated Dermal Blood Vessels (With Perivascular Inflammatory Cells)
    - 3) Dermal Inflammatory Infiltrate (Lymphocytes, Neutrophils, Eosinophils)
    - (Normal Epidermis – No Spongiosis or Hyperplasia)
- **Clinical Features:**
  - Usually on trunk and extremities
  - Individual lesions are transient, usually resolve in 24 hr, but entire episode may last for days
  - All ages, more in 20 – 40y
- **Management:**
  - Stop causative drug
  - Cold pack or wet cold compresses may provide relief
  - Antihistamine (Eg: Cetirizine / Loratadine / Desloratadine / Fexofenadine)
  - Systemic short course prednisone/prednisolone if antihistamines fail
  - IM Adrenaline if threatened anaphylaxis or throat swelling

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HERITABLE DERMATOSIS

HERITABLE DERMATOSIS

ICHTHYOSIS VULGARIS

- **Aetiology:**
  - Inherited or acquired
  - (Autosomal semidominant if inherited)
  - Loss-of-function mutations in the gene encoding for protein Filaggrin (FLG)
- **Epidemiology:**
  - The most common form of inherited ichthyoses
  - Most common in Europeans
- **Pathophysiology:**
  - Defective production of Filaggrin (which normally binds keratin fibres in epidermal cells to form an effective skin barrier)
  - → Defective skin barrier → excess trans-epidermal water loss → Xerosis (dryness)
  - → Excessive scale → hyperkeratosis
- **Clinical Features:**
  - Excessive dry, scaly skin
  - Symptoms worse in cold/dry climates
  - Typically occurs around 2mths after birth
  - Symptoms may worsen until puberty
  - Xerosis worse on extensor surfaces/scalp/face/trunk; skin folds usually spared
- **Management:**
  - Emollients with high lipid content (Eg: Lanolin cream)
  - Bathe in salt water
  - Creams or lotions with salicylic acid
  - Oral Retinoids (Eg: Isotretinoin) if severe

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NEUROFIBROMATOSIS 1

- **Aetiology:**
    - Autosomal Dominant Inherited
    - Or Acquired mutation
- **Epidemiology:**
    - 1 in 3000 births
- **Pathophysiology:**
    - Mutation in the Neurofibromin gene (Chromosome 17)
- **Clinical Features:**
    - 6 or more café-au-lait macules (flat light brown birthmarks)
    - Freckling in skin folds
    - Lisch nodules in the iris of the eye
    - **Multiple neurofibromas (tumours hanging off the skin):**
        - Circumscribed, superficial, soft button-like brown/pink coloured nodules
        - Pathognomonic buttonhole invagination when pressed with a finger
    - Malformation of long bones & scoliosis
    - Short stature/growth hormone deficiency
    - Learning difficulties
    - Tumours on the optic nerve → visual loss
    - Hearing defects
- **Management:**
    - No cure
    - Elective excision of problematic lesions
    - Risk of isolation/loneliness

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VITILIGO

- **Aetiology:**
  - Systemic autoimmune disorder involving the TYR (Tyrosinase) enzyme in melanocytes
- **Epidemiology:**
  - 0.5-1% of the population
- **Pathophysiology:**
  - Depigmenting process due to loss of melanocytes
- **Clinical Features:**
  - Well-defined milky-white patches of skin (leukoderma)
  - Pigment loss commonly begins before age 20
  - Higher risk of autoimmune conditions (Eg: Diabetes, thyroid disease, pernicious anaemia, Addison’s disease, SLE, rheumatoid arthritis, psoriasis, etc)
- **Management:**
  - Minimise skin injury
  - Cosmetic camouflage (Eg: Make-up/tans/tattooing)
  - Sun protection
  - Corticosteroid creams
  - Experimental use of Ruxolitinib for facial vitiligo
  - Phototherapy
  - Systemic (Eg: Oral steroids / methotrexate / Ciclosporin / minocycline)
  - Depigmentation therapy in severely affected dark-skinned individuals

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BACTERIAL INFECTIONS

BACTERIAL INFECTIONS

IMPETIGO (SCHOOL SORES)

- **What is it?**
  - Superficial Bacterial Skin Infection
  - Most Common in school kids
  - Very Contagious – (Spread by Close Contact & Poor Hygiene)
  - Usually resolves slowly
- **Organism:**
  - **Mostly *Staphylococcus Aureus***
  - **Sometimes *Streptococcus Pyogenes***
    - Can lead to Glomerulonephritis or Rheumatic Fever if it's Strep
  - Staphylococcus Aureus (Bullous) - (Pic 1)
  - Streptococcus (Non-bullous) – (Pic 2)
- **Presentations:**
  - Occur most commonly on face
  - Fragile vesicles rupture & crust
  - Can be confused with HSV
  - **1) Nonbullous/Crusted Impetigo:**
    - (Most common)
    - Yellow crusts and erosions
    - Itchy/Irritating (but not painful)
  - **2) Bullous impetigo:**
    - Always due to *Staph-Aureus*
    - →Mildly irritating blisters that erode rapidly leaving a brown crust
  - **3) Ulcerative lesions**:
    - Always due to *S-Pyogenes*
    - Most common in Aboriginal Communities
- **Very Infectious**
  - Epidemic in young children
  - Transmitted through skin contact
  - Outbreaks associated with poor hygiene / crowded living conditions
- **Treatment:**
  - Cover Affected Areas
  - Abstain from School
  - Systemic or Topical Antibiotics

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ERYSIPELAS

- **Aetiology:**
  - Group A Strep (GAS) / Staphylococcus Aureus
  - Breaks in skin (Eg: Insect bites/ulcers/cracking skin conditions/eczema)
- **Epidemiology:**
  - Mostly infants & older people
- **Pathophysiology:**
  - Potentially serious bacterial infection of the skin
  - → Infection of the upper dermis → extends to superficial cutaneous lymphatics
- **Clinical Features:**
  - A superficial form of Cellulitis
  - 'St Anthony's fire' = Intense rash associated with erysipelas
  - Rapid onset fevers, chills, rash
  - Affected skin has sharp, raised border
  - Bright red, firm & swollen
  - May blister or become necrotic
  - may spread to deeper lymphatics (lymphangitis)
- **Management:**
  - Wound care
  - Oral/IV penicillin antibiotic
  - Erythromycin/Roxithromycin if Penicillin allergic
  - Vancomycin if MRSA

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CELLULITIS

- **What is it?**
    - Bacterial infection of the Dermis and Sub-Cutaneous Tissues
- **Organism:**
    - **Adults:** 90% due to *Staphylococcus Aureus*/GAS
    - **Children:** *H-Influenzae* B
    - **Associated with cat/dog bite:** *Pasturella multocida*
- **Presentation:**
    - Painful, raised and Oedematous Erythema (Most commonly on Lower Leg)
    - Possible Blistering
    - Lymphadenopathy - & Malaise & Fever
    - Note: Orbital Cellulitis is a medical emergency!
- **Distribution:**
    - **Children** – Periorbital Area
    - **Adults** – Lower Legs
- **There's typically an underlying cause:**
    - Lymphedema
    - Tinea, Herpes simplex infection, Chronic sinus infection
    - Chronic dermatitis
    - Poor lower leg circulation
    - Wounds
- **Treatment:**
    - Oral/IV penicillin antibiotic
    - Erythromycin/Roxithromycin if Penicillin allergic
    - Vancomycin if MRSA

Source: https://www.nhs.uk/conditions/cellulitis/

FOLLICULITIS

- **Aetiology:**
  - Acute pustular infection/inflammation of a hair follicle
  - Commonly after Waxing/Shaving
  - Commonly *Staphylococcus Aureus;* may also be caused by yeasts
- **Pathophysiology:**
  - Inflammation of hair follicle/s from any of many causes
- **Clinical Features:**
  - An Erythematous Pustule centred on a Hair Follicle
- **Management:**
  - Careful hygiene
  - Appropriate antimicrobials (Eg: Antibiotics if bacterial/Antifungals if fungal)
  - Avoid close shaving/discard old razors

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FURUNCLES (BOILS) & CARBUNCLES

- **Aetiology:**
  - Deep form of bacterial folliculitis
  - Typically staphylococcus aureus
- **Pathophysiology:**
  - Begin as superficial infections of hair follicles (folliculitis)
    - →Organisms travel down Hair Follicles following Disruption (Eg: After Shaving)
    - →Development of boils (furuncles)
  - Number of boils cluster together = “Carbuncle” (aka Abscess)
- **Clinical Features:**
  - Tender, red nodule which enlarges & may later discharge pus
  - Painful
  - Cellulitis may occur
  - Fever/illness is possible
- **Management:**
  - Treated aggressively with antibiotics and drainage

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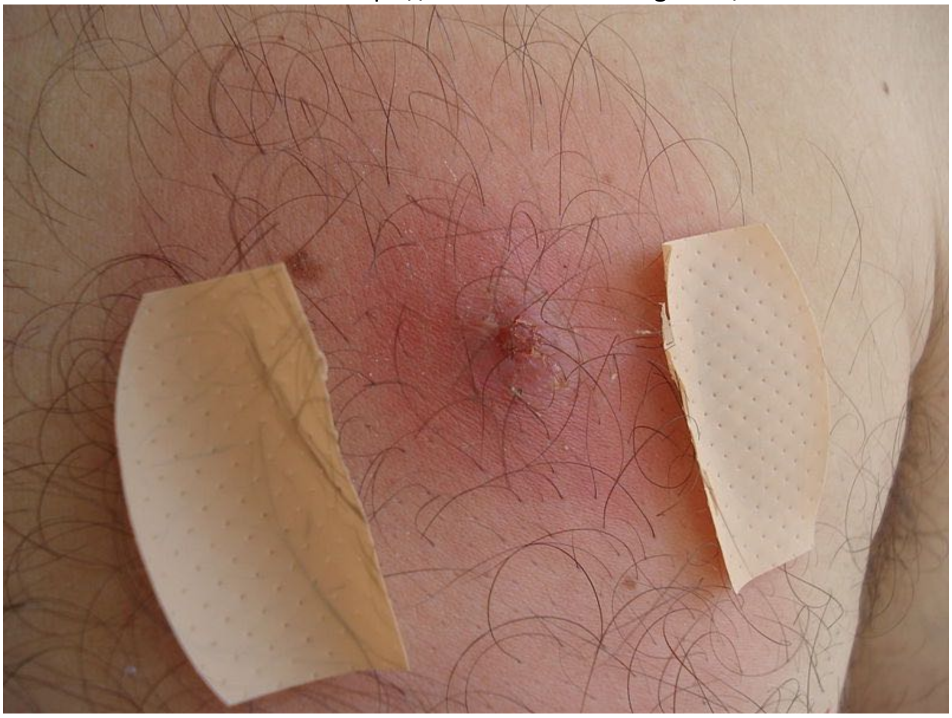

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FUNGAL INFECTIONS

FUNGAL INFECTIONS

FUNGAL INFECTIONS:

- Fungal Nail Infections (**Dermatophytes, Candida Albicans, or Moulds**)
- Superficial (skin, nails and hair):
- **Dermatophytes (Tinea):**
    - Eg: Ringworm Fungi
        - **2 Common Genuses** - (*Tricophyton* & *Microsporum*)
        - Infection Restricted to the Superficial Keratinised Layers of the skin
            - Note: Keratin is the Fungi's food source
        - Can Evoke Cellular immune responses
- **Pathogenesis:**
    - **Fungi ONLY Metabolizes Keratin:**
        - :. Only infect the Stratum Corneum
        - Note: Can Also Invade Hair Shafts
- **Presentation:**
    - *Annular* eruption with *Irregular edge*
    - Central clearing Peripheral scaling
    - Focal hair loss due to infection of Hair Follicle
    - Focal pityriasis (Skin Flaking)
    - Usually not pruritic
    - "Tinea Versicolor" (Depigmentation of the Skin)
- **Conditions Named Based On Location of Infection:**
    - Tinea Corporis (On Body)
    - Tinea Capitis (On Head)
    - Tinea Crura (Pubic Area)
- **Diagnosis:**
    - Clinical Diagnosis
    - Woods lamp – only *Microsporum canis* fluoresces
    - Microscopy of hairs/nail shavings/skin shavings
- **Treatment:**
    - **Topical Antifungals:**
        - Clotrimazole
        - Miconazole
    - **Oral Antifungals:**
        - Fluconazole

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PITYRIASIS (TINEA) VERSICOLOR

- **Aetiology:**
  - Yeast infection of the skin
- **Epidemiology:**
  - Frequently young adults
  - More common in hot, humid climates
  - Common in people who perspire heavily
- **Pathophysiology:**
  - Mycelial growth of fungi (Malassezia) → fungal metabolites diffuse into epidermis and impairs melanocyte function → depigmentation
- **Clinical Features:**
  - Trunk, neck and/or arms
  - Either brown/pale/pink lesions
- **Management:**
  - Topical Azole cream (Eg: Ketoconazole)
  - Selenium sulphide
  - Terbinafine gel
  - Oral antifungals (Eg: Itraconazole/fluconazole can be used if extensive or topicals fail)

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PARASITIC INFECTIONS

PARASITIC INFECTIONS

SCABIES

- **Organism:**
  - ***Sarcoptes scabii*** (Scabies Mite)
- **Epidemiology:**
  - Human infestations originating from pigs, horses and dogs are mild and self-limiting
  - Scabies infestations from other humans never cure without intervention
- **Ecology:**
  - Mites live in stratum corneum (Don't get any deeper)
  - Eat stratum corneal Keratinocytes
  - Make "tunnels" by eating
  - Mating occurs on the hosts skin
  - Fertilized Female Mites Burrow into the Stratum Corneum (1 mm deep)
  - Salivary Secretions contain Proteolytic Enzymes →Digest Keratinocytes
- **Transmission:**
  - High prevalence in children (50%) and adults (25%) in tropical remote communities
  - Spread by close physical contact
- **Presentation:**
  - Itch (Exacerbated at night and after hot showers)
  - Itchy, Excoriated Rash on Trunk, associated with Scaly Burrows on the fingers and wrists
  - Often vesicles and pustules on the palms and soles and sometimes on the scalp
- **Diagnosis:**
  - **Clinical Diagnosis:**
    - Chronic itch with Symmetrical Rash
    - Burrows
  - **Skin Scraping - Look for Scabies Mites:**
    - Intact larvae, nymphs or adults
    - Unhatched or hatched eggs
    - Moulted skins of mites
    - Fragments of moulted skins
    - Mite faeces
- **Treatment:**
  - **Topical** Permethrin
  - **Or Oral** Ivermectin (But not on PBS – Very Expensive)
  - **Environmental Measures:**
    - Mites can contaminate bedding, chairs, floors, and even walls
      - (Usually only a problem with crusted scabies)
    - Wash, sun, vacuum, surface insecticide
  - **Community Prevention:**
    - Treat all close contacts – Especially in Indigenous Communities
    - Simultaneous Effective Treatment
  - ***TREAT AGAIN IN 7 DAYS***

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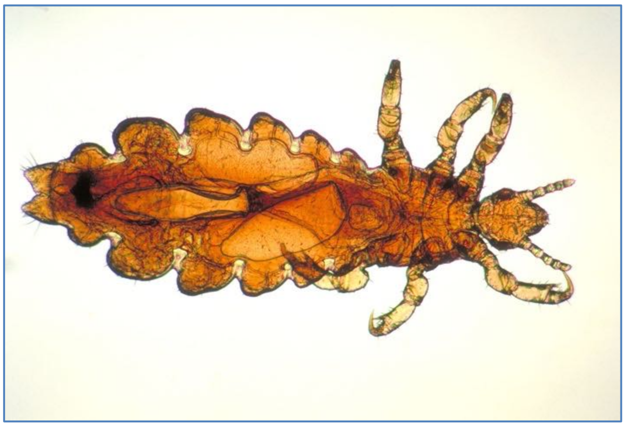
LICE (PEDICULOSIS)

- **3 Types:**
    - **1) Head Lice: *Pediculus Humanus Capitis***
        - **Epidemiology:**
            - Common in Primary School Children in the Tropics
            - Higher prevalence in Aboriginal Children
        - **Diagnosis:**
            - Conditioner + Fine-Tooth Come
            - Wipe combings on white tissue paper
    - **2) Body Lice: *Pediculus Humanus Corporis***
        - Live on clothes, and come to the body to feed
    - **3) Pubic Lice: *Phthirus Pubis***
        - Largely sexually transmitted
        - Blood Feeder
        - Can infect any Body Hair (Pubic/Trunk/Legs/Axilla/Beard) but rarely head
- **Lifecycle:**
    - Eggs laid in hair (knits)
    - Larvae grow into adults
    - Adults – **blood sucking** (live in hair)
- **Transmission:**
    - head-head contact
- **Presentation:**
    - Scalp and Neck can be Itchy
    - Nits are noticeable on the hairs
- **Diagnosis:**
    - **Best Method = *'Conditioner & Comb Technique'*:**
        - Very Practical for parents
        - Cost Effective
        - High Sensitivity
        - Conditioner 'Stuns' the lice by suffocating them → Prevents them from running away
- **Management/Treatment:**
    - Conditioner & Nit Comb
    - Physical Removal
    - Cut Hair
    - Topical Insecticidal Cream
    - Good idea to wash pillows and hats though – Hot Wash
    - (Treat all body hair – for Pubic lice)
    - **Reasons for Treatment Failure:**
        - Inadequate application of the product
        - Lice are resistant to insecticide
        - Failure to retreat to kill nymphs emerged from eggs
        - Reinfection



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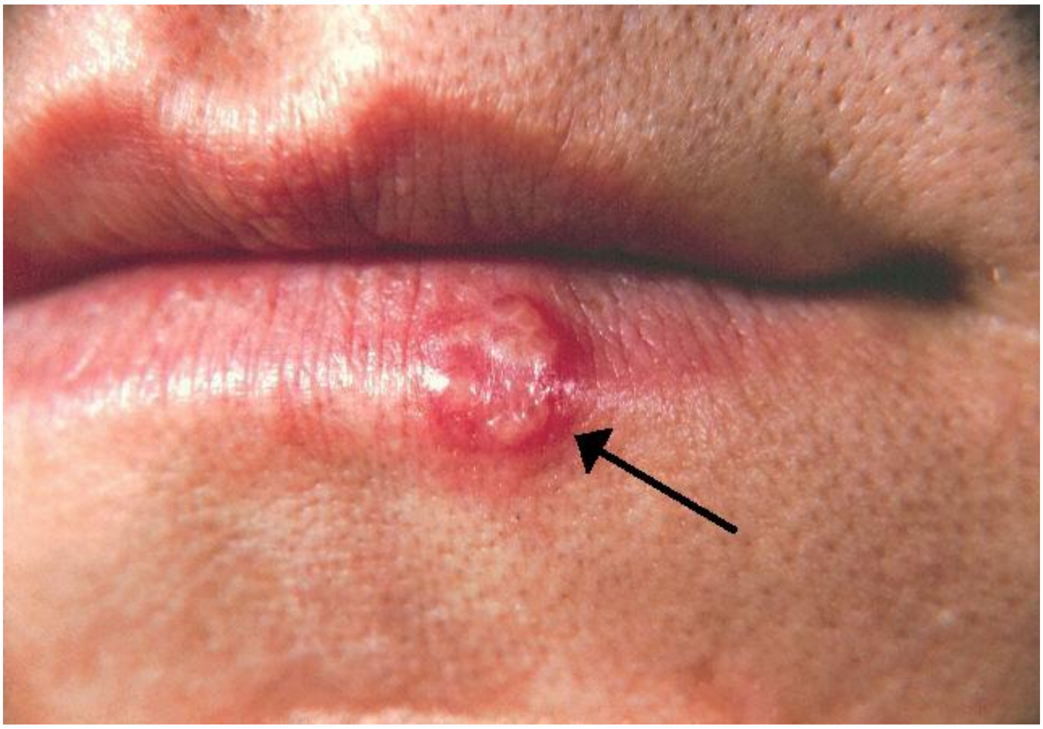
https://www.pcds.org.uk/clinical-guidance/pediculosis

VIRAL INFECTIONS

VIRAL INFECTIONS

HERPES SIMPLEX

- **What is it?**
  - Common Mucosal Viral Infection that presents with localised blistering
  - Can reside in a latent state
- **2 Types:**
  - **Type 1: Typically facial/oral infections** (Cold sores/fever blisters)
    - Occur mainly in infants & young kids
  - **Type 2: Mainly Genital**
    - Occur after puberty (often transmitted sexually)
- **Presentation:**
  - **Stages of Infection:**
    - 1) Prodromal Stage Vesicle or "blister" stage
    - 2) Ulcer stage
    - 3) Crust stage
  - The virus grows down the nerves and out into the skin → Localised Blistering
  - Neuralgia
  - Lymphadenopathy
  - High Fever
- **Recurrences can be triggered by:**
  - Minor trauma/Other infections/UV radiation/Hormonal factors/Emotional stress/Operations/procedures on face
- **Treatment:**
  - Mild cases require no treatment
  - Sun protection to prevent
  - Oral Antiviral Drugs (Stop the virus multiplying)
- **Complications:**
  - Encephalopathy
  - Trigeminal Neuralgia (Neurogenic Pain)



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CHICKEN POX (VARICELLA ZOSTER):

- **What is it?**
  - Highly contagious disease
  - Typically childhood disease (before 10yrs)
  - One infection thought to confer lifelong immunity
- **Organism:**
  - Varicella zoster virus (HHV3) (AKA: Chicken Pox Virus, Varicella, Zoster)
- **Transmission:**
  - Highly Infectious
  - From person to person
  - Aerosol Droplets
  - Direct contact with fluid from open sore
- **Pathophysiology:**
  - Incubation Period ≈ 2wks
  - **(Chicken Pox)** Initial Mucosal Infection → Viremia → Epidermal Lesions
    - May lead to → Latent infection of Dorsal Ganglion Cells of Sensory Nerves
  - **(Shingles)** Reactivation of latent Varicella Zoster Virus in Peripheral Nerves
- **Signs/symptoms:**
  - Itchy rash or red papules
  - Begins on the Trunk →Face and Extremities
  - May cover entire body
  - High fever/headache/cold-like symptoms/vomiting/diarrhoea
- **Diagnosis:**
  - Clinical Diagnosis
  - Immunofluorescence
  - Test for Elevated VZV-Specific Antibodies
    - (IgM – Primary Infection; IgG – Second Infection)
- **Treatment:**
  - Symptomatic
  - Resolves on its own
- **Complications:**
  - **Varicella During Pregnancy can → Congenital Varicella Syndrome:**
    - **Spontaneous Abortion** (3-8% in 1st Trimester) or IUGR
    - **Skin:** Cutaneous Defects, Hypopigmentation
    - **Neuro:** Intrauterine Encephalitis, Brain Damage, Seizures, Developmental Delay
    - **Eye:** Chorioretinitis, Cataracts, Anisocoria
    - **MSK:** Limb Hypoplasia
    - **Systemic:** cerebral cortical atrophy
    - **Renal:** Hydronephrosis, Hydroureter
    - **GI:** GORD
    - **CVS:** Congenital Heart Defects
  - **Perinatal Varicella Infection:**
    - severe → mortality rate of 30%


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HERPES ZOSTER (SHINGLES)

- **What is it?**
  - Reactivation of Latent Herpes Varicella Zoster Virus
- **Pathophysiology:**
  - Incubation Period ≈ 2wks
  - **(Shingles)** Reactivation of latent Varicella Zoster Virus in Peripheral Nerves
- **Presentation:**
  - Painful blistering rash along 1/more Dermatomes
  - Virus is seeded to nerve cells in spinal cord
  - Fever, malaise and headache
  - Lymph nodes draining affected area are often enlarged/tender
  - Can also result in nerve palsy
- **Diagnosis:**
  - Clinical Diagnosis
  - Test for Elevated VZV-Specific Antibodies
  - PCR
- **Transmission:**
  - Shingles are infectious
  - From person to person
  - Direct contact with fluid from open sore
- **Treatment:**
  - Antiviral treatment
  - Rest & analgesia
  - Oral Antiviral

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MOLLUSCUM CONTAGIOSUM

- **Aetiology:**
    - **Poxvirus**
- **Transmission:**
    - Skin to Skin Contact
- **Epidemiology:**
    - Most Common in Children
    - Also Sexually-Active People
- **Lesions:**
    - Dome Shaped Papules
    - ***Dimpled Centre (Centrally Umbilicated)**
    - Not Painful, but may Itch
- **Treatment:**
    - Often Unnecessary - May Resolve on its Own
    - – May Require Antibiotic Treatment if Secondary Bacterial Infection Occurs
    - Cryotherapy,Curettage,Laser,Acid
- **Prognosis:**
    - Once Resolved, the virus is GONE
        - (No Latent Stage like Herpes Viruses)
    - However, there is NO permanent Immunity → Can catch it again

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HPV WARTS (VERRUCA VULGARIS)

- **What are They?**
  - = Benign Tumours of the skin
  - Common in children
  - Infectious – (Spread by direct contact)
- **Organism:**
  - Typically from HPV (human papilloma viruses)
  - HPV 6 & 11 → Genital & Cutaneous Warts
  - HPV 16 & 18 → Cervical & Penile Ca
- **Pathogenesis:**
  - HPV → Cellular Mutation → Hyper/Neo-Plasia
  - → Wart Formation (Mostly Benign)
- **Morphology:**
  - **Gross:**
    - Verrucous Warts
    - Intradermal Hard Cysts
  - **Microscopic:**
    - Inward Growth Deep into the Skin
    - Koilocytic Keratinocytes (Dysplastic Squamous cells)
    - Central Blood Vessels
- **Presentation:**
  - Common on back of fingers, toes and knees
  - Common Warts (Skin/Plantar/Palmar)
  - Genital Warts (Cervix, Vulva, Penis)
    - Note: Cervical Papillomas → Can cause cervical cancer
  - Laryngeal Papilloma
- **No Reliable Treatment:**
  - 50% of childhood warts disappear within 6mths; 90% are gone in 2 years
  - Many don't bother with treatment
  - Surgical Excision/Chemical Treatment/Cryotherapy/Electrosurgery (Cauterise)
- **Clinical Features:**
  - Contagious
  - Central blood vessels → Bleed profusely when the surface is broken

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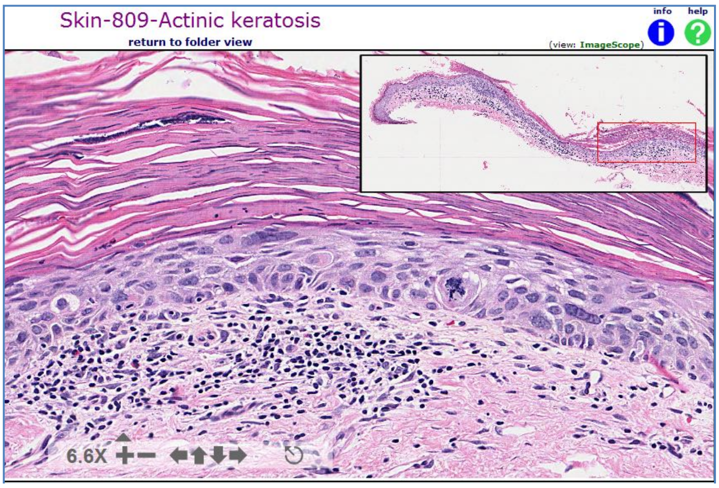
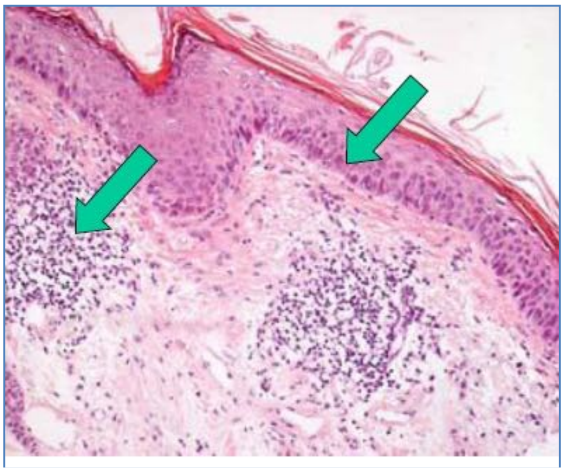
PREMALIGNANT & MALIGNANT SKIN CONDITIONS

PREMALIGNANT & MALIGNANT SKIN CONDITIONS

ACTINIC KERATOSIS (SOLAR KERATOSIS)

- **Aetiology:**
  - Sun Damage
- **Pathogenesis:**
  - = Damage to the skin from UV (wrinkling, pigmentary change, actinic keratoses and cancer)
  - Sun Damage due to Chronic Exposure to UV Radiation
- **Morphology:**
  - **Gross:**
    - Red/Tan, Irregular, Scaly Plaques
    - Hyperkeratosis
    - Inflammation/Ulceration/Crusting
  - **Microscopic:**
    - Dysplasia (Pre-Cancerous) & Atrophy
    - Inflammation
    - Hyperkeratosis *AND* Parakeratosis
    - Loss of Papillary Dermis & Rete Ridges
    - Atrophy of the Epithelium
    - Very Large & Irregular Nuclei of Epidermal Cells
    - Thick & irregular collagen bundles





Unattributable

- **Clinical Features:**
  - The earliest sign = Freckling
  - A Pre-Cancerous Skin-Growth
  - They are most common on the face, back of the hands and forearms
  - Solar keratoses are NOT cancer, however they are a sign of significant sun damage and are therefore an indicator of higher risk for non-melanoma and melanoma skin cancer
- **Management:**
  - Cryotherapy
  - Shave/curettage/electrocautery
  - Excision
  - Imiquimod cream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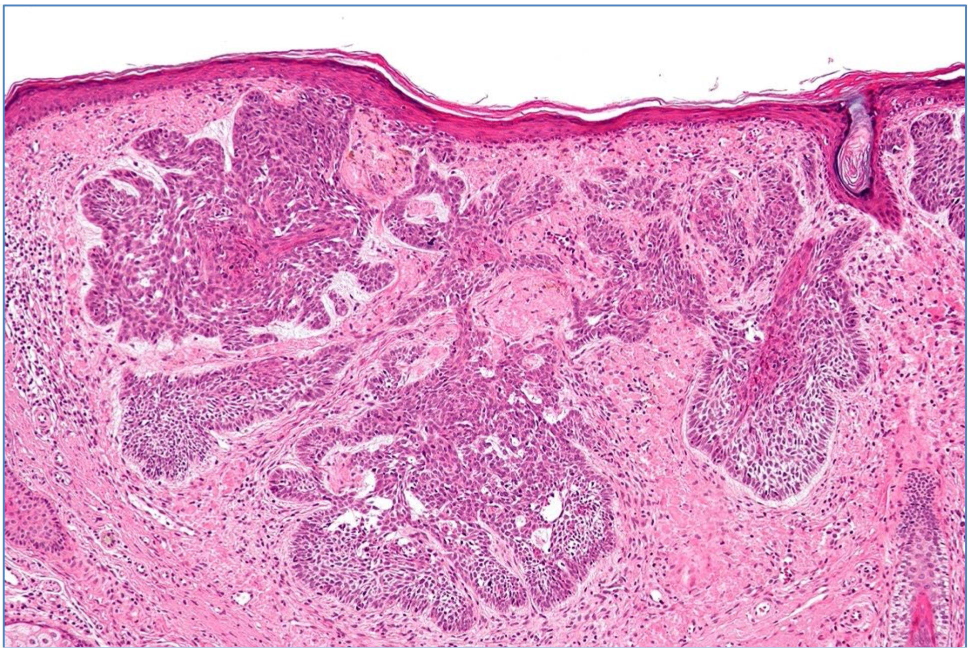
BASAL CELL CARCINOMA

- **Aetiology:**
  - Sun Exposure
- **Pathogenesis:**
  - Dysplasia & Tumorigenesis of Basal Epithelial Cells
  - Slow Growing
  - Rarely Metastasises (Locally Infiltrative, but don't often Metastasise)
- **Morphology:**
  - **Superficial BCC's**
    - Flat Red/Pink Shiny Macules or patches
    - Indistinct Borders
  - **Nodular BCC's**
    - Pearly/red/translucent Nodules
  - **Other Gross Features:**
    - Blood Vessels
    - Pearly/Shiny crust over the lesion
    - Overlying telangiectasia
    - Large Tumours may Ulcerate
  - **Microscopy:**
    - Clusters of blue Proliferating Basal Cells
    - Burrows Deep into the Dermis
    - Basal cells are pleomorphic
    - "Palisading" appearance around the edge of the clusters of Basal cells (like the walls of a palace)



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- **Clinical Features:**
  - #1 Commonest skin cancer
  - Good prognosis if treated early
- **Management:**
  - **Excisional surgery** : Still the gold standard treatment of BCC's
  - Liquid nitrogen cryotherapy
  - Curettage and Cautery
  - Photodynamic therapy (PDT)
  - Radiation treatment

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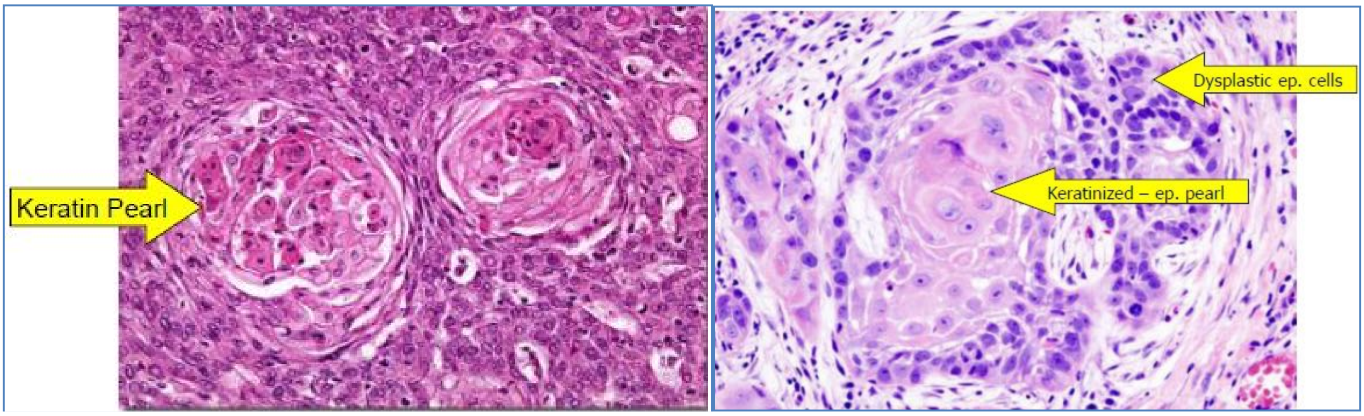
SQUAMOUS CELL CARCINOMA

- **Aetiology:**
  - Chronic Sun Damage
  - Industrial Carcinogens
  - Tobacco
- **Pathogenesis:**
  - Dysplasia & Tumorigenesis of Squamous Epithelial Cells
  - Squamous Cell Mutation → Dysplasia → Neoplasia
- **Morphology:**
  - **Gross:**
    - Typically Tender Papules/Nodules with *ROUGH, ADHERENT SCALE*
    - Hyperkeratosis
    - Sometimes Erythematous base
    - Often Ulcerate Centrally
    - Often Present as Ulcers on Lips/Mucosa
  - **Microscopy:**
    - Dyskeratosis (↑Keratin production by Pleomorphic Squamous Cells)
      1. ***Presence of Keratin "Pearls"
      2. Pink on microscopy due to lots of keratin
    - Whorls & Nests (Clusters) of malignant cells
- **Clinical Features:**
  - 2^{nd} commonest skin cancer after BCC
  - They frequently grow rapidly and may be painful
  - Malignant (Early Treatment via Wide Excision is Essential)
- **Treatment:**
  - Wide excision
  - Cryotherapy
  - Adjuvant Radiotherapy



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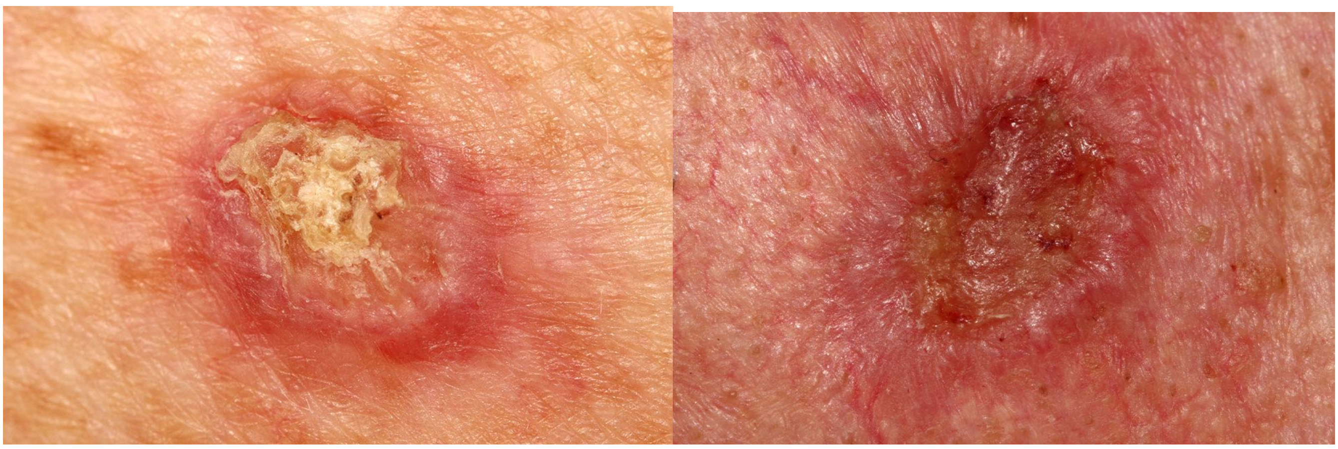


Credit: https://ballaratskincancer.com.au/squamous-cell-carcinoma-scc/

KERATOACANTHOMA

- **Aetiology:**
  - Sun damage
- **Epidemiology:**
  - Elderly / chronic sun-exposure / outdoor workers
  - Fair skin
- **Pathophysiology:**
  - Tumorigenesis of hair follicle skin cells
- **Clinical Features:**
  - Erupting skin lesion
  - Hard keratin core
  - Irritating
- **Management:**
  - Treated surgically → Excision → Pathology
  - Cryotherapy (if <0.5cm)

Image Credit: Peter Mullineux / Shutterstock

MALIGNANT MELANOMA

- **Aetiology:**
  - Sun Damage
  - Congenital
  - incidence is higher in the elderly
  - **3rd Most Common Cancer in Both Men & Women**
- **Risk Factors:**
  - Family history
  - Fair complexion
  - Freckles
  - Tendency to burn (Fitzpatrick type I/II skin)
  - Severe Childhood Sunburn (Solar Skin Damage)
  - Immunosuppression
  - High number of common acquired naevi
  - Large (>10cm) Congenital Naevi
  - Dysplastic Nevus Syndrome (5 or more dysplastic naevi)
- **Pathogenesis:**
  - Tumorigenesis of Melanocytes
  - Most develop in an existing Naevus
  - Superficial Spreading melanomas grow Laterally Initially, then begin to grow downwards
  - **Begin with a Radial Growth Phase (RGP)**
    - Grow wider (↑Diameter)
  - **Vertical growth Phase Melanoma (VGP)**
    - VGP will eventually supervene within most RGP melanomas and typically presents as a raised area that is progressively growing
- **Types of Melanoma:**
  - **"Superficial Spreading Malignant Melanomas"** - linked to intermittent intense sun exposure (sunburns)
    - Presents with a flat, usually pigmented, asymmetric macule that is changing is size, shape or colour
    - **Slow Growing**

https://www.nhs.uk/conditions/melanoma-skin-cancer/

  - **"Lentigo maligna melanoma"** – linked to high doses of cumulative sun exposure
    - presents as an unevenly pigmented, asymmetric facial freckle that is changing in size, shape or colour

https://www.nhs.uk/conditions/melanoma-skin-cancer/

- **"Nodular Melanoma" –** Due to Medium Sun Exposure
    - Tends to grow more rapidly in depth than other melanomas
    - The majority of deeply invasive, High-risk Melanomas

https://www.nhs.uk/conditions/melanoma-skin-cancer/

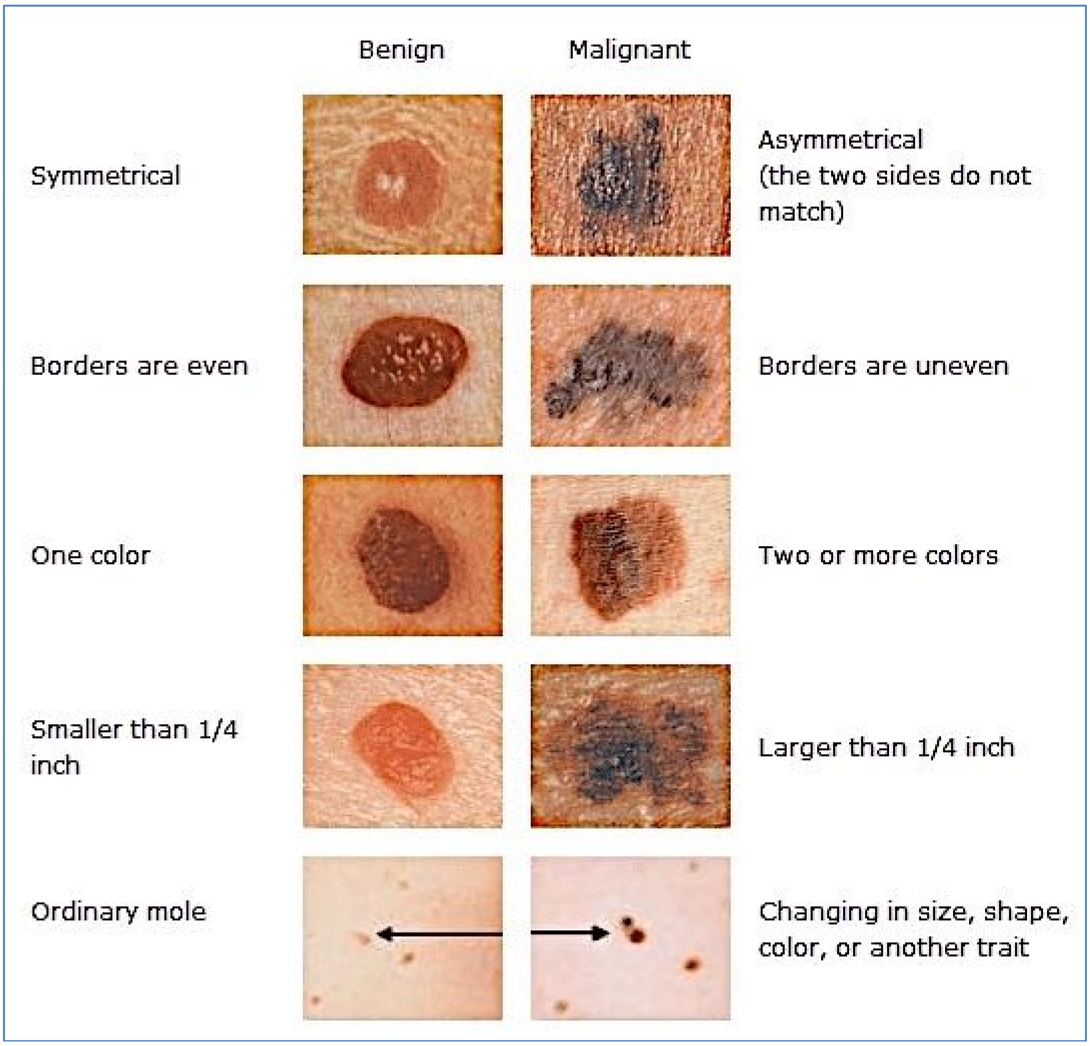
- **"Acral Lentiginous Melanoma"** – NOT due to sun exposure
    - Affects soles of the feet, palms of the hands, toes, and fingers, and occurs in the nail apparatus

https://www.nhs.uk/conditions/melanoma-skin-cancer/

- "**Amelanotic Melanoma**"- Red Patches on the Skin
    - Many melanomas are partially non-pigmented and up to 20% are almost completely without pigmentation. Amelanotic melanomas generally present as a red, changing lesion though they can be skin coloured

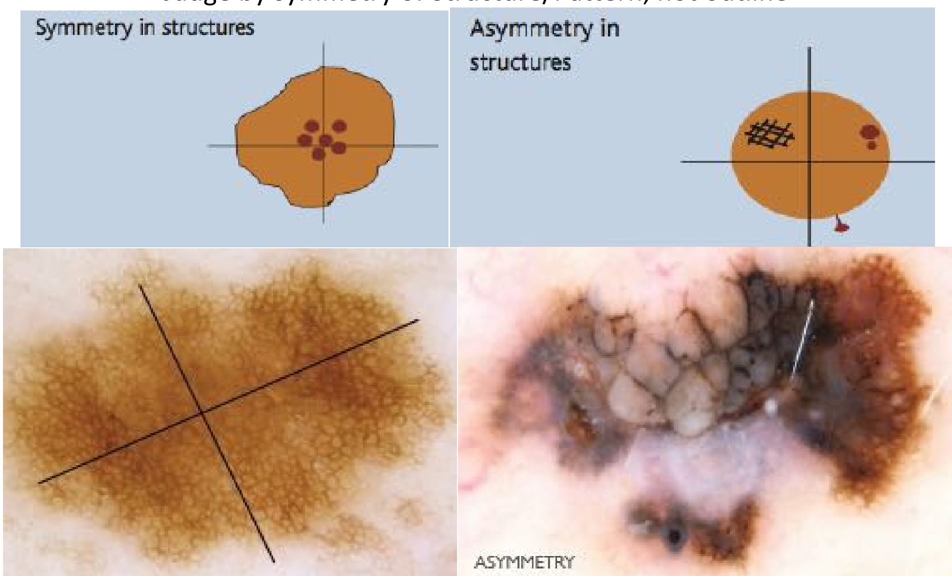
https://www.nhs.uk/conditions/melanoma-skin-cancer/

- **Morphology:**
  - **Gross:**
    - Grow & Change Colour
    - Maculo-Papular Lesion
    - ***Irregular Shape, Colour & Depth – Distinguishing Features***
    - Pigmented
  - **Microscopic:**
    - **Diagnostic Feature: Atypical (Dysplastic) Melanocytes *ABOVE The Epidermis***
    - **(**Also Invade down into the Dermis)
    - Nests/Clusters of Atypical Melanocytes
    - Inflammatory Cells
    - Mitotic Figures (Due to High Replication Rates)
- **Clinical Features:**
  - Malignant
  - Metastasize
- **Prognostic Factor:**
  - **Breslow Thickness**
  - **Clark Levels**
  - **TNM Staging:**
    - **T = Size/Depth**
    - **N = Nodal Involvement**
    - **M = Distant Metastasis?**
- **Diagnosis:**
  - **Dermoscopy is ESSENTIAL!!**
  - Skin exam without dermoscopy will only pick 60% of melanomas
  - **ABCD's of melanoma:**
    - **A – Asymmetry o**f Shape and Structure
      1. Ie: One half doesn't match the other half (border/colour/structure within the lesion)
    - **B – Border Irregularity**: The edges are ragged, blotched, or blurred
    - **C – Colour Variability**: The pigmentation is not uniform. Shades of tan, brown, and black may be present and red, white, grey and blue may add to the mottled appearance
    - **D – Diameter Increasing**: A width >6mm (about the size of a pencil eraser)
      1. Note: Any growth of a mole should be of concern
    - **E –**
      1. **Evolving** - any lesion that is changing or enlarging
      2. **Elevated** - Different elevations/contours *OR* ANY change from flat to elevated is of concern
    - **F – Firm**: A lesion that is firm, and also ones that are friable (easily damaged) are more likely to be malignant. Whilst benign moles are raised, soft and "wobble" like jelly with pressure
    - **G – Growing**: Signs of growth (Slow/Rapid) should raise concern

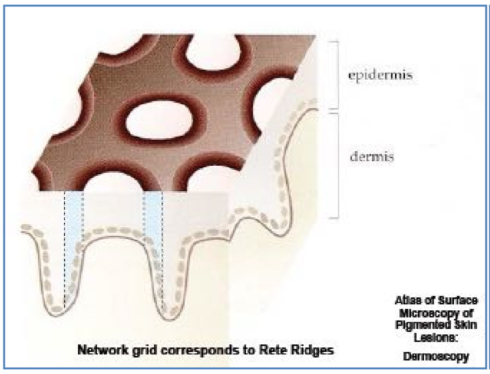


Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Images-showing-comparison-between-benign-and-malignant-skin-lesion-using-the-ABCD-rule_fig1_338262001

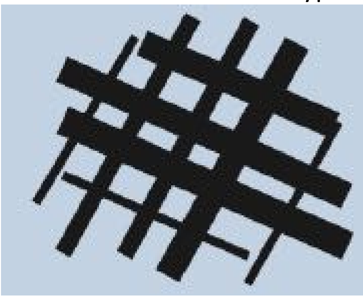
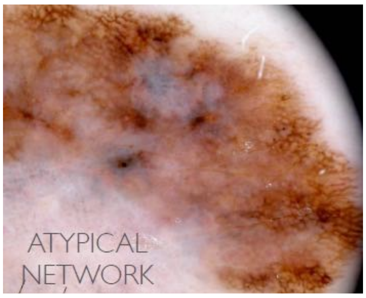
- **3 Point Checklist (>2 = Malignant):**
    - (A Screening tool for Malignancy)
    - High Sensitivity, But Low Specificity (Many benign lesions score higher than 1)
    - **1) Asymmetry of Colour and /or Structure**
        - Judge by symmetry of Structure/Pattern, not outline

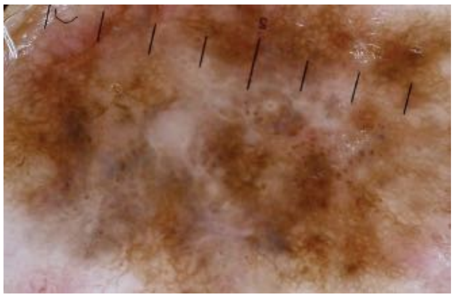


- **2) Atypical Pigment Network:**
    - Note: Typical Pigment Network Grids correspond to Rete Ridges in the Dermis

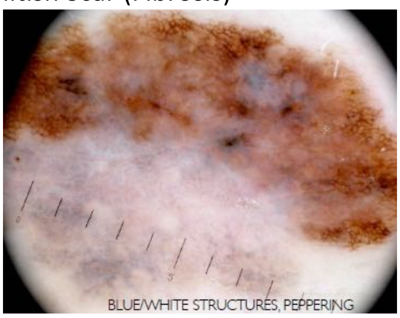


    - Atypical Networks = Lines are of different thickness & sizes of 'holes' are varied

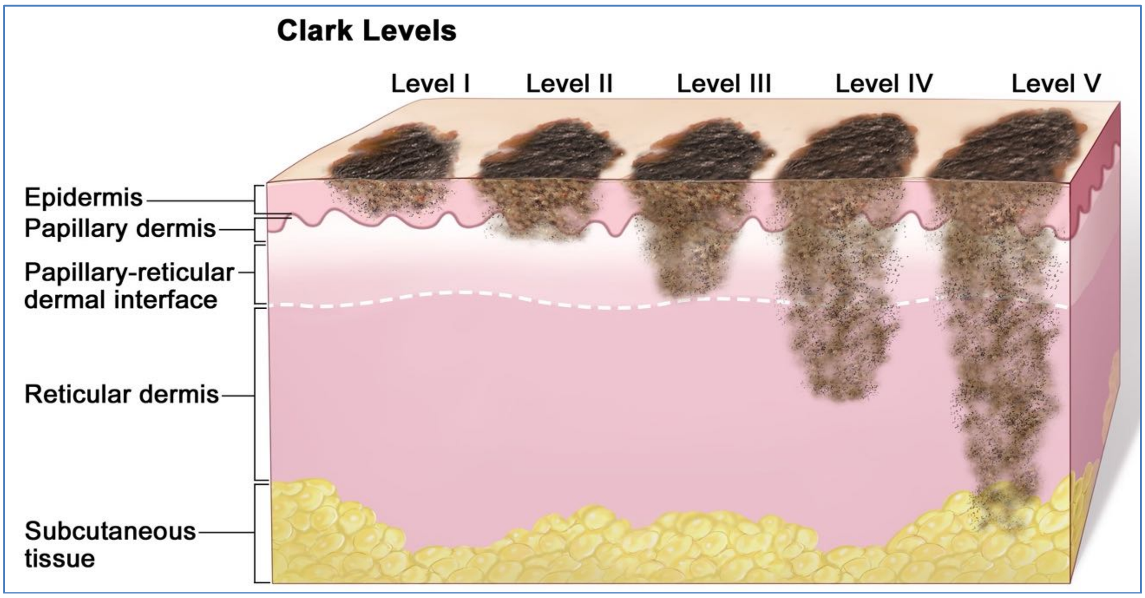


- **3) Blue-White Structures:**
    - Any White and/or Blue colour present in the lesions
    - Include:
        - 'Peppering' (Melanophages)
        - Blue-white Veil (Orthokeratotic Hyperkeratosis overlying the Pigmented Tumour)
        - Whitish Scar (Fibrosis)

- **Treatment:**
    - **Early Detection & Removal** = The Only Reliable Curative Treatment
    - **Follow-up** = Those who have had a melanoma, have a 13% risk of another

- **Prognostic Factor:**
    - **Breslow Thickness:**
        - Tumour thickness measured from the granular layer of the epidermis to the deepest identified tumour cell
        - Best Available predictor of Prognosis
    - **Poor Prognostic Factors:**
        - Ulceration
        - High Mitotic Rate
        - Signs of Regression = Bad
    - **Clark Levels:**
        - Relateds to the Deepest invasive tumour cells
        - Levels 1 – 5
        - (Not the same as TNM Staging)
    - **TNM Staging:**
        - **T = Size/Depth**
        - **N = Nodal Involvement**
        - **M = Distant Metastasis?**



https://www.cancer.gov/publications/dictionaries/cancer-terms/def/clark-level-iv-skin-cancer

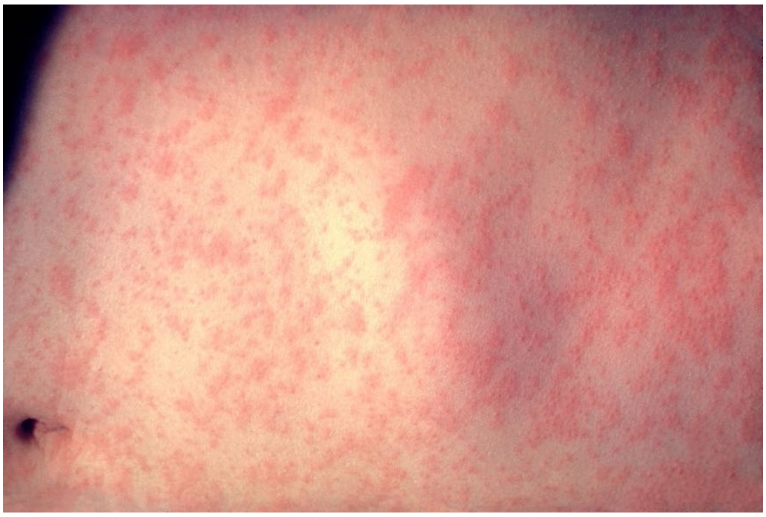
Note: Just realise that despite subtle changes in appearance, the *actual* size and infiltration of the Melanoma can be quite significant.

PEDIATRIC EXANTHEMS

PEDIATRIC EXANTHEMS

MEASLES VIRUS:

- **Organism:**
  - Morbillivirus
- **Transmission:**
  - Respiratory Route (Aerosol)
  - Contact with fluids from infected person's nose/mouth
- **Pathogenesis:**
  - Typically a Respiratory Infection
  - →Produces a Viremia → Rash
- **Presentation:**
  - Fever
  - URTI - Cough, Rhinorrhoea, Red Eyes
  - Maculopapular Erythematous (Morbilliform) Rash
  - "Koplik's Spots" – Seen on the Inside of the Mouth
- **Complications Include:**
  - Croup
  - Otitis Media
  - Enteritis with diarrhoea
  - Febrile convulsions
  - Encephalitis (Serious)
  - Subacute Sclerosing Panencephalitis (very rare)
    - (Chronic, progressive Encephalitis caused by persistent infection with immune-resistant Measles Virus)
    - No Cure
    - Fatal
- **Diagnosis:**
  - Clinical Diagnosis (Generalised Maculopapular Rash + Fever)
  - Presence of Measles IgM Antibodies
  - PCR of Respiratory Specimens
- **Treatment:**
  - No Specific Treatment
  - Prevented by MMR Vaccine
- **Prevention:**
  - Attenuated MMR Vaccine (Admin at 12mths & 4yrs)
  - Developing Countries: Low Herd Immunity → Higher Prevalence
    - Relatively High Death-Rates in Non-Immune



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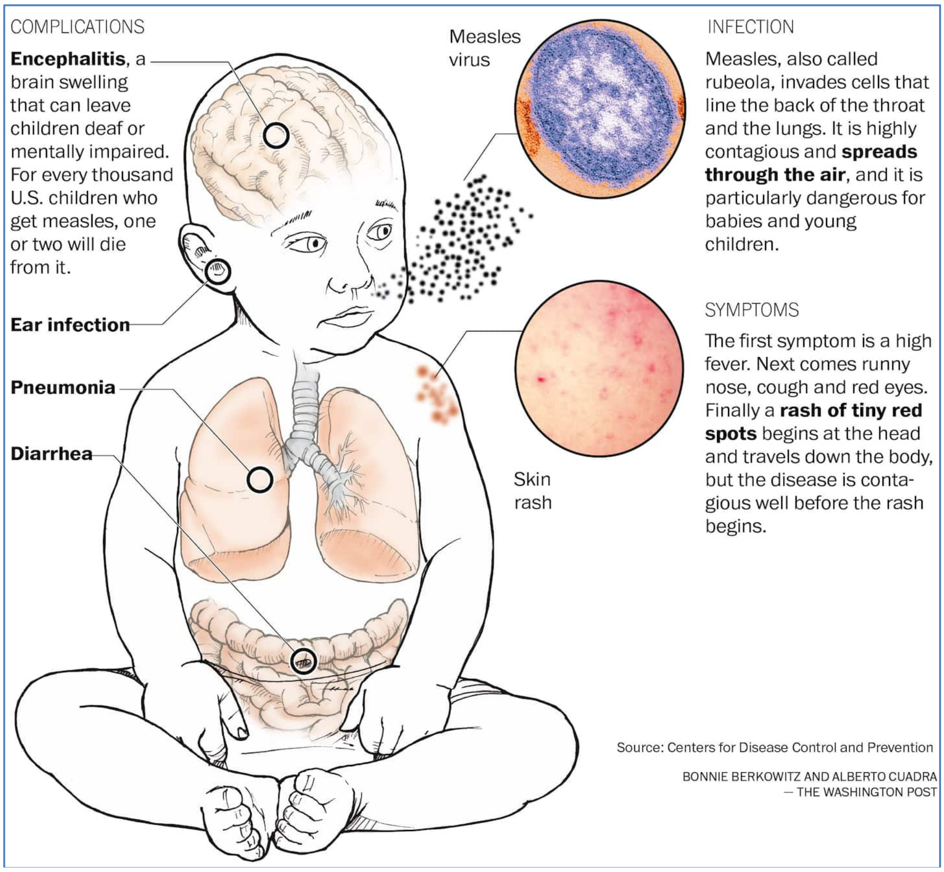

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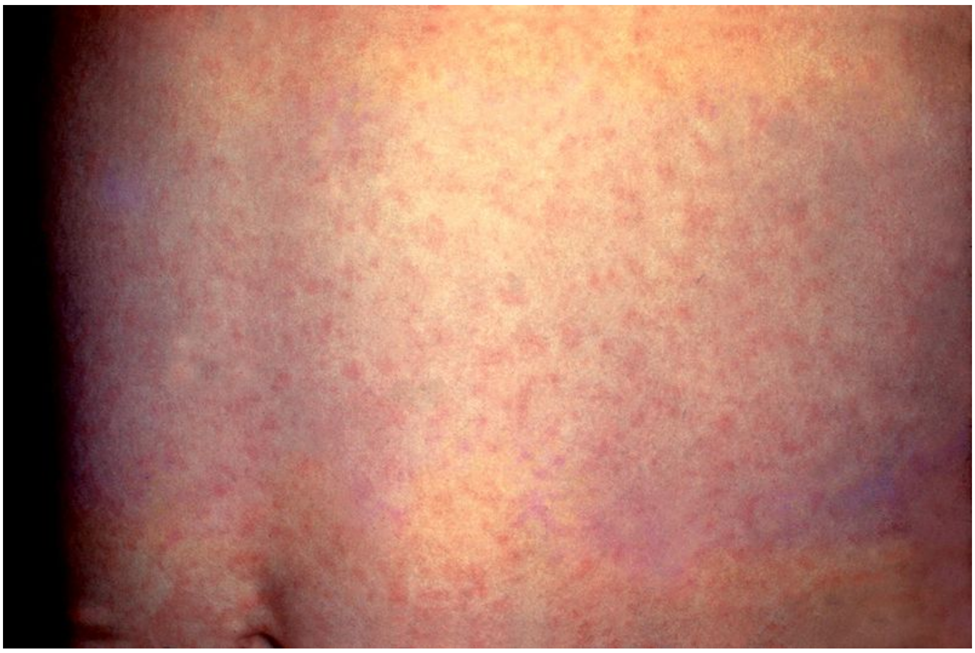
RUBELLA VIRUS ("GERMAN MEASLES):

- **Organism:**
    - Rubella Virus
- **Transmission:**
    - Respiratory Route
    - (Human Reservoir Only)
- **Presentation:**
    - Initial Flu-Like Symptoms
    - * Rash on Face → Spreads to Trunk & Limbs
        - Pink-Red, Itchy
    - Low-grade Fever, Lymphadenopathy, Joint Pains, Headache, Conjunctivitis
- **Prognosis:**
    - Typically Benign
    - Typically Lasts 1-3 Days (Children Recover Quicker)
    - Complications may include arthritis, thrombocytopenia purpura, and encephalitis
    - ***HOWEVER, Maternal Infection During PREGNANCY can be SERIOUS!!**
        - If Infected in the 1^{st} 20wks of Pregnancy → ***Congenital Rubella Syndrome***
            - →Abortion
            - →Cardiac/Cerebral/Ophthalmic/Auditory Defects
        - Specific Foetal Damage Depends on Organ Development @ the Time:
            - The 1^{st} Trimester is Worst, as Organ *Development* occurs during this time
            - After 1^{st} Trimester, Organ *Growth* is the main process
- **Diagnosis:**
    - Clinical Diagnosis
    - Presence of Virus-Specific IgM Antibodies
- **Treatment:**
    - No Specific Treatment
    - Controlled in Australia by vaccination (MMR Vaccine)
    - Test pregnant women for immunity early
- **Prevention:**
    - (Note: Rubella *Itself* is relatively Benign, so why bother Vaccinating?)
    - **MMR Vaccine:**
        - **(Live Attenuated)**
        - **#1 Aim:** Prevent Rubella in Pregnant Women → ↓Congenital Rubella Syndrome
        - Aimed at *BOTH* Males & Females to ↓Male Transmission to Pregnant Females



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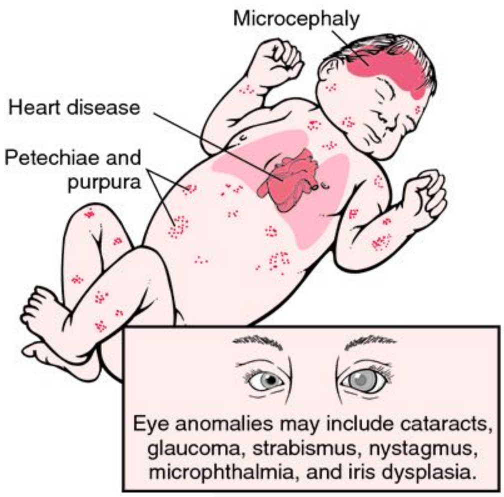
https://www.cdc.gov/rubella/about/photos.html



http://phil.cdc.gov/phil_images/20030724/28/PHIL_4284_lores.jpg

HUMAN PARVOVIRUS B19 ("5TH DISEASE")

- **Organism:**
  - Parvovirus B19
- **Transmission:**
  - Respiratory Droplet
  - Blood-Borne
- **Pathophysiology:**
  - **Virus Replicates in Rapidly-Dividing Cells (Eg: Bone Marrow RBC Precursors)**
    - → RBC Haemolysis
    - → Severe Anaemia
    - → Can Result in ***Haemolytic Crisis***
  - The receptor for the virus is a globoside, which is abundant on tissues of mesodermal origin
  - **Can cross the placenta into the foetus**
    - → Foetal Anaemia
- **Presentation:**
  - Fever/Malaise
  - Characteristic Rash
    - Teenagers: 'Papular Purpuric Gloves & Socks Syndrome'
    - Children: 'Slapped Cheek Syndrome'
- **Note: Foetal Infection →**Foetal Damage or Abortion

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SCARLET FEVER:

- **Organism:**
  - Certain strains of ***Strep pyogenes*** *(Which carry a Bacteriophage* – A virus infecting the bacteria → Produce an Eruthrogenic toxin)
- **Epidemiology:**
  - Mostly occurs in kids aged 4-8yrs
- **Pathogenesis:**
  - GAS infection of Tonsils/Pharynx/Skin
  - ***Exotoxin*** Released by Strep-Pyogenes → Local effect on Tonsils/Pharynx/Skin
  - →Abnormalities of tongue
    - Initially covered with white exudate
    - Exudate is shed
    - inflammation of underlying tissue
  - → Diffuse, Erythematous Exanthem
- **Treatment:**
  - **Antibiotics** (Usually penicillin for 10days; or single IM dose; or erythromycin if penicillin allergic)
  - Antipyretics
  - Fluids
  - Oral antihistamines to relieve the rash
- **Longer-Term Complications:**
  - Rheumatic fever
  - Glomerulonephritis


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SCALDED SKIN SYNDROME (SSS):

- **Organism:**
  - *Staph-Aureus* →Releases two Exotoxins (epidermolytic toxins A and B) from toxigenic strains
- **Pathogenesis:**
  - Toxins Degrade Desmosomes → Acantholysis
- **Presentation:**
  - Fever, Irritability, Rash
  - 24-48hrs later → Widespread Red Blistering Skin that looks like a burn or scald
  - occurs mostly in children younger than 5 years, particularly neonates
- **Diagnosis:**
  - **Tzanck smear**
  - Skin biopsy
- **Treatment:**
  - IV Antibiotics to eradicate staph infection (Eg: Flucloxacillin/Cephalosporin/clindamycin)
  - Vancomycin if MRSA suspected

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TOXIC SHOCK SYNDROME (TSS):

- **Organism:**
  - TSST-1-producing *Staphylococcus aureus*
  - or TSST-1-producing *Strep-pyogenes*
- **Pathogenesis:**
  - Gram Positive Organisms release Exotoxins (Superantigens)
  - → Mass, Non-Specific activation of T-Cells → Overproduction of Cytokines → Systemic Vasodilation → Shock
  - Cutaneous Response to toxins = Sunburn-like Rash + Desquamation
- **Presentation:**
  - Rapid Onset Fever
  - Generalised Skin/Mucosal Erythema
  - Shock → Hypotension & Multi-System Failure
  - Shedding of skin in large sheets (especially palms and soles) 1-2wks after onset of illness
- **Treatment:**
  - Prompt treatment is required to avoid failure of vital organs (Eg: Liver, lungs & heart)
  - Remove source of infection
  - IV Antibiotics (Flucloxacillin/cephalosporin/vancomycin)
  - IV Fluids & supportive management
  - Dialysis may be needed

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PROCEDURAL DERMATOLOGY

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Local Anaesthetic:

- **Mechanism of Action:**
  - Use-Dependent blockade of Voltage-Gated Sodium Channels on Nerves
    - → Prevents Action Potential Conduction along Sensory Nerves
  - Note: Very Quick Onset of Action
  - Note: Also causes vasodilation → Short Duration of Action (If without adrenaline)
- **Lignocaine – Most Common:**
  - Rapid Onset (1-5mins)
  - Medium (30-120mins)
  - **Max Dose:** 4mg/kg
    - ≈ 50mL @ 1% Lignocaine
  - (Note: 1% Lignocaine = 1g/1000mL)
- **+ Adrenaline:**
  - → Improves Haemostasis
    - → ↓Bleeding
  - → ↓Systemic Absorption
    - → ↓Risk of systemic toxicity
    - → Prolonged Effect
  - **Max Dose:** 7mg/kg (Higher Dose than without adrenaline)
  - **Note: DO NOT USE IN DIGITS OR PENIS (Or anywhere else with "End Arteries")**
  - **Note: DO NOT USE in LONG QT-Syndrome**
- **Naropin (Ropivicaine):**
  - Longer onset of action
  - Longer acting than lignocaine
  - **Max Dose:** 2-3mg/kg
- **Topical Anaesthetic:**
  - **Xylocaine Gel -** Useful for Mucosal Surfaces (Eg: Oral Mucosa)
  - **ELMA Cream –** Useful for topical anaesthesia of skin
- **Nerve Blocks:**
  - **Digital Block (Ring Block)**
    - (Most Common)
    - Blocks Digital Nerves
    - Use 2% Lignocaine (Because you want to inject as little as possible)
  - **Wrist Blocks:**
    - Blocks Radial, Median, & Ulnar Nerves
    - → Totally anaesthetizes the hand

Curettage & Electrocautery:

- **Indications:**
  - Warts
  - Keratoses
  - Molluscum contagiosum
  - Curettes can also be used for curette biopsy
- **Wound Healing**
  - The curette wound can take some time to heal
  - If it is just a light curette such as tiny seborrheic keratoses with no cautery they can heal in a few days
  - A full curettage and electrodessication on the trunk might take 4 to 6 weeks to heal

Cryotherapy:

- Cryotherapy is a very useful form of treatment for a number of benign skin lesions and few pre-malignant and malignant skin lesions
- Liquid Nitrogen = The gold standard
- **Indications:**
    - **Benign lesions:**
        - Molluscum contagiosum
        - Seborrheic keratosis
        - Skin tags
        - Warts
    - **Premalignant lesions:**
        - Actinic keratoses
        - Actinic Chelitis
    - **Malignant lesions:**
        - Superficial BCC
- **Effects of Cryotherapy – 3 Main Groups:**
    - **1) Those responses we expect to happen:**
        - Pain on treatment and for a period afterwards
        - Oedema and swelling of the treated site and the surrounding tissue (Eg: periorbital swelling after treatment of lesions on the forehead)
        - Vesicle and bulla formation
        - Exudation weeping and crust formation
    - **2) Temporary Adverse Outcomes:**
        - Hypopigmentation
        - Hyperpigmentation
        - Secondary Infection
    - **3) Permanent Adverse Outcomes:**
        - Permanent Hypopigmentation
        - Scarring + Possible Retraction
        - Allopecia
        - Nail Dystrophy

Biopsy Techniques:

- Many different Techniques
- Technique depends on the Nature of the Lesion & Information Required
- **Types:**
    - **Punch biopsy**
        - Leaves Minimal Scarring
        - Best for Cosmetically Sensitive Areas – (Eg: Face)
    - **Shave Biopsy**
        - Fast and easy to perform
        - Requires little equipment
        - A shave biopsy is excellent for nodular bulky lesions which are easy to shave off for histology
    - **Curette biopsy**
        - Fast and easy to perform
        - Requires little equipment
        - A shave biopsy is excellent for nodular bulky lesions which are easy to shave off for histology
    - **Incision Biopsy**
        - A biopsy set is required with scalpel, forceps, needle holders and fine scissors as well as a suture to perform a small incisional ellipse. Be sure to go through the dermis to get a full thickness specimen
        - Advantages:
        - Provides the best specimen for the pathologist to assess the tissue adequately
    - **Excision biopsy**
        - Similar to an incisional biopsy but the whole lesion is excised
        - Good pathology specimen

Wound Design:

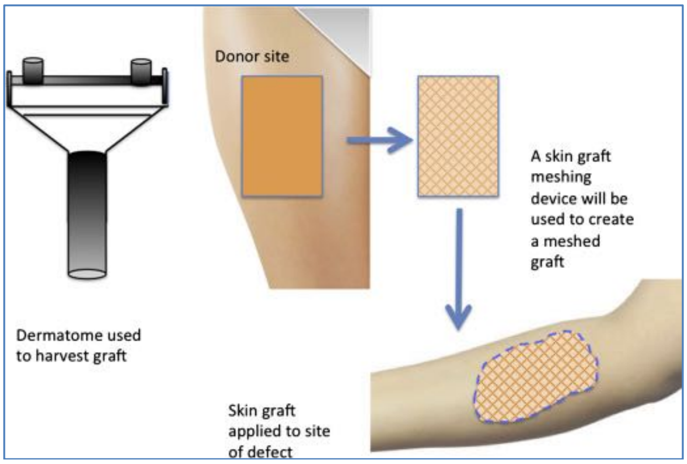
- Wounds should be designed so that scars sit in or parallel with the relaxed skin tension lines or the cosmetic junction lines
- Excision margins are important to ensure complete removal of the lesion. For benign lesions the margins are usually 2mm to 5 mm. For malignant lesions the margins can vary from 3mm up to 2 cm depending on the nature of the pathology

Surgical Wound Repair:

- **Suture Materials:**
  - **Synthetic Non-Absorbable:**
    - Nylon
    - Polypropylene
    - (Used on external surfaces where they are easily removed)
  - **Synthetic Absorbable:**
    - Monocryl
    - Vicryl (Polylactin)
    - (Broken down by the body via Hydrolysis &/or Proteolytic Enzymatic Degradation)
  - **Non-Synthetic Non-Absorbable:**
    - Silk
  - **Non-Synthetic Absorbable:**
    - Catgut (Being Phased out)
- **Suture Sizes:**
  - Scale from 1-6 (1 = Thinnest (0.4mm); 2 = Thickest (0.8mm)
- **Surgical Wound Repair:**
  - **Primary Repair:**
    - 6-8hrs
  - **Delayed Primary Repair**
    - In 72hrs
    - In surgical conditions
  - **Secondary Repair:**
    - After 72hrs
    - In surgical conditions
- **Important Pre-Requisites:**
  - **Wound needs to be CLEAN!**
    - Eliminate Dead Tissue/Dead Space/Haematoma/Foreign Bodies
  - **Finest Suture Material**
  - **Minimise Tension**
  - **Suture Removal @ Appropriate time for site of injury**
    - 1-2 wks

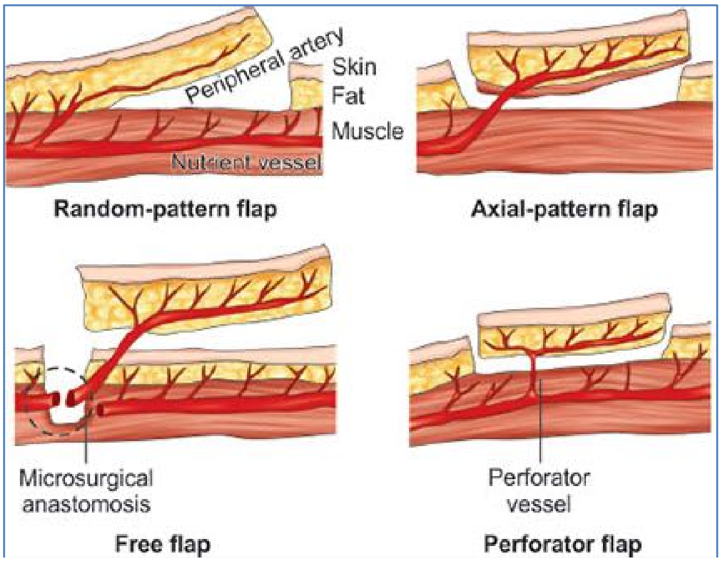
Skin Grafts Vs Skin Flaps:

- **Skin Grafts:**
  - Has been totally dethatched from its original location
  - **:. Has NO intrinsic Blood Supply**
    - :. Must be grafted onto Vascularised Tissue
    - It will initially survive via Simple Diffusion of Nutrients
    - Eventually, Neovascularisation → New Blood Supply
  - **Types:**
    - **Full Thickness Graft (FTG)**
      - All of the Dermis & Epidermis
    - **or Split Skin Graft (SSG)**
      - All of the Epidermis & some of the Superficial Dermis



Unattributable

- **Skin Flaps:**
  - Has NOT been detached from its original location
  - :. Still Has Intrinsic Blood Supply
  - **Indications:**
    - Poor Vascularity of Location (Eg: Bare bone, Bare tendons)
    - Vital Structures (Eg: Exposed Vessels/joints)
    - Cosmetics (Eg: Face) – Usually gives a better result than a skin graft
  - **3 Basic Types:**
    - Rotation
    - Advancement
    - Transposition
- **"Free" Flaps:**
  - = Basically a Skin *Graft*, but the Blood Supply is Reconstituted using Microsurgery to reconnect the Artery and Vein



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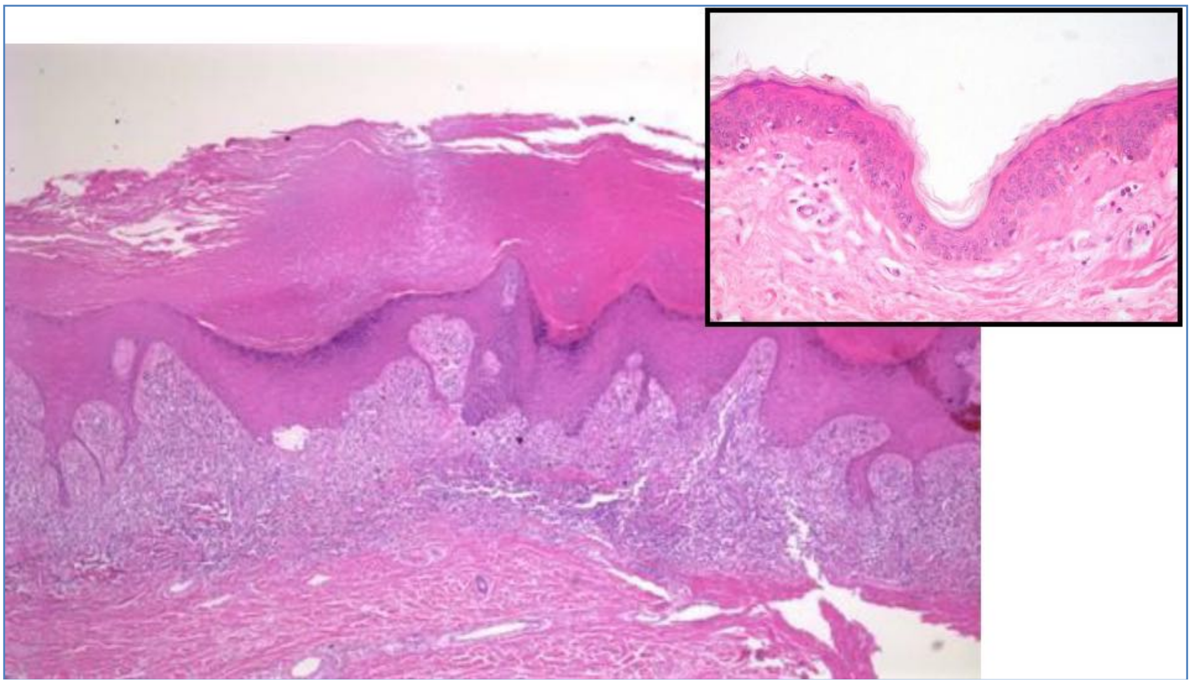
SKIN PATHOLOGY – UNDER A MICROSCOPE

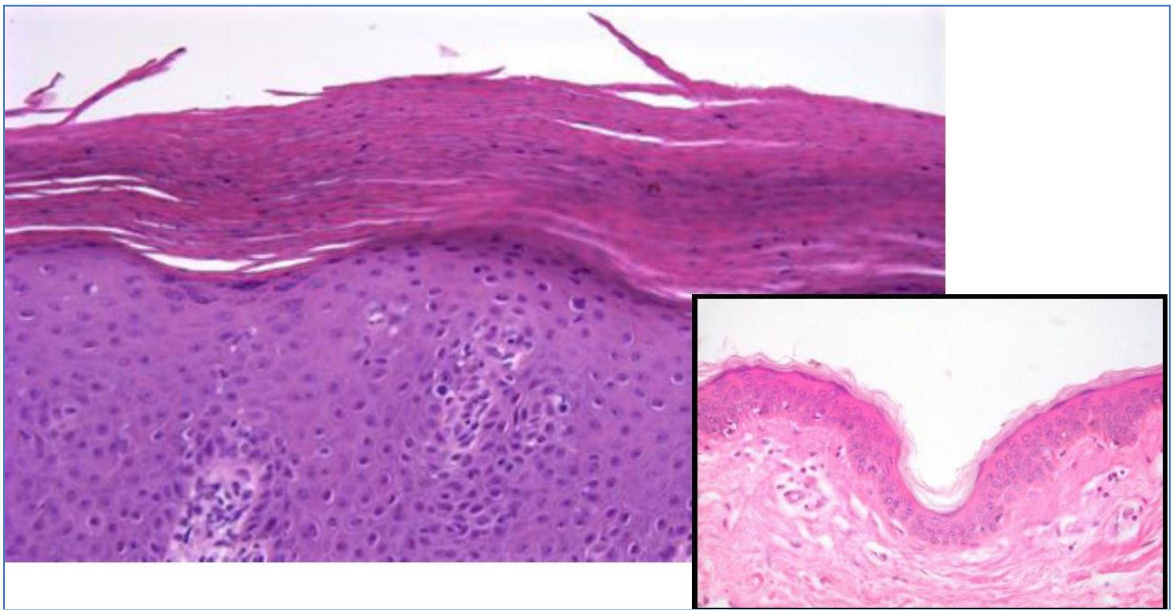
SKIN PATHOLOGY – UNDER A MICROSCOPE

Common Histopathological Terms:

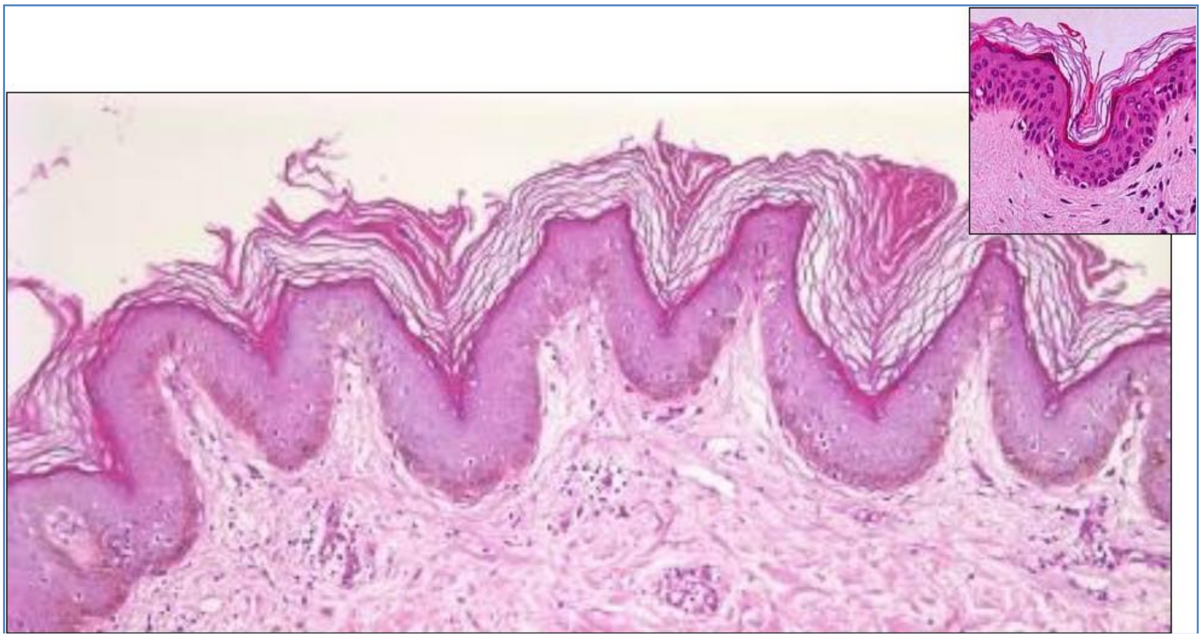
- **Hyperkeratosis:**
  - Hyperplasia of the Stratum Corneum
  - Note: Also Thickening of the Dermis & Rete ridges
  - Clinical appearance = scaling
    - Psoriasis, eczema



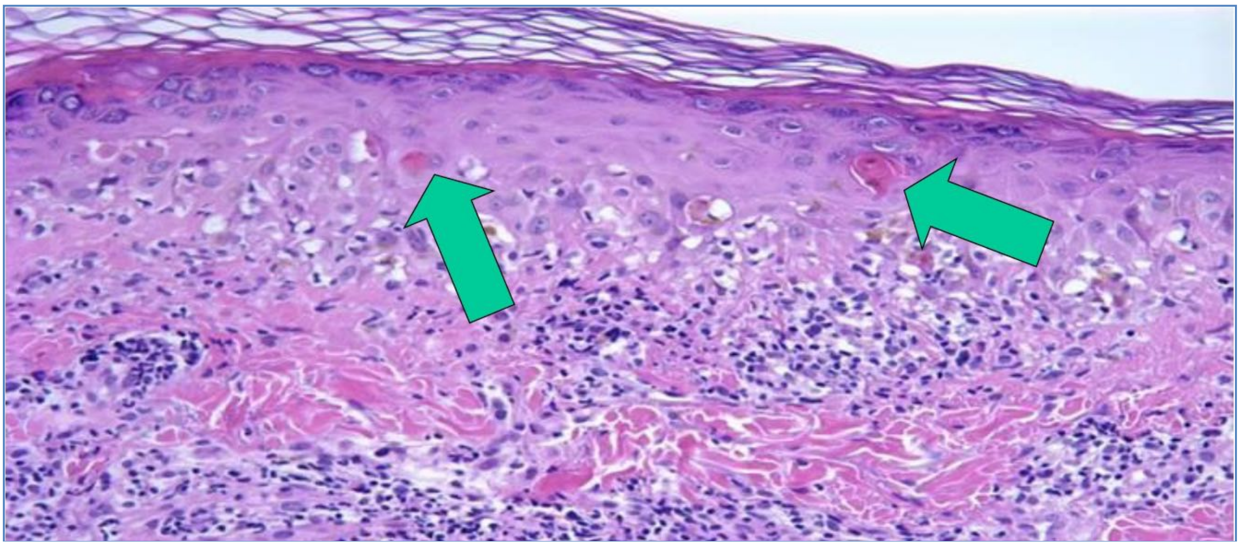
- **Parakeratosis:**
  - A Pattern of Keratinization Characterized by **Retention of Nuclei** in the stratum corneum (due to very very rapid keratinisation –no chance to mature & lose the nucleus)
    - (Note: This is normal on mucous membranes)
  - Seen in many cases of hyperkeratosis
  - Clinical appearance = Scaling



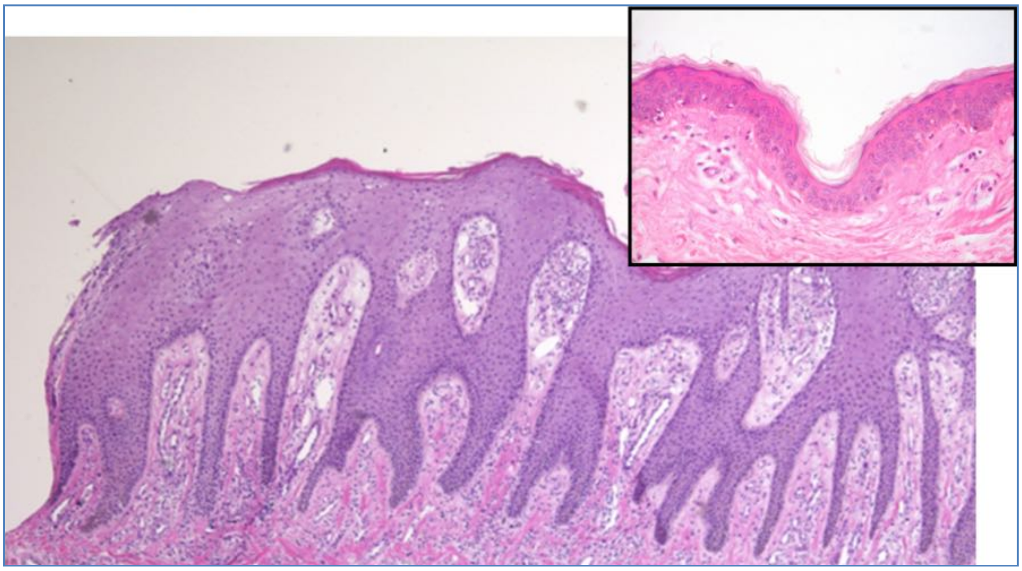
- **Papillomatosis:**
  - Hyperplasia of the Papillary Dermis
  - (The whole skin becomes folded – Papillary Projections – Typical in infections)
  - Caused by Neoplastic-verrucous ca, venous stasis, viral



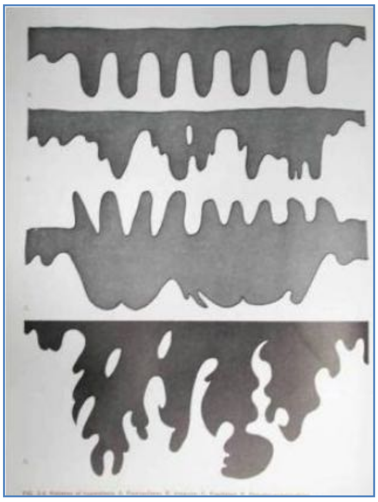
- **Dyskeratosis:**
  - Abnormal Keratinization due to Carcinoma
    - Very typical of malignancy
  - Cells containing high amounts of keratin beneath the Stratum Corneum
    - → Keratin Cysts
  - Note: Also many inflammatory cells



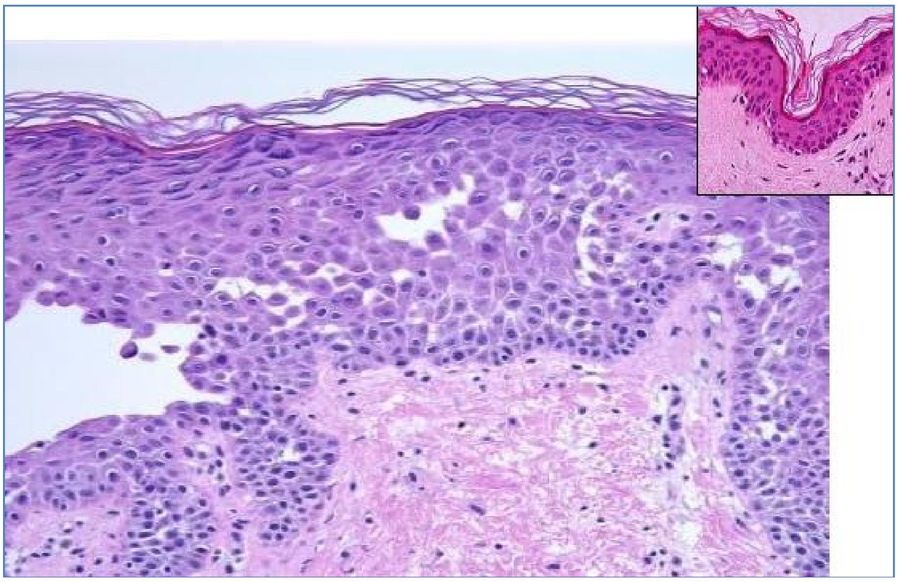
- <u>**Acanthosis:**</u>
    - Epidermal Hyperplasia (Increased number of epidermal cells, deepening of rete ridges) → Thickening of the Epidermis
    - Seen in: Eczema, psoriasis



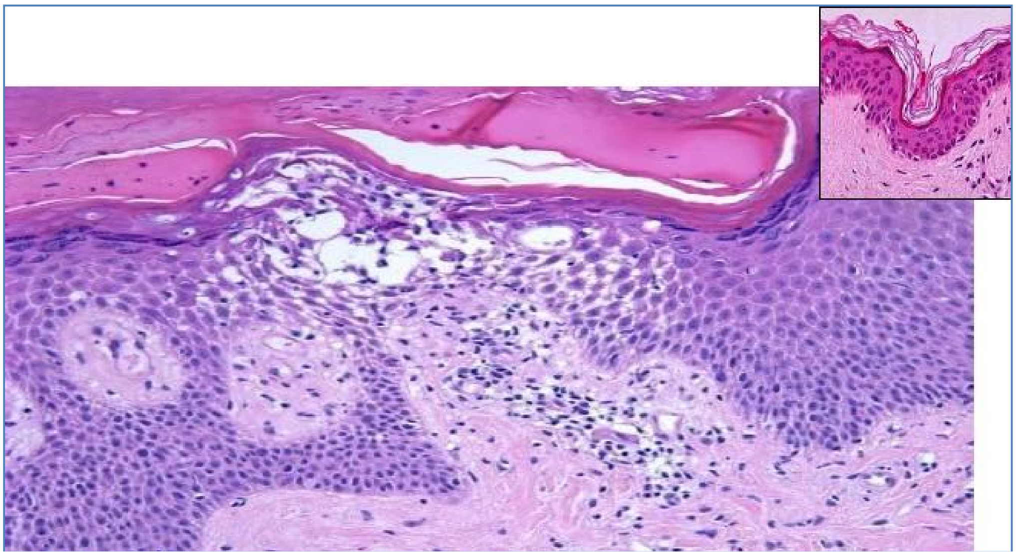
    - **Types of Acanthosis:**
        - **Psoriasiform (Regular) Acanthosis**
        - **Irregular Acanthosis**
        - **Papillated Acanthosis** (Epidermal Growth goes above the normal level)
        - **Pseudoepitheliomatous/pseudocarcinomatous** (Epidermal Growth Infiltrates into the dermis)



- **Acantholysis:**
  - Loss of Intercellular Connections
    - Antibody-Mediated Destruction Against Desmosomes
    - → "Breaking down of brick wall" → Blisters
  - **Seen in:** Pemphigus (a group of Blistering Autoimmune Diseases)



- **Spongiosis:**
  - Inflammation and Intercellular Epidermal Oedema
  - →Separation of cells without breakdown of cell-junctions (Not like Acantholysis)
  - **Aetiology:** The initial response to any damage to the epidermis (scratching/irritants/Etc)
  - **Seen in:** Eczema, Pemphigus, Seborrheic dermatitis



EXAMPLES OF EXAM QUESTIONS:

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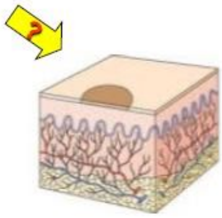
1. **Lesions following viral fever & sore throat. Lesions have a central clearing (Target-eyed Lesions), are red. Diagnosis?**
    a. **Erythema Multiforme**
2. **Itchy, lesion on wrist for 3 days following wearing new bracelet. Microscopy shows spongiosis (intraepidermal oedema) Diagnosis?**
    a. **Acute Eczema/Contact Dermatitis (Due to allergy against material in bracelet)**
    b. **(Not Urticaria – would present within hours, and doesn't have spongiosis)**
3. **84yr old, scalp lesions. (It is Actinic Keratosis) What is the diagnostic feature you would see under the microscope?**
    a. **Solar Elastosis (Very specific to solar damage)**
    b. **Also have Parakeratosis (but is also seen in other conditions)**
4. **41 yr old male, generalised very itchy rash, swollen lips. Diagnosis?**
    a. **Urticaria**
5. **What is the characteristic microscopic feature of Urticaria?**
    a. **Dermal Oedema (As opposed to Spongiosis, which occurs after 3-4 days) due to mast cell degranulation**
6. **Rash 6 days after giving blood @ the site of the Band-Aid. Diagnosis?**
    a. **Contact Dermatitis**
7. **What is the microscopic characteristics of Acanthosis:**
    a. diffuse epidermal hyperplasia.[1] Acanthosis implies increased thickness of stratum spinosum
8. **82 yr male, lesions on hand and ear. Microscopy shows irregularity of cells, and the lesion is shiny, erythromatous plaques. Diagnosis?**
    a. **Squamous Cell carcinoma**
9. **48yr female, large flaccid blisters. Some broken with crusting. Diagnosis?**
    a. **Pemphigus (As opposed to Bullous Pemphigoid)**
10. **28yr male, asymptomatic lesions for 3 weeks on chest & back. Diagnosis?**
    a. **Tinea Versicolor**
11. **81yr male, surfing enthusiast, lesions on cheek or 3 years. Slowly increasing. Diagnosis?**
    a. **Actinic Keratosis (Sun damage)**
12. **21yr male, lesions on face since 4 weeks. Slowly increasing. Itchy, but no pain. Diagnosis?**
    a. **Impetigo (Staphylococcal)**
13. **71yr female, recurrent flaccid blisters. Diagnosis?**
    a. **Pemphigus Vulgaris**
14. **Asymptomatic mole. Growing. Diagnosis?**
    a. **Melanoma**
15. **3yrs lip licking → Crusting. Diagnosis?**
    a. **Impetigo**
16. **13 yr girl. Allopecia, Erythema. Diagnosis?**
    a. **Tinea Capitis Infection**
17. **83yr nodular lesion on face for 2 years. Diagnosis?**
    a. **BCC**
18. **Recurrent small itchy vesicles. Lesions exacerbate following eating bread.**
    a. **Dermatitis Herpetiformis**

What is the type of lesion?

1. Macule
2. Papule
3. Plaque
4. Nodule
5. Pustule
6. Wheal (hive)
7. Scales
8. Crust
9. Erosion/Ulcer



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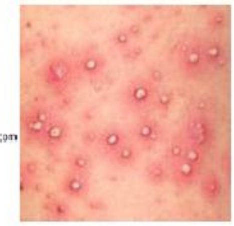

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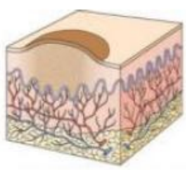

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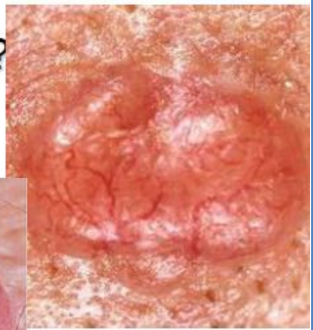
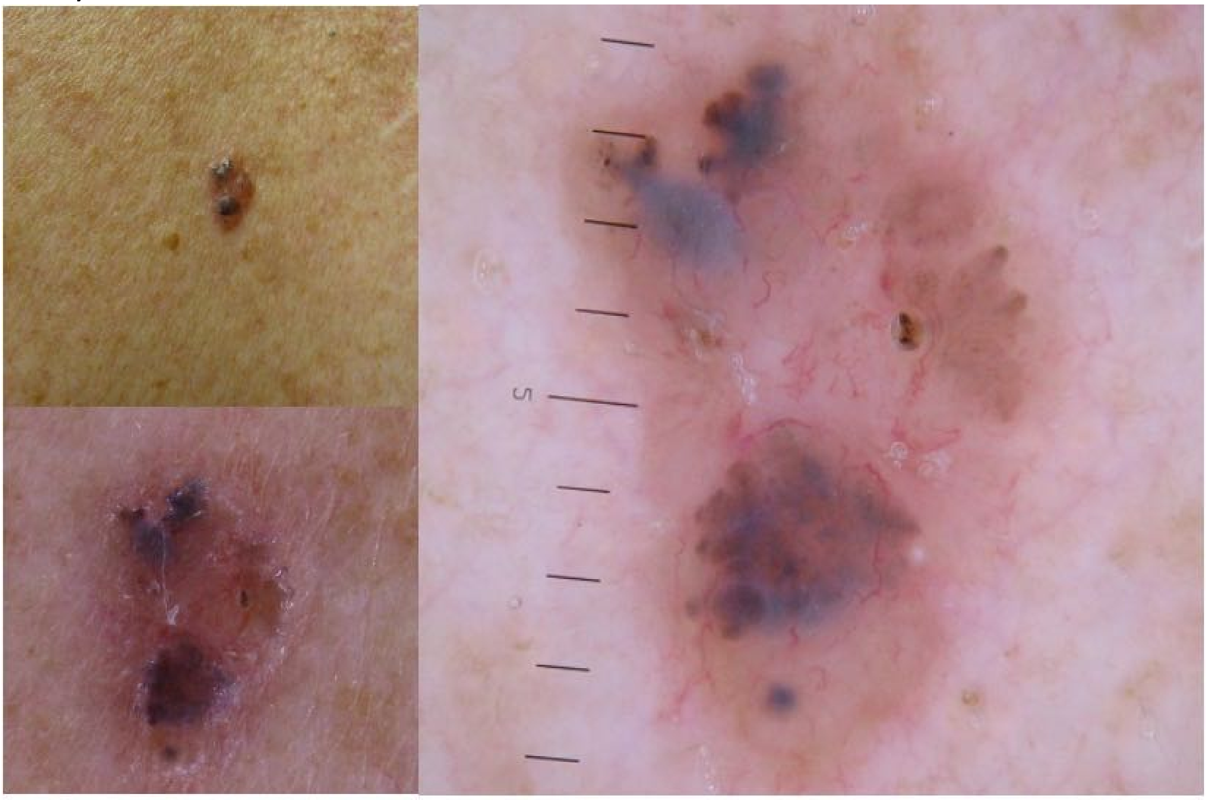


What is the type of lesion?

1. Macule
2. Papule
3. Plaque (Large areas of raised lesions)
4. Nodule
5. Pustule
6. Wheal (hive)
7. Scales
8. Crust
9. Erosion/Ulcer

Case 1)



CASE 1. Male 45 lesion over scapular, patient unaware of it, palpable but not firm or hard.

Clinical Description:

- Small, brown papule
- 9mm diameter
- Irregular shape

ABCD (EFG) Criteria:

- Asymmetrical
- Border Irregularity
- Colour Variability
- Diameter increasing unknown
- Don’t know if Evolving
- Elevated

Three Point Checklist Score (Malignant):

- Asymmetry?
  - Yes
- Atypical Network?
  - No
- Blue-white Structures?
  - Yes

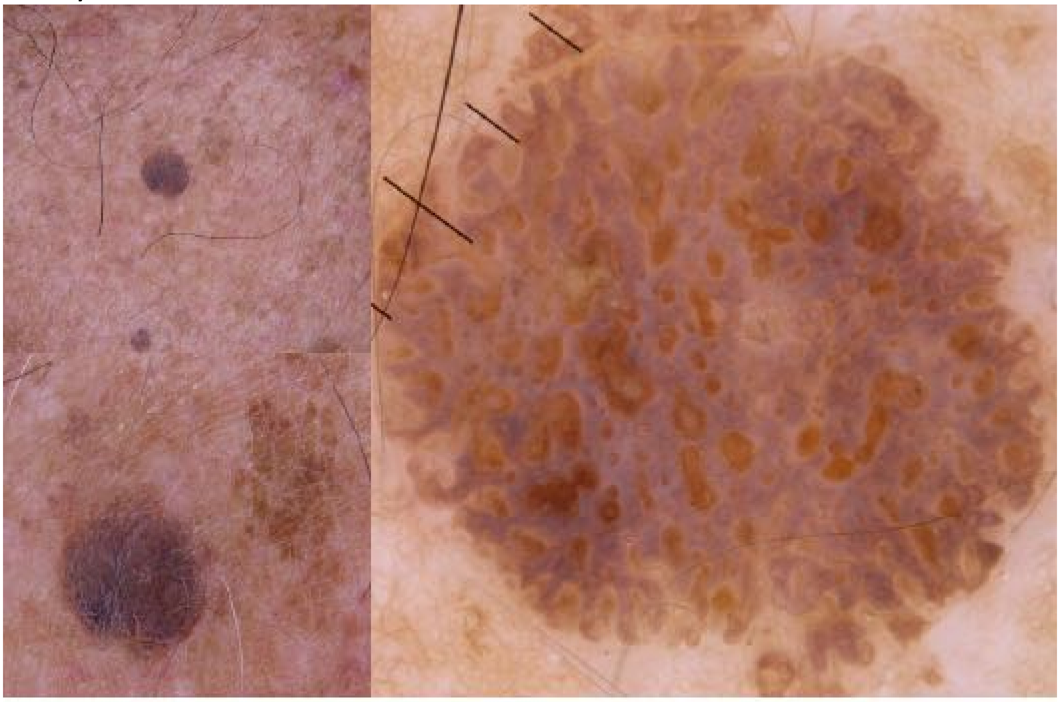
Provisional Diagnosis:

- Potentially malignant BCC

Management:

- Excision

Case 2)



CASE 2. Male 57 palpable plaque on back.

Clinical Description:

- 5mm diameter
- Papular-Plaque
- Brown
- Well defined border
- Regular shape & Pigment distribution
- Has ridges & fissures/como-like openings

ABCD (EFG) Criteria:

- Symmetrical
- Regular borders
- Regular Colour
- Diameter not increasing
- Not evolving
- Elevated
- Not growing

Three Point Checklist Score:

- Asymmetry?
  - No
- Atypical Network?
  - No
- Blue-White Structures?
  - No

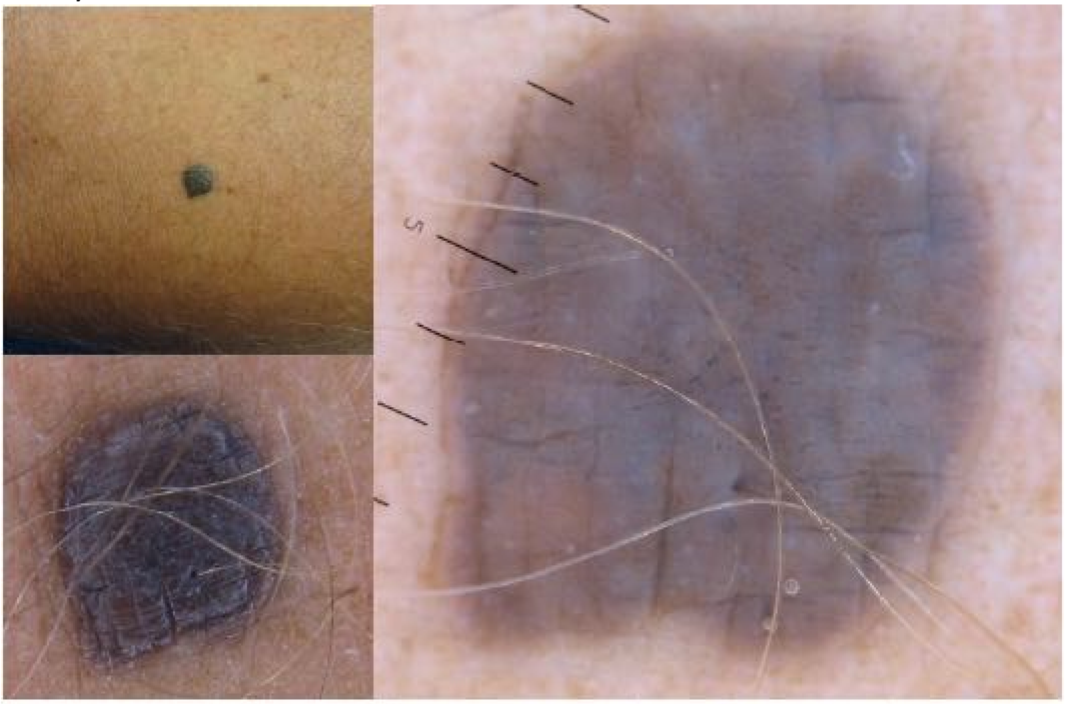
Provisional Diagnosis:

- Benign Seborrheic Keratosis

Management:

- Monitor
- No management needed

Case 3)



CASE 3. Male 32 years long standing lesion on medial calf no history of change, palpable plaque soft.

Clinical Description:

- 1cm wide plaque
- Well defined border
- Regular shape
- Regular pigmentation

ABCD (EFG) Criteria:

- Symmetrical
- Well defined border
- Regular Colour
- No history of change (diameter/evolution)

Three Point Checklist Score:

- Asymmetry?
    - No
- Atypical Network?
    - No
- Blue-Grey Structures?
    - Yes

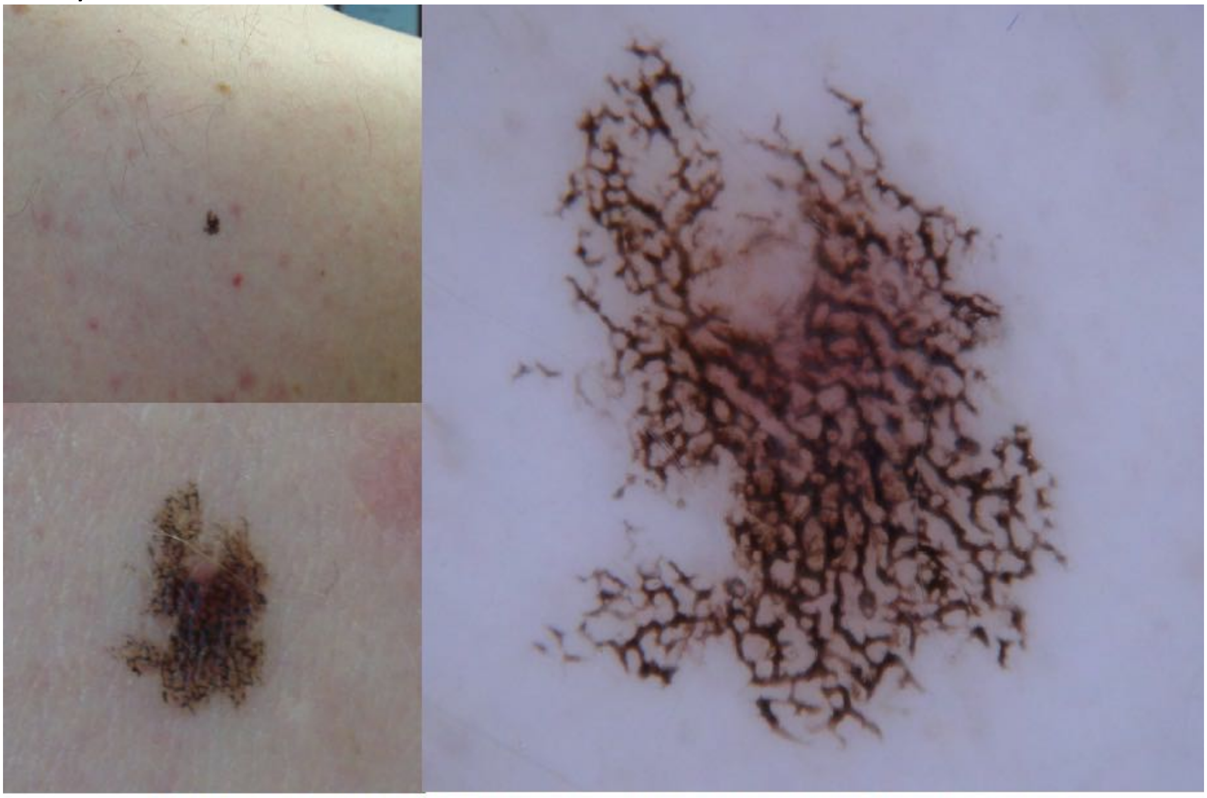
Provisional Diagnosis:

- Benign
- Blue Naevus

Management:

- Monitor
- No Management Needed

Case 4)



CASE 4. Male 18, Fitzpatrick type 1 skin, one of several similar maclure lesion on upper trunk and proximal upper limbs

Clinical Description:

- Small pigmented lesion
- Macule (not raised)
- Irregular border

ABCD (EFG) Criteria:

- Symmetrical pattern, asymmetrical shape
- Irregular Border
- Consistent Colour
- No history of increasing Diameter/Evolution

Three Point Checklist Score:

- Asymmetry?
  - No
- Irregular Network?
  - No
- Blue-Grey Structures?
  - No

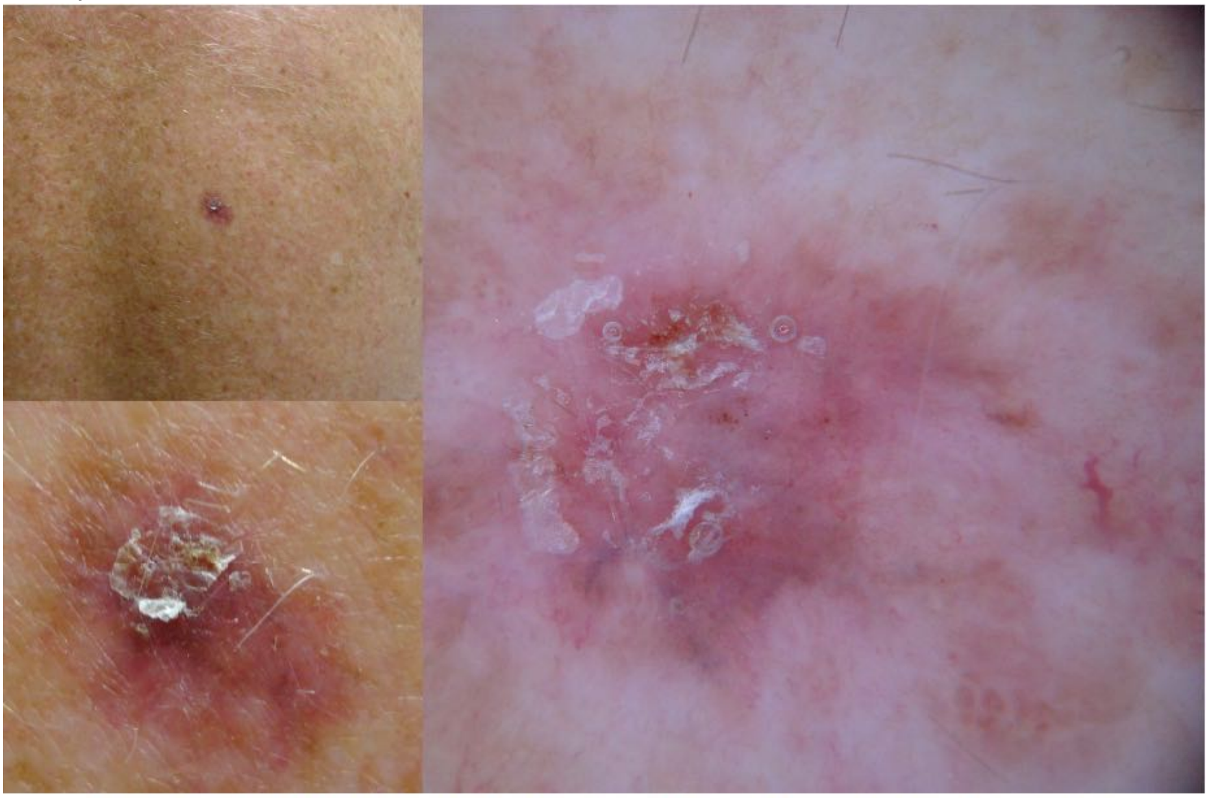
Provisional Diagnosis:

- Benign
- Ink-Spot Lentigo

Management:

- No management needed

Case 5)



CASE 5. Male 56 years, enlarging lesion on back noted by partner 6 months previously, recently itchy, macule with surface scale.

Clinical Description:

- Scaly raised, erythematous Macule
- Poorly defined border
- Erythematous base
- scaling

ABCD (EFG) Criteria:

- Asymmetrical
- Irregular border
- Consistent Colour
- Increasing Diameter
- Evolving

Three Point Checklist Score:

- Asymmetry?
  - Yes
- Irregular Network?
  - No
- Blue-Grey Structures?
  - Yes

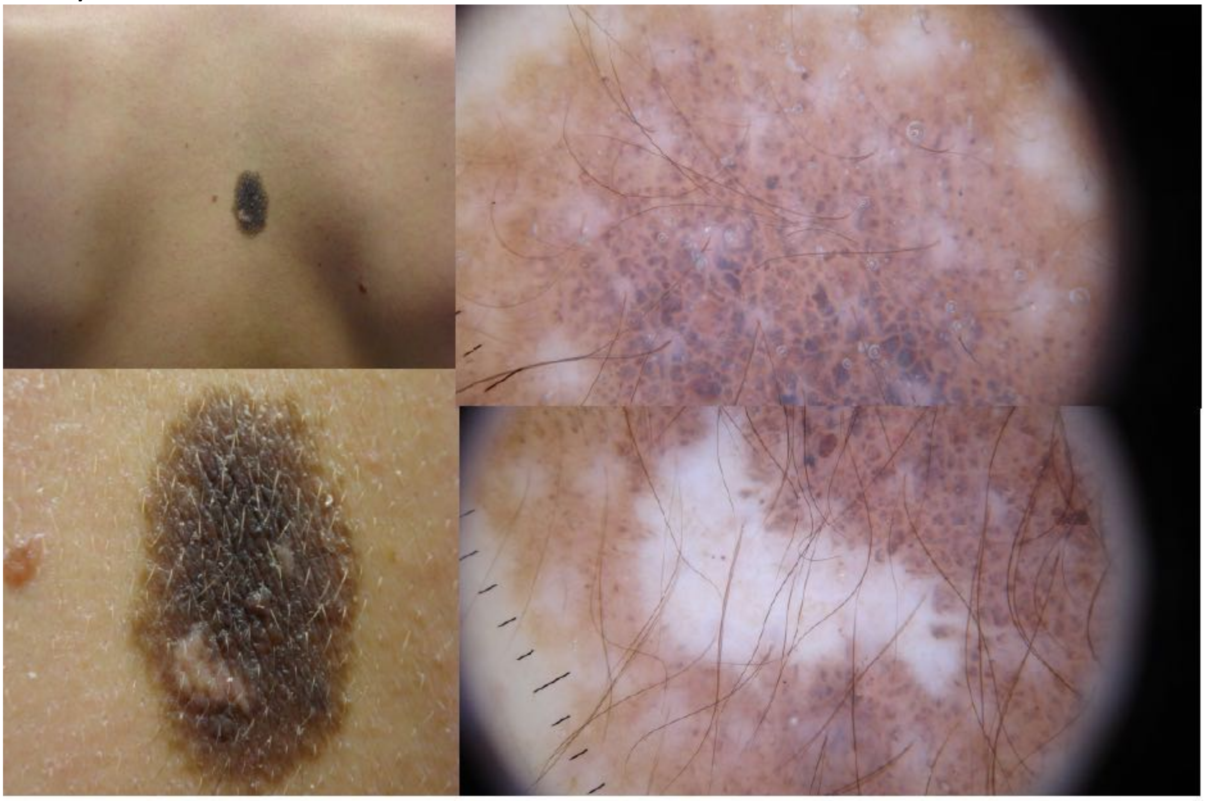
Provisional Diagnosis:

- Malignant
- Amelanotic Melanomas

Management:

- Excision

Case 6)



Case 6. Male 36 lesion present since infancy no history of change, plaque with warty sufrace

Clinical Description:

- Large pigmented plaque
- Warty surface
- No history of change
- Well defined border

ABCD (EFG) Criteria:

- Symmetrical in shape but asymmetrical in structure
- Well defined Border
- Consistent Colour except for area of white regression
- Constant diameter
- No history of evolution

Three Point Checklist Score

- Asymmetry?
  - No
- Irregular Network?
  - No
- Blue-White Structures?
  - Yes

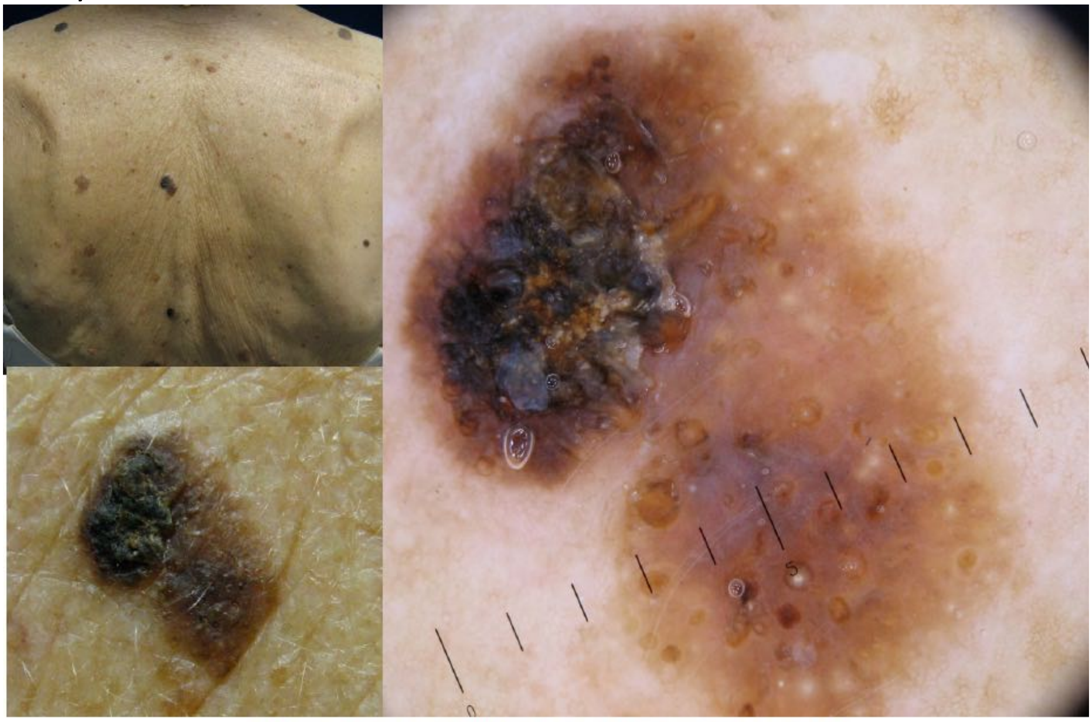
Provisional Diagnosis:

- Benign
- Birth Mark (congenital Naevus)

Management:

- None needed

Case 7)



CASE 7. Female 75 one of multiple similar lesion on trunk, plaque with stiff surface

Clinical Description:

- Asymmetrical macula-papular lesion
- Crusted surface
- Large
- Pigmented
- Milia like cysts, comedo-like openings

ABCD (EFG) Criteria

- Asymmetrical
- Well-Defined Borders
- Regular Colour
- Diameter ≈ 1cm

Three Point Checklist Score

- Asymmetry?
    - Yes
- Irregular Network?
    - No
- Blue White Structures?
    - Yes

Provisional Diagnosis:

- Seborrheic Keratosis
- Benign

Management

- None Needed

Case 8)

CASE 8. Male 45 years lesion chest, increasing in size over 12 months, palpable, firm.

Clinical Description:

- Papular lesion on chest
- 7mm diameter
- Well defined borders
- Regular shape
- Increasing in size over 12 mths

ABCD (EFG) Criteria:

- Symmetry in shape, but Asymmetry in colour

Three Point Checklist Score:

- Asymmetry?
    - Yes
- Irregular Network?
    - No Network (Many melanomas don't have network)
- Blue White Structures?
    - Yes

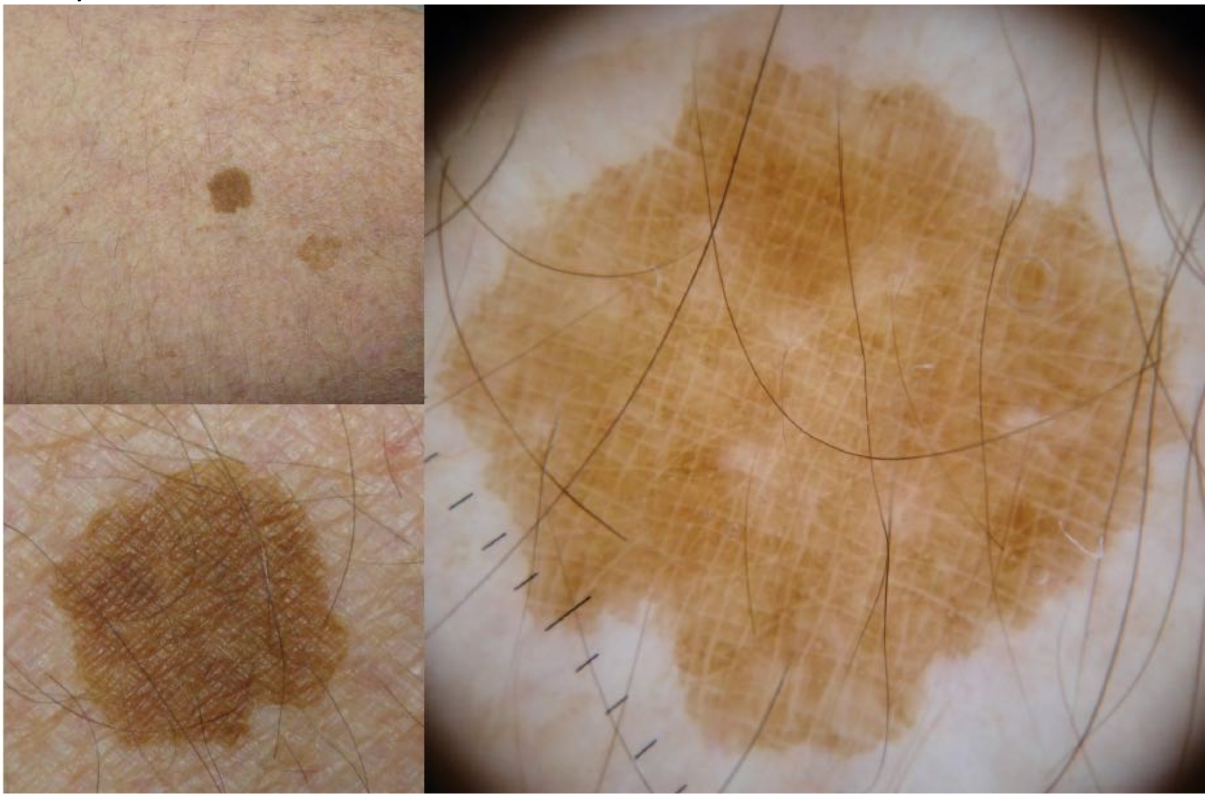
Provisional Diagnosis:

- Malignant
- Invasive Melanoma

Management:

- Excision

Case 9)



Clinical Description:

- Pigmented macule
- Symmetrical
- Well defined, regular border
- Enlarging
- 1cm diameter

ABCD (EFG) Criteria:

- Symmetrical
- Regular, well defined borders
- Regular consistent colour
- Diameter 10mm
- Evolving? Growing

Three Point Checklist Score:

- Asymmetry?
  - No
- Irregular network?
  - No
- Blue-white structures?
  - No

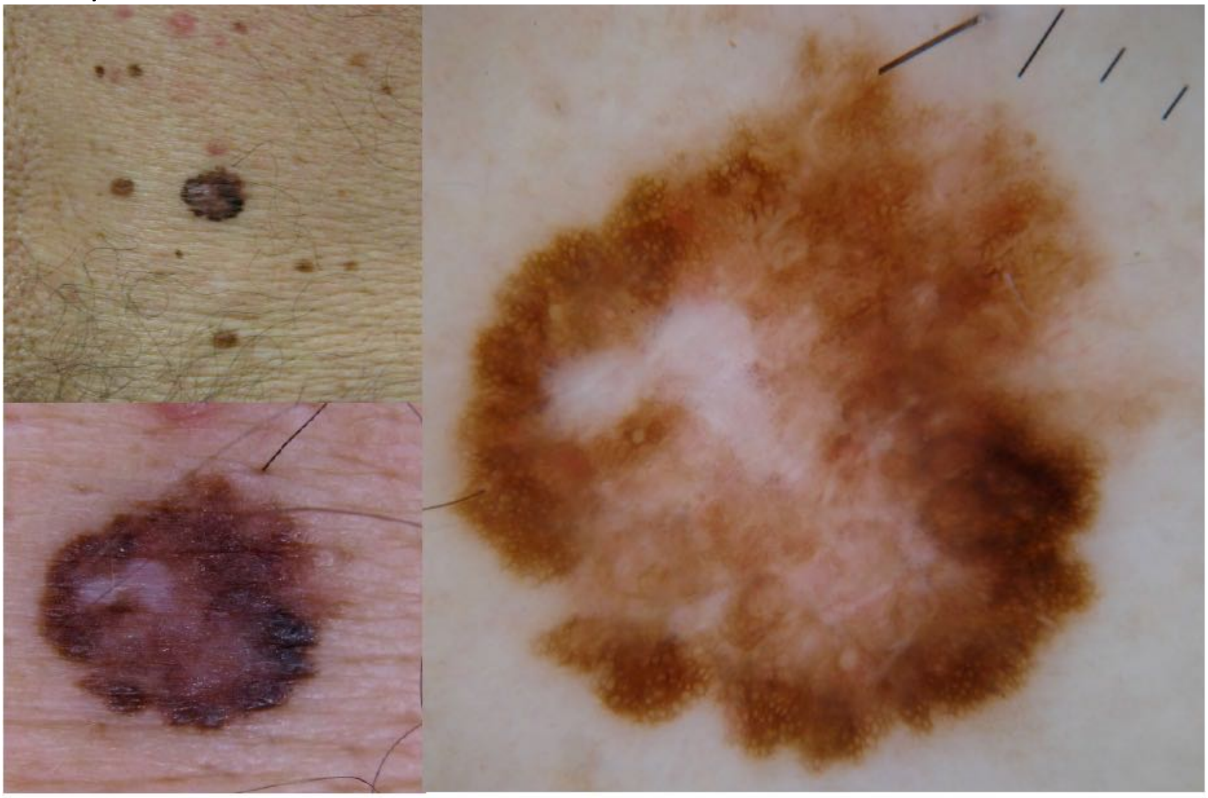
Provisional Diagnosis:

- Solar Lentigo

Management

- None Needed

Case 10)



CASE 10. Male 48 longstanding pigmented macule lower back.

Clinical Description:

- Pigmented Macule
- Irregular Colour & Border
- Asymmetrical
- Long-Standing
- Regression in the centre → white area

ABCD (EFG) Criteria:

- Asymmetrical
- Irregular Border
- Irregular Colour
- Diameter 9mm
- Evolving? NO

Three Point Checklist Score

- Asymmetrical?
  - Yes
- Irregular Network?
  - Yes
- Blue white structures?
  - Yes

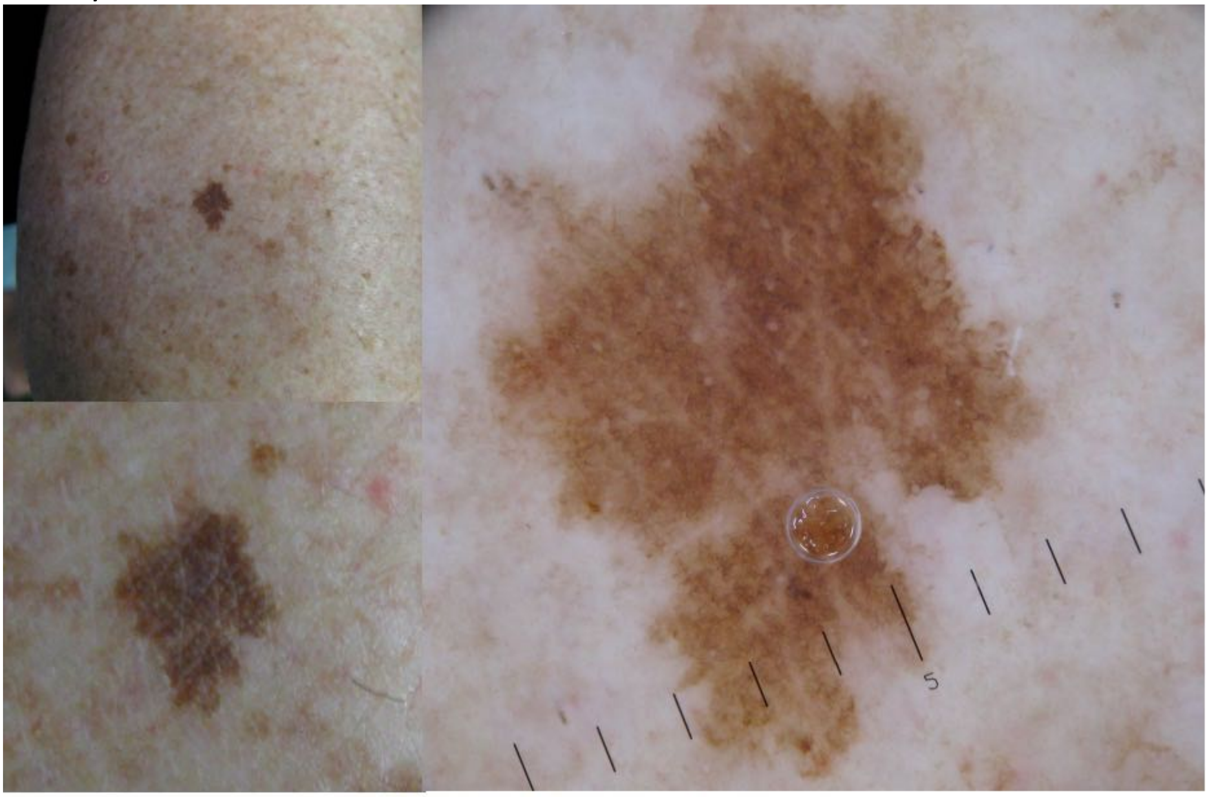
Provisional Diagnosis:

- Melanoma evolving inside a dysplastic naevi

Management:

- Excision

Case 11)



CASE 11. Female 71, enlarging pigmented macule shin.

Clinical Description:

- Enlarging Macule
- Irregular Border
- Consistent Colour
- Symmetrical
- Faint Network

ABCD (EFG) Criteria:

- Symmetrical
- Irregular Borders
- Consistent Colour
- 1cm diameter

Three Point Checklist Score

- Asymmetrical?
  - Yes
- Blue white Structures?
  - No
- Irregular Pigment Network?
  - No

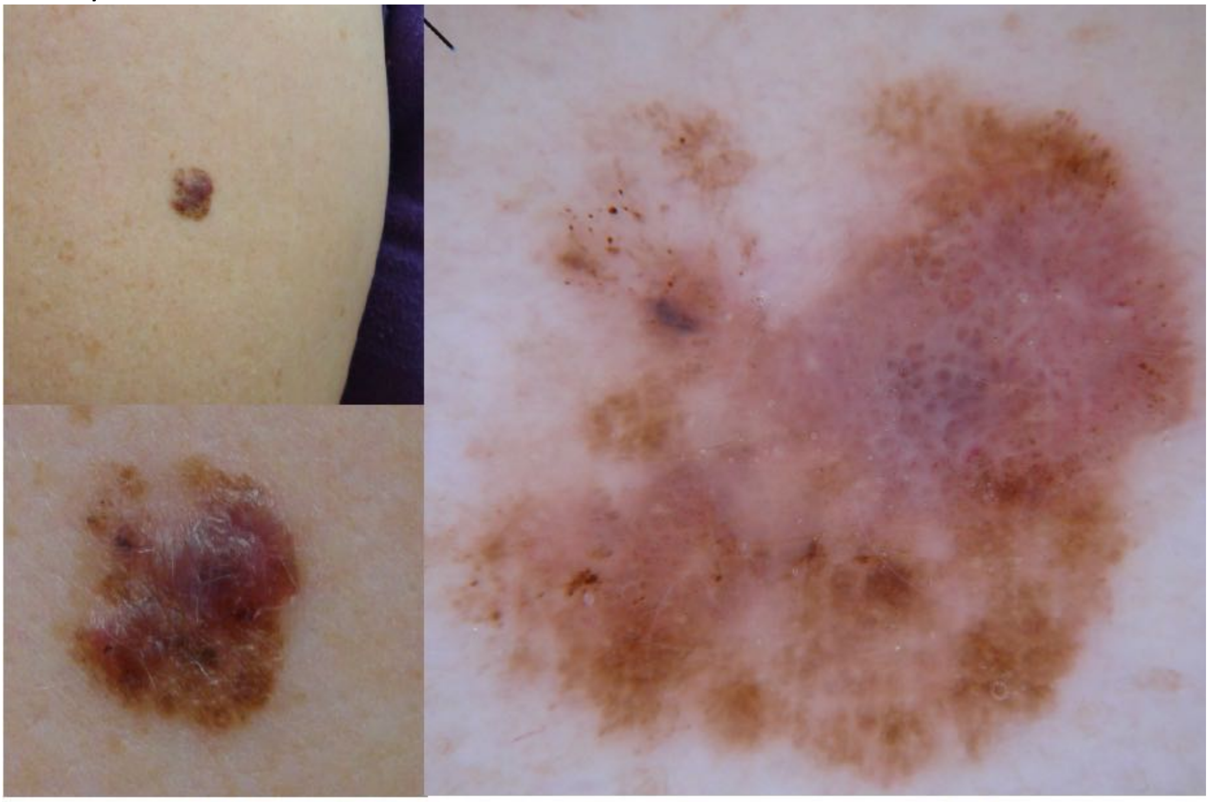
Provisional Diagnosis

- Solar Lentigo

Management

- None

Case 12)



CASE 12. Female 28 lesion over biceps longstanding recently traumatised, palpable but not firm/ hard.

Clinical Description

- Pigmented papule
- Irregular Border
- Asymmetrical
- Irregular colour pattern

ABCD (EFG) Criteria

- Asymmetrical
- Irregular Border
- Irregular Colour
- Diameter – Not Growing
- Evolving? no

Three Point Checklist Score

- Asymmetrical
    - Yes
- Blue white structures
    - Yes
- Atypical Network
    - yes

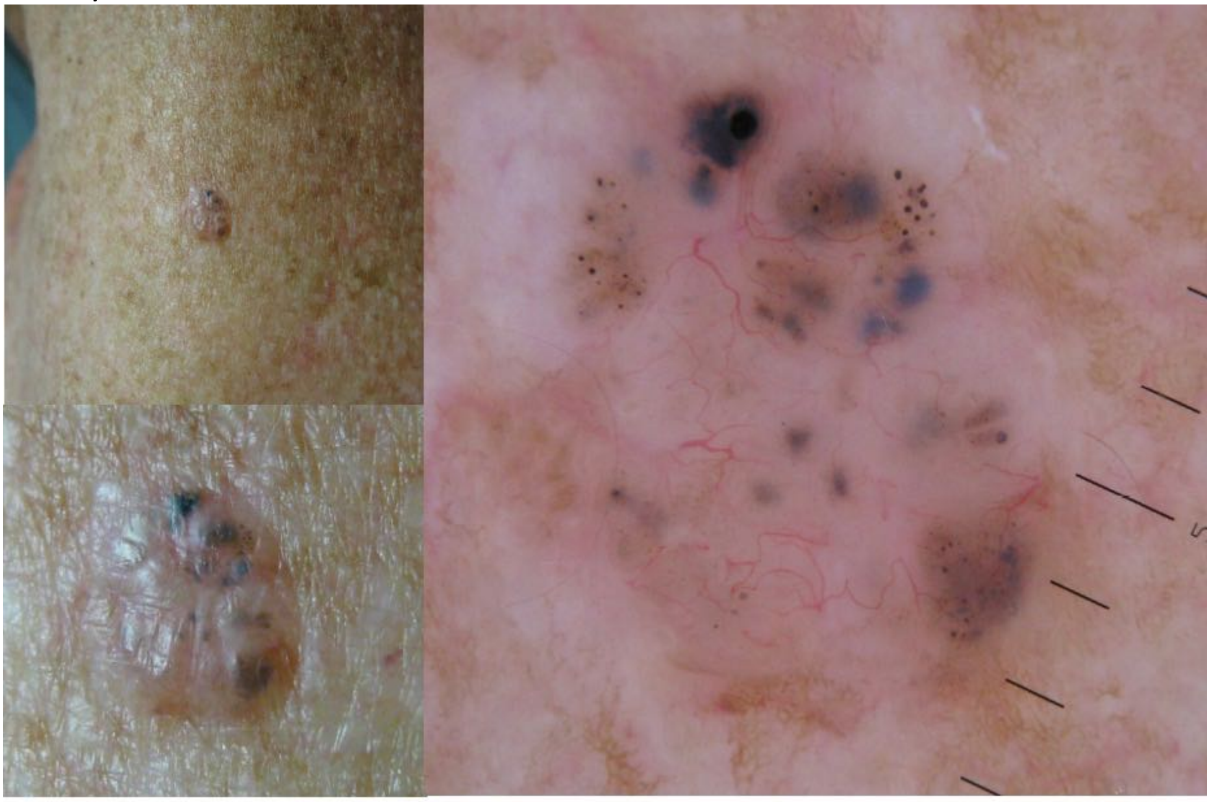
Provisional Diagnosis

- Melanoma

Management:

- Excision

Case 13)



CASE 13. Male 75 nodular lesion over supraspinatous.

Clinical Description:

- Telangiectasia

ABCD (EFG) Criteria:

- Asymmetrical
- Ill-defined border
- Irregular colours
- Diameter – 5mm
- Evolving - unknown

Three Point Checklist Score

- Asymmetrical:
    - Yes
- Blue White Structures?
    - Yes
- Irregular network?
    - no

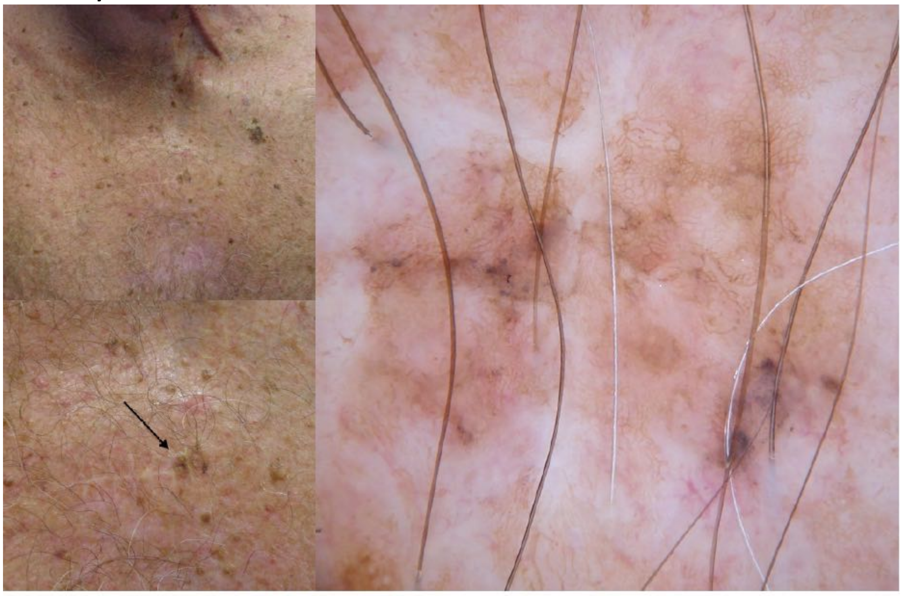
Provisional Diagnosis:

- BCC

Management

- Excision

Case 14)



CASE 15. Male 62 pigmented macule near scar from previous BCC excision.

Clinical Description

- Ugly duckling
- Poorly defined border

Three Point Checklist Score

- Irregular Network
    - yes

Provisional Diagnosis

- Invasive Melanoma

Case 15)

CASE 16. Female 67 scaly plaque on leg.

Clinical Description:

- Diffuse,
- Asymmetrical
- Pigmented lesion

Three Point Checklist Score

- Asymmetrical
- Irregular network
- White structures

Provisional Diagnosis:

- Melanoma Insitu

Case 16)

CASE 16. Female 67 scaly plaque on leg.

Clinical Description:

- Glomerular Vessels

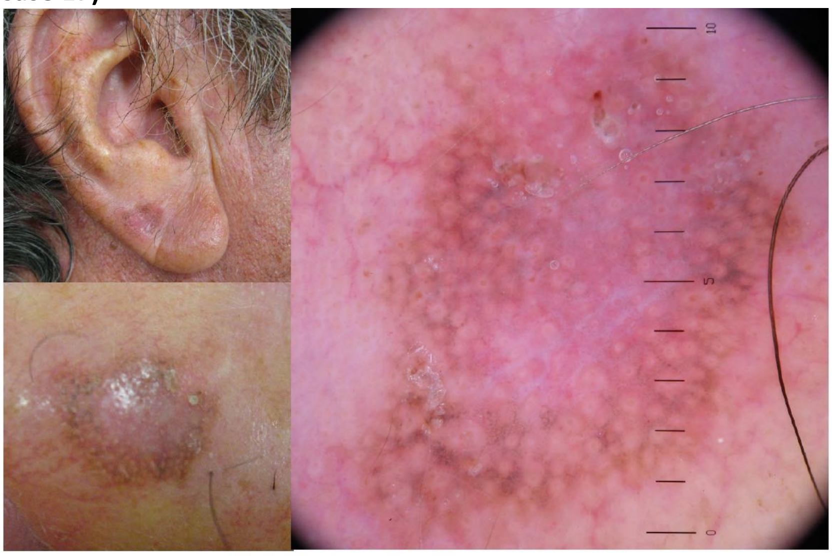
Three Point Checklist Score:

- Asymmetry

Provisional Diagnosis:

- Bowens Disease (Intraepidermal Carcinoma)

Case 17)



CASE 17. Male 68, lesion on the ear, history unknown, scaly macule.

Three Point Checklist Score:

- Irregular network
- Asymmetrical
- Inverse pigment Network

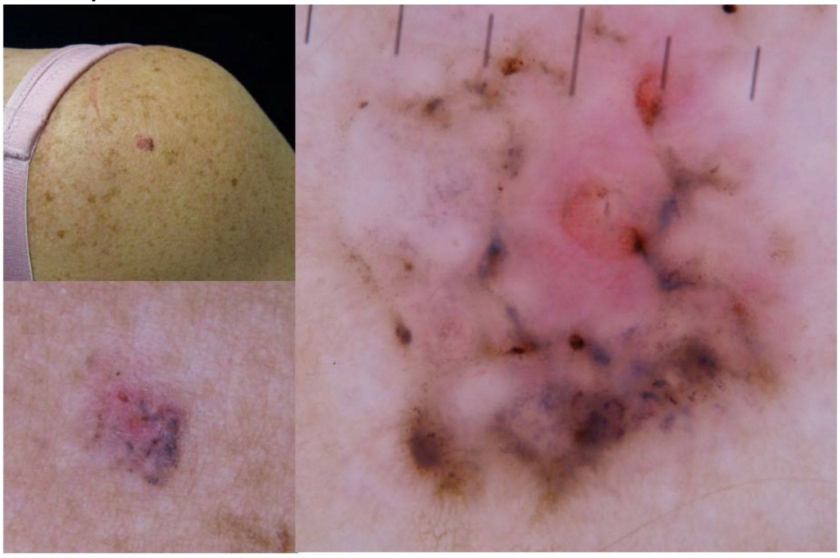
Provisional Diagnosis:

- Melanoma

Management:

- Excision

Case 18)



CASE 18. Female 39 lesion over shoulder.

Clinical Description

- Fine telangiectasia

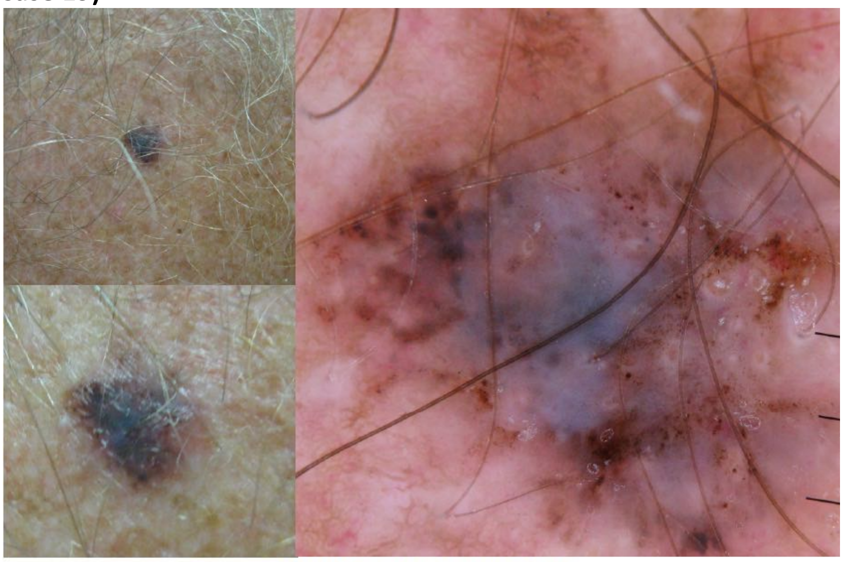
Three Point Checklist Score

- Asymmetrical
- Blue white structures

Provisional Diagnosis:

- BCC

Case 19)



CASE 19. Male 62 lesion upper back, history unkonwn.

Three Point Checklist Score

- Asymmetrical
- Blue white structures
- Irregular Pigment Network

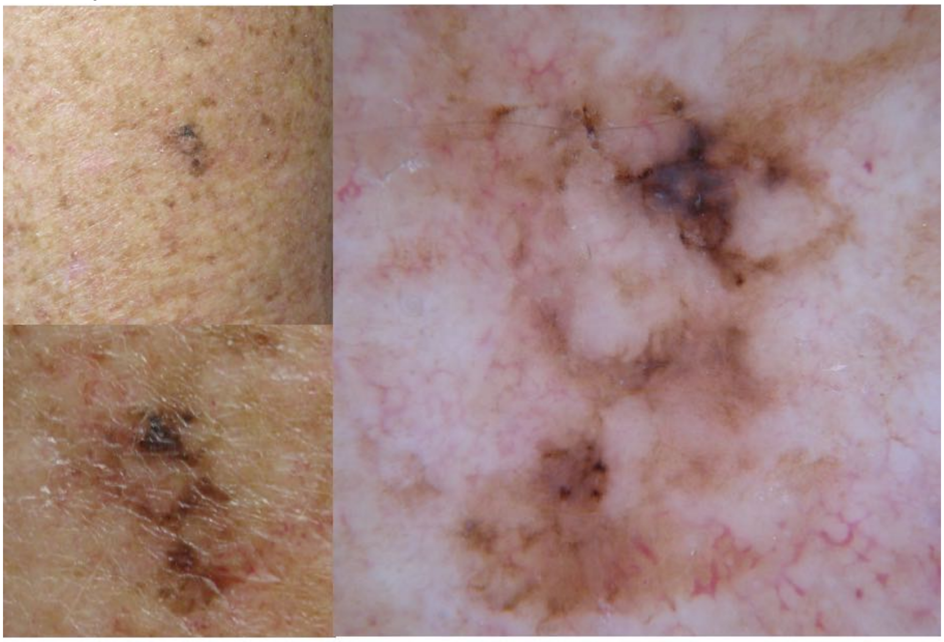
Provisional Diagnosis

- Melanoma

Management:

- Excision

Case 20)



CASE 20. Female 61 pigmented macule triceps.

Three Point Checklist Score:

- Asymmetrical
- White Structures

Provisional Diagnosis:

- Melanoma

Management

- Excision

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